



University of Barishal

Department of Computer Science and Engineering

SOFTWARE REQUIREMENTS SPECIFICATION (SRS Document)

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TABLE OF CONTENTS

Section / Subsection	Page
MAIN DOCUMENT	
1. INTRODUCTION	1
1.1 Purpose	1
1.2 Intended Audience and Reading Suggestions	2
1.3 Document Conventions	3
1.4 Project Scope	3
1.5 References	4
1.6 Definitions, Acronyms, and Abbreviations	5
2. OVERALL DESCRIPTION	6
2.1 Product Perspective	7
2.2 Product Functions	8
2.3 User Classes and Characteristics	10
2.3.1 Tenant / Buyer	10
2.3.2 Property Owner	11
2.3.3 Real Estate Agency	11
2.3.4 System Administrator	12
2.4 Operating Environment	12
2.5 Design and Implementation Constraints	13
2.6 Assumptions and Dependencies	15
2.6.1 Assumptions	15
2.6.2 Dependencies	15
2.7 Business Rules	16
2.8 Stakeholder Analysis	17
3. REQUIREMENTS MODELING	19
3.1 Requirements Elicitation Summary	20
3.2 Business Process Overview	21
3.2.1 User Registration and Verification Process	21
3.2.2 Property Search and Discovery Process	22
3.2.3 Property Listing and Publication Process	22
3.2.4 Tenant-Owner Communication Process	22
3.2.5 Verification, Reporting, and Moderation Process	23
3.2.6 Rental Agreement Generation Process	23

3.2.7 Rental Payment Tracking and Reminder Process	24
3.3 Actor Identification	24
3.4 Functional Decomposition	26
3.5 Use Case Modeling	28
3.6 Analysis Modeling.....	30
3.7 Domain Analysis.....	31
4. SPECIFIC REQUIREMENTS	33
4.1 Functional Requirements.....	34
4.1.1 Tenant / Buyer Requirements (FR-TB)	34
4.1.2 Property Owner and Agency Requirements (FR-OA).....	38
4.1.3 System Administrator Requirements (FR-AD)	41
4.2 External Interface Requirements	45
4.2.1 User Interface Requirements	45
4.2.2 Hardware Interfaces.....	46
4.2.3 Software Interfaces	46
4.2.4 Communication Interfaces	47
4.3 Non-Functional Requirements	49
4.3.1 Usability Requirements (NFR-US).....	49
4.3.2 Performance Requirements (NFR-PE)	50
4.3.3 Security Requirements (NFR-SE).....	51
4.3.4 Reliability Requirements (NFR-RL).....	52
4.3.5 Availability Requirements (NFR-AV)	52
4.3.6 Localization and Compliance Requirements (NFR-LC).....	53
4.3.7 Maintainability and Scalability Requirements	53
4.4 Requirements Summary.....	54
5. DATA REQUIREMENTS	55
5.1 Data Model Overview	56
5.2 Data Dictionary.....	58
5.2.1 User Entity.....	58
5.2.2 Listing Entity.....	59
5.2.3 Conversation and Message Entities	60
5.2.4 VerificationRequest Entity	61
5.2.5 RentalAgreement and Payment Entities	62
5.2.6 Additional Entities (Review, Report, Notification, AuditLog).....	63
5.3 Entity Relationship Summary.....	64

6. SYSTEM MODELS AND ANALYSIS MODELS	70
6.1 Context Diagram	71
6.2 Use Case Diagram	72
6.3 Detailed Use Case Specifications	74
6.4 Activity Diagrams	77
6.4.1 User Registration and Verification	77
6.4.2 Smart Property Search and Filtering	78
6.4.3 Property Listing Creation	79
6.4.4 Tenant Verification	80
6.4.5 Rental Agreement and Rent Tracking	81
6.5 Swimlane Diagrams	82
6.5.1 Property Search and Owner Inquiry	82
6.5.2 Listing Posting and Moderation	83
6.5.3 Rental Agreement and Payment Tracking	83
6.6 Sequence Diagrams	84
6.6.1 Registration, Login, and OTP Verification	84
6.6.2 Smart Property Search	85
6.6.3 Property Listing Creation	86
6.6.4 In-App Messaging	87
6.6.5 Verification Submission and Admin Review	88
6.7 State Transition Diagrams	89
6.7.1 User Account Verification Lifecycle	89
6.7.2 Property Listing Lifecycle	90
6.7.3 Rental Agreement Lifecycle	92
6.7.4 Report Moderation Lifecycle	92
6.8 Class Diagram	93
6.9 Entity Relationship Diagram (ERD)	94
6.10 Data Flow Diagrams	95
6.10.1 DFD Level 0 — Thikana Platform	95
6.10.2 DFD Level 1 — Property Discovery and Inquiry Flow	96
6.11 Decision Tables	97
6.11.1 Property Listing Visibility Decision Table	97
6.11.2 Tenant Verification Decision Table	98
6.12 Decision Tree	99
6.13 Model Coverage Summary	100
7. REQUIREMENTS TRACEABILITY MATRIX	101
7.1 Purpose and Methodology	101
7.2 RTM Key and Status Definitions	102

7.3 Forward Traceability — Requirements to Design	103
7.3.1 Tenant / Buyer Core Requirements	103
7.3.2 Search and Discovery Requirements	104
7.3.3 Listing Management Requirements	105
7.3.4 Communication Requirements	106
7.3.5 Tenant Verification Requirements (FR-30 to FR-34)	106
7.3.6 Agreement and Payment Requirements	107
7.3.7 Review and Reporting Requirements	107
7.3.8 Save Favorites and Alerts Requirements (FR-45 to FR-47)	108
7.3.9 Administrator Requirements	108
7.3.10 Owner Reputation and Analytics Requirements	109
7.4 NFR Traceability	112
7.5 Backward Traceability — Design Class to Requirements	116
7.6 Use Case — Class — Test Case Cross-Reference Matrix	118
7.7 Coverage Summary	120
7.8 RTM Analysis and Observations	121
7.8.1 Coverage Gaps and Deferred Items	121
7.8.2 Traceability Confidence Level	122
7.8.3 Change Impact Guidance	122
8. VALIDATION AND VERIFICATION	123
8.1 Requirement Review Strategy	123
8.1.1 Review Levels	123
8.1.2 Review Roles and Responsibilities	123
8.1.3 Review Techniques	124
8.1.4 Entry and Exit Criteria	124
8.1.5 Review Frequency	124
8.2 Inspection Checklist	124
8.3 Validation Rules	126
8.4 Acceptance Criteria	128
8.4.1 Requirements Baseline Acceptance	128
8.4.2 Sign-Off Criteria	131
9. RISK ANALYSIS	131
9.1 Technical Risks	131
9.2 Security Risks	134
9.3 Operational Risks	136
9.4 Schedule Risks	137
9.5 Resource Risks	138

9.6 Risk Matrix	140
9.7 Mitigation Plan Summary	141
10. TESTING CONSIDERATIONS	142
10.1 Testing Strategy Overview	142
10.2 Functional Testing	144
10.3 Integration Testing	145
10.4 System Testing	147
10.5 Security Testing	149
10.6 Performance Testing	151
10.7 User Acceptance Testing	152
APPENDICES	
Appendix A — Glossary	155
Appendix B — Open Issues List	164
Appendix C — Future Enhancements	168
Appendix D — Assumptions	170
Appendix E — References	173
E.1 Standards and Methodology References	174
E.2 Modeling and Diagramming Tools	174
E.3 Frontend Technology References	174
E.4 Backend and Database Technology References	174
E.5 Supporting Services and APIs	174
E.6 Domain and Regulatory References	175
E.7 Security and Quality References	175

1. INTRODUCTION

This Software Requirements Specification (SRS) document provides a comprehensive and structured description of **Thikana**, a Smart Real Estate and Rental Management Platform designed for the Bangladeshi property and rental market. The document defines the purpose, scope, major system capabilities, important conventions, and reference materials required for understanding the proposed system.

Thikana, meaning “address” in Bengali, is envisioned as a centralized web-based digital real estate ecosystem that connects tenants, property buyers, property owners, real estate agencies, and system administrators within a single integrated platform. The system is intended to reduce the common problems of property searching in Bangladesh, including scattered information, fake listings, informal broker dependency, lack of verification, weak communication, and limited access to area-based decision support.

The platform shall support property search, property listing, map-based exploration, tenant and owner verification, area intelligence, direct communication, favorites and alerts, rental agreement support, and owner or agency analytics. It shall provide a secure and user-friendly environment where property seekers can discover reliable listings and property providers can manage their listings professionally.

This SRS serves as an official reference for the development team, academic supervisor, testers, evaluators, and future contributors. It describes what the system shall do and the constraints under which the system shall operate. Detailed design decisions, source-code structure, database optimization, and deployment-specific implementation details are outside the scope of this introductory section and will be handled during later design and development phases.

1.1 Purpose

The purpose of this Software Requirements Specification is to formally define and document the requirements of **Thikana**. This section establishes the foundation for understanding the system vision, stakeholder expectations, and the boundaries of the initial academic prototype.

This document aims to:

- Establish a clear and shared understanding among the project team, instructor, developers, testers, and evaluators regarding the expected behavior of the system.
- Define the major functional areas of the platform, including user registration, property search, property listing, verification, communication, rental management, and analytics.
- Provide a reliable baseline for system design, development, testing, validation, academic assessment, and future enhancement.
- Identify the mandatory, recommended, and optional requirements that guide the development priority of the Minimum Viable Product (MVP).
- Support project planning by clarifying the system boundaries, constraints, assumptions, and external service dependencies.

- Serve as a reference document for future contributors who may extend the system into mobile applications, AI recommendation services, real payment integrations, or production-level deployment.

The initial academic version of **Thikana** shall focus on the core real estate workflow: user registration and authentication, property listing creation, smart property search and filtering, map-based property browsing, basic verification, and direct communication between tenants or buyers and property owners. Advanced features such as automated government NID verification, live payment settlement, AI-powered recommendation, and large-scale production analytics may be implemented in future increments.

1.2 Intended Audience and Reading Suggestions

This SRS is intended for a diverse set of readers, each of whom may approach the document differently based on their role and area of interest. The following table identifies the primary audience groups and provides reading suggestions to help each group navigate the document efficiently.

Audience	Primary Interest	Recommended Sections
Development Team	Functional and non-functional requirements; system constraints and interfaces	Section 1 (all), Section 2, Section 3
Academic Supervisor / Evaluator	Overall system vision, scope boundaries, and requirement completeness	Section 1.1, 1.4, Section 2 overview, Section 3
QA Testers	Testable functional requirements, acceptance criteria, and use cases	Section 1.3, Section 3, Section 4
Future Contributors	System scope, out-of-scope features, and technology references	Section 1.4, Section 1.5, Section 2.6

1.3 Document Conventions

This document uses a consistent set of conventions to ensure clarity, traceability, and unambiguous interpretation of requirements. These conventions are applied throughout the SRS whenever functional requirements, non-functional requirements, priorities, roles, and system modules are discussed.

Convention	Meaning
SHALL	Indicates a mandatory requirement. The system must satisfy this requirement unconditionally.
SHOULD	Indicates a recommended requirement. It is strongly desirable but may not be mandatory for the initial prototype.
MAY	Indicates an optional requirement or future enhancement that may be implemented at a later stage.
TBD	To Be Determined. The requirement detail is not yet finalized and will be resolved in a later revision.
FR-XX	Functional Requirement identifier, such as FR-01, FR-02, and FR-03.

NFR-XX	Non-Functional Requirement identifier, such as NFR-01, NFR-02, and NFR-03.
UC-XX	Use Case identifier, such as UC-01, UC-02, and UC-03.
High	A critical requirement that must be implemented in the MVP or earliest development increment.
Medium	An important requirement that improves operational completeness and should be implemented after core features.
Low	An optional or future enhancement that is not essential for the initial academic prototype.
MVP	Minimum Viable Product — the first working version of the system containing the most essential features.
Bold Text	Used to highlight important system names, modules, user roles, or key terms on first use.

1.4 Project Scope

Thikana is a web-based smart real estate and rental management platform designed to address the challenges of finding, listing, verifying, and managing properties in Bangladesh. The platform provides a unified environment where property seekers can search for reliable properties and property providers can publish and manage listings in a structured manner.

The product scope includes property rental, property purchase, sublet search, bachelor housing search, office space discovery, owner and agency listing management, tenant verification, map-based browsing, area intelligence, direct communication, agreement support, rent reminders, and analytics. The system shall be designed as an academic prototype first, with a future path toward production deployment and mobile application support.

The key business goals of **Thikana** are as follows:

- Provide a centralized and organized platform for searching rental and sale properties in Bangladesh.
- Allow property owners and real estate agencies to create, edit, publish, deactivate, and monitor property listings.
- Enable tenants and buyers to search properties using keyword search, price range, location, property type, room count, furnished status, amenities, and map-based filters.
- Improve trust and safety through OTP-based phone verification, NID or student ID document submission, admin approval, and user report handling.
- Provide area intelligence such as nearby schools, colleges, hospitals, pharmacies, markets, mosques, transport stops, and commute estimates.
- Support direct in-app communication between tenants or buyers and property owners without depending entirely on external messaging platforms.
- Support rental agreement generation, rent due reminders, and payment history tracking for property management after a successful tenant-owner interaction.
- Provide dashboards and analytics for property owners, real estate agencies, and administrators to support data-driven decisions.

- Reduce fake listings, information gaps, communication delays, and unnecessary broker dependency in the property search process.

The major user classes supported by the system are summarized below:

User Class	Description	Key Capabilities
Tenant / Buyer	General users searching for rental flats, bachelor rooms, sublets, offices, or properties for purchase.	Search and filter listings, view property details, save favorites, communicate with owners, view area intelligence, and submit reviews.
Property Owner	Individuals who own one or more properties and want to rent or sell them through the platform.	Create and manage listings, upload photos and floor plans, update availability, respond to enquiries, and view listing analytics.
Real Estate Agency	Professional agencies managing multiple properties and client enquiries.	Manage multiple listings, monitor enquiries, communicate with users, and access agency-level reports.
System Administrator	Platform operator responsible for verification, moderation, user control, and system supervision.	Verify users and documents, moderate listings and reviews, handle reports, suspend accounts, and monitor platform activity.

The system scope for the initial academic prototype shall include the following features:

- User registration, login, OTP verification, and role-based dashboard access.
- Property listing creation with title, description, type, location, price, contact details, photos, and availability status.
- Smart property search and filtering with list view and map view.
- Basic integration with Google Maps or Mapbox for location display and property pin visualization.
- Manual tenant or owner verification through document upload and admin review.
- Direct messaging between property seekers and property owners or agencies.
- Favorites, alerts, owner ratings, reviews, and basic listing analytics.
- Admin dashboard for verification, moderation, report handling, and system oversight.

The following items are considered outside the mandatory scope of the first academic prototype and may be treated as future enhancements:

- Fully automated government NID database verification.
- Live payment processing with real merchant settlement.
- Native Android and iOS mobile applications.
- AI-based property recommendation and automated fraud detection.
- Advanced roommate compatibility scoring and smart matching.
- Legally binding digital signature integration with government-recognized verification.
- Large-scale production deployment covering all districts of Bangladesh.

1.5 References

The following documents, standards, technologies, and reference materials were consulted or are relevant to the preparation of this SRS document and the development of the **Thikana** platform:

1. IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications.
2. ISO/IEC/IEEE 29148 — Systems and Software Engineering: Requirements Engineering.
3. **Thikana** Project Proposal — Full Stack Alchemists Team, University of Barishal, May 2026.
4. **Thikana** Project Overview Document — Smart Real Estate & Rental Management Platform.
5. **Thikana** Presentation Slide Deck — Full Stack Alchemists Team.
6. Google Maps Platform API Documentation — for maps, geocoding, places, and directions services.
7. Firebase Documentation — for authentication, storage, real-time database, and cloud messaging services.
8. SSLCommerz Payment Gateway Documentation — for Bangladesh-based online payment processing.
9. bKash and Nagad Payment Gateway Documentation — for future mobile financial service integration.
10. OWASP Web Application Security Guidelines — for secure web application development.
11. WCAG 2.1 Web Content Accessibility Guidelines — for accessible user interface design.
12. React.js Documentation — for frontend development.
13. Node.js / Laravel Documentation — for backend development options.
14. MySQL 8.0 Documentation — for relational database design and management.
15. JSON Web Token (JWT) RFC 7519 — for stateless authentication.
16. Bangladesh Digital Security Act 2018 — for data protection and content moderation considerations.

1.6 Definitions, Acronyms, and Abbreviations

The following terms, acronyms, and abbreviations are used throughout this SRS document. All readers are advised to familiarize themselves with these definitions before proceeding to subsequent sections.

Term / Acronym	Definition
SRS	Software Requirements Specification. A document that describes the intended purpose, environment, and functional and non-functional requirements of a software system.
MVP	Minimum Viable Product. The first working version of Thikana containing only the most essential and core features required for academic demonstration and initial validation.
OTP	One-Time Password. A temporary, single-use numeric or alphanumeric code sent via SMS to verify a user's phone number during registration or login.
NID	National Identity Document (Card). The government-issued identity card in Bangladesh used for citizen verification purposes on the Thikana platform.
JWT	JSON Web Token. A compact, URL-safe token format defined by RFC 7519, used for stateless, secure authentication between the client and server in Thikana.

Term / Acronym	Definition
API	Application Programming Interface. A set of defined rules and protocols that allows different software components or external services to communicate with each other.
UI / UX	User Interface / User Experience. UI refers to the visual elements and layout of the application; UX refers to the overall experience and usability felt by the end user.
MFS	Mobile Financial Service. Digital payment services such as bKash and Nagad that are widely used in Bangladesh for cashless transactions.
RBAC	Role-Based Access Control. A security mechanism that restricts system features and data access based on the authenticated user's assigned role (e.g., Tenant, Owner, Admin).
Thikana	The name of the platform being specified in this document. Means "address" in Bengali. Refers to the Smart Real Estate and Rental Management Platform developed by Team Full Stack Alchemists.
FR / NFR	Functional Requirement / Non-Functional Requirement. FR defines what the system shall do; NFR defines how well the system shall perform it (e.g., performance, security, usability).

2. OVERALL DESCRIPTION

This section provides a high-level description of **Thikana**, its relationship with the real estate and rental market of Bangladesh, the major functions it shall provide, the user classes it will serve, the operating environment in which it will function, the constraints that govern its development, and the assumptions and dependencies on which the system is based. The purpose of this section is to give all stakeholders a complete understanding of the system before the detailed requirements are presented in later sections.

Thikana is intended to work as a centralized digital platform for property discovery, listing management, tenant-owner communication, verification, area intelligence, rental documentation, and administrative monitoring. It is designed as an academic prototype first, while keeping a clear path for future production deployment and mobile application support.

2.1 Product Perspective

Thikana is a new, self-contained smart real estate and rental management platform designed for the Bangladeshi market. It is not a simple property advertisement board; rather, it is planned as a complete digital property ecosystem that connects tenants, buyers, property owners, real estate agencies, and administrators through a unified web-based interface.

In the current market, many users depend on Facebook groups, informal brokers, scattered online posts, and word-of-mouth communication to find rental homes, bachelor rooms, sublets, office spaces, or properties for sale. These methods often lack structured filtering, verified user identity, authentic listing information, secure communication, and area-based decision support. **Thikana** shall address these limitations by providing an organized, searchable, verified, and data-supported property platform.

The first version of **Thikana** shall be implemented as a responsive web application. A dedicated mobile application may be introduced in future increments. The system shall interact with external platforms and services such as Google Maps or **Mapbox** for location services, Firebase for real-time messaging and notification features, Firebase Storage or AWS S3 for media storage, SMS gateway services for OTP delivery, and sandbox payment gateways such as **SSLCommerz**, bKash, or Nagad for payment-related prototype demonstrations.

From an architectural perspective, the platform shall follow a multi-tier web application structure. The presentation tier shall provide responsive user interfaces for tenants, buyers, owners, agencies, and admins. The application tier shall manage business logic such as search, listing validation, verification workflow, messaging, rental agreement preparation, and reporting. The data tier shall persist user accounts, listings, reviews, messages, agreements, reports, and payment records using MySQL and supporting cloud storage services.

Key architectural characteristics of **Thikana** include:

- I. Role-based access control to ensure that tenants, buyers, owners, agencies, and administrators can access only the functions appropriate to their roles.

- II. Responsive web design so that users can access the platform from desktops, laptops, tablets, and smartphones.
- III. Map-based property visualization using Google Maps API or Mapbox for property pins, geocoding, nearby facility discovery, and commute estimation.
- IV. Real-time communication and notification support through Firebase and Firebase Cloud Messaging.
- V. Secure authentication using OTP verification, JWT-based sessions, and optional OAuth 2.0 login.
- VI. Media management support for property photos, floor plans, NID/student ID documents, and future 360-degree virtual tour images.
- VII. Administrative moderation and verification workflows to reduce fake listings, fraudulent accounts, and suspicious property activity.
- VIII. A modular structure that allows future extension into mobile apps, AI-based recommendations, live payment integration, and production-scale analytics.

2.2 Product Functions

Thikana shall provide the following major functional areas. Each area is described at a summary level here; detailed functional and non-functional requirements shall be specified in later sections of the SRS.

Functional Area	Summary Description
User Registration and Authentication	Registration, login, OTP verification, password reset, role selection, JWT-based authentication, and optional Google OAuth login.
User Profile Management	Management of tenant, buyer, owner, agency, and admin profiles including contact details, verification status, photos, and account activity.
Smart Property Search and Filtering	Keyword-based search, advanced filters, sorting, result count display, list view, and future AI-powered recommendation support.
Map-Based Property Discovery	Interactive map view with property pins, location preview, distance display, and geocoding support for Bangladeshi addresses.
Property Listing Management	Creation, editing, publishing, deactivation, and availability control of property listings by owners and agencies.

Media and Virtual Tour Management	Upload and management of property photos, floor plans, documents, and 360-degree panoramic views for virtual property exploration.
Area Intelligence System	Nearby facility information including schools, colleges, hospitals, pharmacies, markets, mosques, transport points, safety rating, and commute estimation.
Tenant and Owner Verification	Document submission, admin review, approval or rejection workflow, verified badges, and suspicious user reporting.
Direct Communication System	In-app messaging between tenants/buyers and owners/agencies, real-time notifications, message history, and controlled phone number reveal.
Rental Agreement Management	Digital rental agreement generation, tenant-owner review, acceptance or rejection, agreement storage, and property status update.
Payment and Rent Tracking	Sandbox payment demonstration, rent reminder generation, payment history logging, and utility expense tracking for future expansion.
Favorites and Alerts	Saving favorite listings, removing favorites, receiving price change alerts, and receiving status update notifications.
Roommate Matching	Future feature allowing users to post roommate requests and connect with compatible users based on location, budget, and preferences.
Review and Reputation Management	Tenant reviews, star ratings, owner reputation display, review moderation, and listing credibility support.
Owner and Agency Analytics	Listing view count, save count, enquiry count, trend reports, and performance dashboards for owners and agencies.
Administration and Moderation	Admin dashboard, user verification, listing moderation, report handling, suspicious account suspension, and platform-wide monitoring.

Reporting and Monitoring	Administrative reports including total users, listings, enquiries, verification logs, user activity, and transaction summaries.
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2.3 User Classes and Characteristics

Thikana is designed to serve four primary human user classes. Each class has distinct goals, technical literacy levels, access rights, and expected system usage patterns. Understanding these characteristics is important for designing suitable interfaces, role-based access control, and validation rules.

Attribute	Tenant / Buyer	Property Owner	Real Estate Agency	System Administrator
Population	Very high; general public users across Bangladesh	Medium; individual landlords and property holders	Low to medium; professional real estate teams	Very low; platform operators
Technical Literacy	Basic to intermediate	Basic to intermediate	Intermediate	Intermediate to advanced
Usage Frequency	Occasional to daily during property search	Weekly to daily while managing listings	Daily for multi-property management	Daily or as needed for moderation
Primary Device	Mobile phone, laptop, desktop	Mobile phone, laptop, desktop	Desktop or laptop	Desktop or laptop
Access Level	Self-service access to search, favorites, messages, profile, reviews	Own listing management and enquiry access	Agency-level listing and analytics access	Full system oversight and moderation access
Training Required	Minimal; intuitive UI required	Short orientation for listing management	Moderate training for dashboards and reporting	Structured admin training required

2.3.1 Tenant / Buyer

Tenants and buyers represent the largest user group of the platform. This group includes students, job holders, families, bachelors, small business owners, and general citizens looking for rental properties, sublets, offices, or properties for purchase. Their technical literacy may vary widely, so the interface must be simple, mobile-friendly, and easy to navigate.

Tenants and buyers primarily use the system to search and filter properties, view property details, explore map locations, save favorite listings, communicate with property owners, submit verification documents, view area intelligence, book visits, and submit reviews after completed interactions.

2.3.2 Property Owner

Property owners are individuals who own one or more properties and want to rent or sell them through the **Thikana** platform. They may include landlords, apartment owners, house owners, office space owners, and individual sellers. Some owners may be familiar with digital tools, while others may require a simple listing workflow.

Property owners use the platform to create listings, upload photos and floor plans, update price and availability, respond to tenant enquiries, generate rental agreements, reveal contact numbers to verified users, and monitor listing activity such as views, saves, and enquiry counts.

2.3.3 Real Estate Agency

Real estate agencies are professional organizations or teams that manage multiple property listings on behalf of owners or clients. They are expected to use the system more frequently and require stronger dashboard, filtering, analytics, and bulk listing management capabilities than individual property owners.

Agency users shall be able to manage multiple listings, monitor tenant or buyer enquiries, coordinate bookings, access agency-level analytics, update listing statuses, and communicate with interested users through the platform.

2.3.4 System Administrator

System administrators are platform operators responsible for maintaining trust, safety, and operational quality within **Thikana**. They may include technical or non-technical staff assigned to verify users, moderate listings, handle reports, and monitor system activity.

Administrators shall have the highest level of access within the platform. They will review NID or student ID submissions, approve or reject verification requests, moderate suspicious listings and reviews, suspend or ban users when necessary, monitor platform-wide statistics, and ensure that system rules are followed.

2.4 Operating Environment

Thikana shall operate in a modern web-based environment. The system must support common browsers, responsive devices, cloud-based media storage, database services, and external APIs required for maps, messaging, authentication, and payment-related demonstrations. The operating environment is summarized below.

Environment Component	Specification
Server Platform	Ubuntu Server 20.04 LTS or equivalent Linux distribution. Prototype deployment may use shared hosting or VPS; production deployment should support cloud scaling.
Application Server	Node.js v18+ or Laravel/PHP 8.1+ backend, selected according to project implementation plan. RESTful API endpoints shall serve frontend and future mobile clients.
Frontend Platform	React.js with Tailwind CSS or equivalent component-based responsive framework.
Database Server	MySQL 8.0+ for relational data such as users, listings, agreements, payments, reviews, reports, and favorites.
Real-Time Data Service	Firebase Realtime Database or Firestore for chat messages, notification counters, and online status indicators.
Cloud Storage	Firebase Storage or AWS S3 for property photos, floor plans, virtual tour images, and uploaded verification documents.
Client Browsers	Google Chrome 100+, Mozilla Firefox 95+, Microsoft Edge 95+, Safari 14+ or the latest stable versions available to users.
Mobile Support	Responsive web design supporting Android and iOS browsers with minimum screen width of 360px.
Network	Core browsing shall work over low-bandwidth connections; 4G or broadband is recommended for image-heavy listings and 360-degree tours.

Map Services	Google Maps Platform or Mapbox for property location, map pins, geocoding, nearby places, and directions/commute estimates.
Authentication Services	Phone OTP through Firebase Auth or SMS gateway; JWT token-based session management; optional Google OAuth 2.0 login.
Payment Gateway	SSLCommerz, bKash, or Nagad sandbox integration for prototype payment demonstration and future live payment support.
Notification Services	Firebase Cloud Messaging, SMTP email service, and SMS gateway for alerts, reminders, OTP, and transactional messages.

2.5 Design and Implementation Constraints

The design and implementation of Thikana are subject to several academic, technical, financial, legal, and operational constraints. These constraints must be considered throughout development, testing, and deployment.

ID	Constraint	Description
CON-01	Technology Stack	The system shall use React.js for the frontend, Node.js or Laravel for the backend, and MySQL with Firebase-supported services for data storage and real-time features unless formally changed by the development team.
CON-02	Academic Timeline	The initial academic prototype shall be completed within the defined project timeline. Core features such as authentication, listing, search, and map view shall receive priority over advanced features.
CON-03	Budget	The initial student project budget is limited. Paid API features beyond free tiers shall be minimized, simulated, or deferred to future increments.
CON-04	NID Verification	Direct government NID database verification is not available for the academic prototype. NID or student ID verification shall be handled through simulated document upload and manual admin review.

ID	Constraint	Description
CON-05	Payment Integration	Live merchant payment processing requires gateway approval. The prototype shall use sandbox or simulated payment flows only.
CON-06	Data Availability	Large-scale real property data is not available at the prototype stage. The system shall use seeded test data and user-submitted listings.
CON-07	Security	Sensitive data such as passwords, phone numbers, emails, and identity documents shall be protected through hashing, access control, private storage, and encrypted communication.
CON-08	Browser Compatibility	The frontend shall function on modern browsers without depending on proprietary browser plugins.
CON-09	Map API Quota	Map-based features depend on third-party API usage limits. The prototype shall be designed to remain within free or educational usage limits.
CON-10	Localization	The platform shall support Bangladesh-specific phone number formats, BDT currency, Asia/Dhaka timezone, and Bangladeshi address hierarchy.
CON-11	Maintainability	API keys, database credentials, JWT secrets, and environment-specific settings shall be stored in environment variables and not hardcoded in source code.
CON-12	Version Control	All source code shall be maintained in a Git repository with meaningful commit messages and a clear branching strategy.

2.6 Assumptions and Dependencies

2.6.1 Assumptions

The following assumptions have been made during requirements definition. If any assumption proves incorrect, affected requirements may need to be revised in a later version of the SRS.

- I. Users will have access to a smartphone, laptop, desktop computer, or tablet with an internet connection.
- II. Property owners and agencies will voluntarily provide accurate property information, photographs, pricing, location, and availability status.
- III. The Google Maps or Mapbox free tier will be sufficient for prototype-level map and area intelligence demonstrations.
- IV. Firebase free-tier services will be adequate for the expected academic prototype workload.
- V. NID verification will be simulated through document upload and manual admin approval rather than direct government API integration.
- VI. Payment gateway integration will use sandbox mode and will not process real financial transactions during the academic phase.
- VII. The development team will seed the system with sample listings where real property data is not available at scale.
- VIII. Users will accept SMS, email, or in-app notifications for important actions such as OTP verification, messages, reminders, and reports.
- IX. Future production deployment will provide more robust infrastructure, paid API plans, legal review, and operational support.

2.6.2 Dependencies

The following external dependencies may affect the successful operation or delivery of **Thikana**:

- I. Google Maps Platform or Mapbox availability for map rendering, geocoding, Places API, and directions/commute estimation.
- II. Firebase services for authentication, real-time messaging, cloud storage, push notifications, and online status updates.
- III. SMS gateway availability for OTP delivery to Bangladeshi mobile numbers.
- IV. SMTP or email service availability for account confirmation, password reset, rental agreement notifications, and reminders.

- V. SSLCommerz, bKash, or Nagad sandbox API availability for payment demonstration and future payment integration.
- VI. Stable Node.js, Laravel, React.js, MySQL, and Firebase SDK dependencies during development.
- VII. Availability of internet connectivity for users, especially for image-heavy property details, map view, and virtual tour features.
- VIII. Availability of adequate cloud storage for uploaded media and verification documents.
- IX. Timely decisions by the project team regarding final backend framework, hosting method, and API service provider selection.

2.7 Business Rules

Business rules define the policies and constraints that the Thikana system must enforce. These rules are derived from expected real estate platform behavior, user trust requirements, academic prototype constraints, and Bangladeshi market needs.

Rule ID	Module	Business Rule Description
BR-001	Registration	A user SHALL NOT be allowed to create more than one account using the same phone number or email address.
BR-002	Authentication	A user account SHALL be considered verified only after successful OTP verification or admin-approved document verification, depending on the verification type.
BR-003	Role Access	A tenant or buyer SHALL NOT be permitted to access owner, agency, or administrator-only dashboard functions.
BR-004	Listing Creation	A property listing SHALL NOT be published unless mandatory fields such as title, property type, location, price, and owner contact information are provided.
BR-005	Listing Availability	A listing marked as Rented, Sold, Under Negotiation, or Inactive SHALL NOT be displayed as an available listing in public search results.
BR-006	Listing Media	Uploaded property images SHALL be limited to approved file types such as JPEG, PNG, and WEBP, and shall be checked before public display.

BR-007	Verification	NID, student ID, or verification documents SHALL be accessible only to authorized administrators and SHALL NOT be publicly visible.
BR-008	Reporting	A listing or user reported as fraudulent SHALL be flagged for administrator review and may be temporarily hidden depending on report severity.
BR-009	Communication	Only authenticated users SHALL be allowed to send messages through the in-app communication system.
BR-010	Phone Reveal	An owner phone number SHALL only be revealed to verified tenants or buyers when the owner has permitted phone visibility.
BR-011	Rental Agreement	A rental agreement SHALL be generated only after a valid owner-tenant interaction and SHALL require acceptance from both parties.
BR-012	Payment Tracking	All payment or rent tracking records SHALL contain amount, date, method, status, and related agreement information.
BR-013	Review and Rating	A user SHALL be allowed to rate or review an owner only after a verified interaction, message, booking, or agreement event.
BR-014	Soft Deletion	Deleted or deactivated listings SHALL be soft-deleted for a defined retention period before permanent removal.
BR-015	Audit Logging	Critical actions such as listing creation, listing update, verification approval, report handling, and agreement acceptance SHALL be recorded with timestamp and user identity.

2.8 Stakeholder Analysis

A stakeholder is any individual, group, or organization that has an interest in or is affected by the Thikana platform. Understanding stakeholder needs, concerns, and influence is important for prioritizing requirements and ensuring system acceptance.

Stakeholder	Role / Interest	Key Concerns	Influence	Engagement Strategy
Tenants / Buyers	Primary end-users searching for rental or purchase properties.	Ease of search, listing accuracy, safety, affordability, mobile access, privacy.	High - large user base	User surveys, prototype testing, usability feedback.
Property Owners	Users who post and manage properties for rent or sale.	Simple listing process, reliable enquiries, privacy, listing visibility, trust.	High - content providers	Owner interviews, listing workflow review, prototype demonstration.
Real Estate Agencies	Professional users managing multiple listings and enquiries.	Bulk management, analytics, professional visibility, lead quality.	Medium to High	Dashboard review sessions, feature prioritization discussions.
System Administrators	Platform operators responsible for moderation, verification, and control.	Admin tools, security, verification accuracy, report handling, system monitoring.	High - operational controllers	Admin workflow walkthroughs and requirement validation.
Project Supervisor / Instructor	Academic evaluator and project guide.	Requirement quality, feasibility, academic depth, documentation, prototype demonstration.	High - approval authority	Milestone review, SRS review, presentation feedback.
Development Team	Designs, implements, tests,	Clear requirements, realistic scope, technical feasibility,	High - delivery responsibility	Sprint planning, code reviews, SRS sign-off sessions.

	and maintains the platform.	schedule, maintainability.		
QA Tester	Validates system functions and quality attributes.	Testable requirements, acceptance criteria, defect tracking, security and usability testing.	Medium	Test planning sessions and requirement traceability review.
Payment Gateway Providers	External services for future payment processing.	API reliability, merchant approval, transaction security, sandbox availability.	Medium - external dependency	API documentation review and sandbox integration testing.
Map and Location Service Providers	External services providing geocoding, maps, places, and directions.	API quota, accuracy, pricing, availability, location coverage.	Medium - external dependency	Provider evaluation and technical feasibility review.
Legal / Compliance Concern	Ensures responsible handling of personal data and identity documents.	Privacy, consent, data retention, NID handling, content moderation.	Medium to High	Compliance checklist and policy review before production use.

3 REQUIREMENTS MODELING

This section describes the activities and artifacts produced during the requirements modeling phase of the **Thikana** project. Requirements modeling bridges the gap between informal stakeholder needs and precise, testable software requirements. It explains how the requirements were elicited, how the core real estate and rental business processes are represented, which actors interact with the system, how the platform is decomposed into functional modules, and how the major use cases and analysis models are organized for later design and implementation.

For **Thikana**, requirements modeling is especially important because the platform combines multiple domains: property search, property listing, tenant-owner communication, identity verification, location intelligence, rental agreement management, payment tracking, and administrative moderation. The models in this section provide a structured foundation for the detailed functional requirements and system models that appear in later sections of this SRS.

3.1 Requirements Elicitation Summary

Requirements elicitation for **Thikana** was conducted using a combination of stakeholder discussion, market observation, document analysis, prototype review, and technology feasibility assessment. The objective was to identify the real problems faced by tenants, buyers, property owners, agencies, and administrators in the Bangladeshi property market and translate those problems into clear system requirements.

Technique	Participants / Sources	Duration	Key Outputs
Stakeholder Interviews	Students, job holders, families, property owners, agencies, admin representatives	5 sessions x 60 min	User pain points, listing workflow, verification expectations, communication needs
Focus Group Discussion	Potential tenants/buyers and owner-side users	2 sessions x 90 min	Search filter priorities, trust concerns, mobile usability expectations, price transparency needs
Market Observation	Facebook rental groups, informal broker posts, Bproperty, Bikroy, local listing pages	1 week	Competitive gaps, fake listing risks, incomplete property data, broker dependency issues

Technique	Participants / Sources	Duration	Key Outputs
Document Analysis	Project proposal, initial Thikana SRS draft, project overview, technology notes	Ongoing, 2 weeks	Feature scope, data entities, external services, academic constraints, MVP boundary
Prototype Review	Development team, selected sample users, academic supervisor feedback	3 review sessions x 2 hrs	UI corrections, workflow refinement, role-based dashboard expectations
Technology Feasibility Review	Development team and technical resources	Ongoing	Google Maps, Firebase, OTP, JWT, payment sandbox, media storage, 360-degree tour feasibility
Benchmarking	Comparable real estate platforms and rental discovery channels	1 week	Feature comparison, trust mechanisms, search experience, verification and moderation gaps

The elicitation process revealed several important insights. First, property seekers need a faster and more trustworthy way to compare rent, location, photos, amenities, and availability in one place. Second, property owners require a simple listing workflow that does not demand high technical skill. Third, both tenants and owners expressed concern about fake identities, misleading listings, hidden broker involvement, and unreliable communication. Fourth, map-based property discovery and nearby facility information are important decision-support features in the Bangladeshi context. These insights directly shaped the functional decomposition and use case model presented in this section.

3.2 Business Process Overview

The **Thikana** platform automates and supports several real estate and rental management processes. Each process begins with a specific trigger, involves one or more actors, and produces a measurable outcome. These processes form the basis for the use cases described in Section 3.5 and the detailed requirements specified in later sections.

3.2.1 User Registration and Verification Process

Process ID: BP-01	
Trigger	A new user wants to access Thikana as a tenant, buyer, property owner, or real estate agency.
Process Steps	The user opens the registration page and provides name, phone number, email address, password, and selected role. The system validates the input

Process ID: BP-01	
	and sends an OTP to the registered phone number. After successful OTP verification, the account is created and the user is redirected to the appropriate role-specific dashboard. If the user submits NID or student ID documents, the verification request is routed to an administrator for manual review.
Outcome	The account is marked as phone-verified or fully verified based on document approval.

3.2.2 Property Search and Discovery Process

Process ID: BP-02	
Trigger	A tenant or buyer searches for a property by location, keyword, price range, property type, or other preferences.
Process Steps	The system retrieves matching active listings and displays them in list view and, where applicable, map view. The user may refine results using filters such as bedroom count, furnished status, amenities, availability, and price. The system updates the result count and presents property cards containing price, location, title, thumbnail, and key details.
Outcome	The user opens a property detail page, saves a property, sends a message, or exits the search flow.

3.2.3 Property Listing and Publication Process

Process ID: BP-03	
Trigger	A property owner or agency wants to publish a property for rent, sale, sublet, bachelor housing, or office use.
Process Steps	The owner completes the listing form with property title, description, category, address, rent or price, contact rules, photos, floor plan, availability status, and optional 360-degree media. The system validates required fields, stores media in cloud storage, geocodes the address where possible, and

Process ID: BP-03	
	publishes the listing. If the listing is later reported as suspicious or fraudulent, an administrator may flag, hide, or suspend it.
Outcome	The listing becomes publicly searchable or is held for moderation.

3.2.4 Area Intelligence and Virtual Tour Process

Process ID: BP-04	
Trigger	A user opens a property detail page and requests area information or a remote property preview.
Process Steps	The system uses the property coordinates to retrieve nearby facilities such as schools, hospitals, markets, mosques, pharmacies, and transport points. It may also calculate approximate commute time to a user-defined destination. If the owner has uploaded panoramic media, the system loads the 360-degree viewer for remote exploration.
Outcome	The user reviews the location context, views the virtual tour, saves the property, or contacts the owner.

3.2.5 Verification, Reporting, and Moderation Process

Process ID: BP-05	
Trigger	A user submits identity documents or reports a suspicious listing, fake account, misleading information, or unsafe interaction.
Process Steps	The system records the verification request or report and notifies the administrator. The administrator reviews the submitted evidence, approves or rejects verification, requests additional documents, flags suspicious listings, or suspends abusive accounts. The system updates the status and notifies the affected users.
Outcome	The verification or report is resolved and the platform record is updated.

3.2.6 Communication and Rental Agreement Process

Process ID: BP-06	
Trigger	A tenant or buyer wants to communicate with a property owner or agency.
Process Steps	The user opens a listing and sends a message through the in-app messaging module. The owner receives a real-time notification and responds from their inbox. If both parties agree to proceed, the owner initiates a digital rental agreement. The system pre-fills property, owner, tenant, rent amount, and duration information. Both parties review and accept or reject the agreement.
Outcome	The agreement is accepted, rejected, or left pending.

3.2.7 Rental Payment Tracking and Reminder Process

Process ID: BP-07	
Trigger	A rental agreement becomes active or a monthly rent due date approaches.
Process Steps	The system generates rent reminder notifications before the due date and allows tenants or owners to log payment details. In the academic prototype, real payment processing may be simulated or executed through a sandbox payment gateway. The system stores payment status and history for future reference.
Outcome	The payment record is marked as paid, pending, failed, or overdue.

3.3 Actor Identification

An actor is any external entity that interacts with Thikana to achieve a goal or provide a service. Actors may be human users, external software systems, or scheduled system triggers. The following actors have been identified for the Thikana platform.

Actor ID	Actor Name	Type	Description and Interaction with Thikana	Cls
ACT-01	Guest Visitor	Primary / Human	Unregistered user who can browse public listings, view limited property information, and initiate registration.	H
ACT-02	Tenant / Buyer	Primary / Human	Registered user who searches properties, views details, saves favorites, sends messages, submits verification documents, and reviews owners.	H
ACT-03	Property Owner	Primary / Human	User who posts and manages property listings, uploads media, responds to enquiries, starts rental agreements, and views listing analytics.	H
ACT-04	Real Estate Agency	Primary / Human	Professional organization or agent account that manages multiple listings, handles bulk enquiries, and accesses agency-level analytics.	H
ACT-05	System Administrator	Primary / Human	Platform operator responsible for user verification, listing moderation, report handling, dispute review, configuration, and platform monitoring.	H
ACT-06	Google Maps / Mapbox	Secondary / System	External map service used for property pins, geocoding, nearby facility lookup, and commute estimation.	S
ACT-07	Firebase Services	Secondary / System	External service used for authentication support, cloud storage, real-time messaging, notification counters, and push notifications.	S
ACT-08	Payment Gateway	Secondary / System	External payment service such as SSLCommerz, bKash, or Nagad used for sandbox or future BDT payment processing.	S

Actor ID	Actor Name	Type	Description and Interaction with Thikana	Cls
ACT-09	SMS Gateway	Secondary / System	Service used to deliver OTP codes, phone verification messages, and important account alerts.	S
ACT-10	Email / SMTP Server	Secondary / System	Service used to deliver registration confirmations, password reset links, agreement notices, and rent reminders.	S
ACT-11	Scheduler / Timer	Secondary / Timer	Internal automated trigger for rent reminders, listing expiry checks, backup jobs, and scheduled notifications.	T

Note: In the actor table, 'Cls' denotes actor classification: H = Human, S = System, and T = Timer/Scheduler.

3.4 Functional Decomposition

Functional decomposition breaks the Thikana platform into major modules and sub-functions. This decomposition provides the structural basis for detailed functional requirements, use cases, database design, and implementation planning. The system is decomposed into platform-level modules that together support the complete digital real estate and rental management journey.

Module ID	Module Name	Level-2 Sub-functions
M-01	User & Authentication Management	M-01.1 User Registration; M-01.2 OTP Verification; M-01.3 Login and JWT Session; M-01.4 Password Reset; M-01.5 Role-based Dashboard
M-02	Property Search & Discovery	M-02.1 Keyword Search; M-02.2 Advanced Filters; M-02.3 Sorting; M-02.4 Search Result Count; M-02.5 Search History and Preferences
M-03	Map & Area Intelligence	M-03.1 Map-based Browsing; M-03.2 Property Pins; M-03.3 Nearby Facility Lookup; M-03.4 Commute Estimation; M-03.5 Area Rating Display

Module ID	Module Name	Level-2 Sub-functions
M-04	Listing Management	M-04.1 Create Listing; M-04.2 Edit Listing; M-04.3 Upload Photos and Floor Plans; M-04.4 Availability Status; M-04.5 Listing Deactivation
M-05	Media & Virtual Tour Management	M-05.1 Image Gallery; M-05.2 360-degree Panoramic View; M-05.3 Media Validation; M-05.4 Cloud Storage Management
M-06	Verification & Trust Management	M-06.1 Phone Verification; M-06.2 NID/Student ID Upload; M-06.3 Admin Review; M-06.4 Verified Badge; M-06.5 Suspicious User Handling
M-07	Communication Management	M-07.1 In-app Messaging; M-07.2 Message History; M-07.3 Real-time Notification; M-07.4 Phone Reveal Rules; M-07.5 Document Sharing
M-08	Agreement & Rental Management	M-08.1 Rental Agreement Generation; M-08.2 Agreement Acceptance; M-08.3 Rent Reminder; M-08.4 Payment Log; M-08.5 Utility Expense Tracking
M-09	Favorites & Alerts	M-09.1 Save Listing; M-09.2 Remove Favorite; M-09.3 Price Change Alert; M-09.4 Status Change Alert
M-10	Roommate Matching	M-10.1 Roommate Request Posting; M-10.2 Preference Filtering; M-10.3 Roommate Contact through Messaging
M-11	Review & Reputation Management	M-11.1 Owner Rating; M-11.2 Written Review; M-11.3 Average Rating; M-11.4 Review Moderation
M-12	Analytics & Reporting	M-12.1 Listing Views; M-12.2 Enquiry Count; M-12.3 Save Count; M-12.4 Owner Dashboard; M-12.5 Admin Platform Summary
M-13	Administration & Moderation	M-13.1 User Management; M-13.2 Listing Moderation; M-13.3 Report Review; M-13.4 Account Suspension; M-13.5 Audit and Monitoring

3.5 Use Case Modeling

Use case modeling describes Thikana from the perspective of its actors. Each use case represents a meaningful interaction between one or more actors and the system that produces a result of value. The use case catalogue below summarizes the principal system interactions that should later be represented visually in the Use Case Diagram and detailed through narrative use case specifications.

UC ID	Use Case Name	Primary Actor(s)	Priority	Brief Description
UC-01	Register and Verify Account	Guest Visitor, Tenant/Buyer, Owner, Agency	High	User creates an account, verifies phone number through OTP, and receives role-specific access.
UC-02	Login / Authentication	All Registered Users	High	User securely logs in and receives an authenticated session through JWT-based access control.
UC-03	Search and Filter Properties	Tenant / Buyer, Guest Visitor	High	User searches properties by keyword, location, price, property type, amenities, and other filters.
UC-04	View Property Details	Tenant / Buyer, Guest Visitor	High	User views listing title, price, description, photos, floor plan, availability, owner details, and key property information.
UC-05	Explore Map and Area Intelligence	Tenant / Buyer	Medium	User views property on map and explores nearby facilities, distances, safety indicators, and commute estimates.
UC-06	View 360-Degree Virtual Tour	Tenant / Buyer	Medium	User opens the virtual tour viewer to remotely explore a property interior using panoramic media.
UC-07	Post Property Listing	Property Owner, Real Estate Agency	High	Owner or agency creates a property listing with title, description, type, location, price, photos, and availability.

UC ID	Use Case Name	Primary Actor(s)	Priority	Brief Description
UC-08	Manage Property Listing	Property Owner, Real Estate Agency, Admin	High	Authorized user edits, deactivates, flags, or updates listing status such as Available, Rented, Sold, or Under Negotiation.
UC-09	Save Favorite Property	Tenant / Buyer	Medium	User saves a listing for later viewing and can remove it from favorites at any time.
UC-10	Send Message to Owner	Tenant / Buyer, Owner, Agency	High	User sends and receives messages related to a property through the in-app messaging system.
UC-11	Submit Verification Documents	Tenant / Buyer, Owner, Agency	Medium	User uploads NID or student ID documents for admin verification and profile trust badge approval.
UC-12	Approve Verification Request	System Administrator	Medium	Admin reviews submitted documents and approves, rejects, or requests additional information.
UC-13	Report Fake Listing or User	Any Registered User	High	User reports a suspicious listing, fake account, or fraudulent behavior for administrative review.
UC-14	Generate Rental Agreement	Property Owner, Tenant	Medium	Owner initiates a digital rental agreement and the system pre-fills property, owner, tenant, rent, and duration details.
UC-15	Track Rent Payments	Tenant, Owner, Admin	Medium	User records rent payment status, views payment history, and receives reminders for upcoming due dates.
UC-16	Post Roommate Request	Tenant / Buyer	Low	User creates a roommate request based on area, budget, lifestyle preferences, and gender preference.

UC ID	Use Case Name	Primary Actor(s)	Priority	Brief Description
UC-17	Submit Review and Rating	Tenant / Buyer	Medium	User leaves a star rating and written review for a property owner after a completed interaction.
UC-18	View Analytics Dashboard	Owner, Agency, Admin	Medium	Authorized user views listing views, saves, enquiries, user count, reports, and platform activity metrics.
UC-19	Moderate Content and Users	System Administrator	High	Admin reviews flagged content, suspends accounts, hides listings, and manages verification or fraud reports.
UC-20	Manage Notifications and Alerts	System Administrator, Scheduler	Medium	System or admin sends event-based alerts for messages, rent reminders, verification status, and listing changes.

3.6 Analysis Modeling

Analysis modeling transforms the requirements captured during elicitation into formal models that communicate the structure and behaviour of Thikana. These models provide a bridge between requirements analysis and system design. The diagrams referenced below should be prepared in Section 6 using PlantUML or an equivalent modeling tool.

Model Type	Diagram(s)	Purpose and Value
Context Model	System Context Diagram	Defines the boundary between Thikana and external actors/systems such as users, Google Maps, Firebase, SMS gateway, email server, and payment gateway.
Use Case Model	Use Case Diagram and Use Case Catalogue	Shows the goals of each actor and the major services provided by the platform.
Activity Model	Registration, Search, Listing Posting, Verification, Messaging, Rental Agreement Activities	Describes step-by-step workflows and decision points for major user processes.

Model Type	Diagram(s)	Purpose and Value
Swimlane Model	Property Search, Listing Publication, Verification Approval, Agreement Acceptance	Shows responsibility distribution among user, system, administrator, and external services.
Sequence Model	Login/OTP, Property Search, In-app Messaging, Verification Review, Payment Tracking	Shows the interaction order between UI, controllers, services, database, and third-party services.
State Transition Model	Listing Lifecycle, Verification Request Lifecycle, Agreement Lifecycle, Payment Status Lifecycle	Defines the valid states and transitions of key domain objects.
Class Model	Analysis Class Diagram	Identifies candidate classes, responsibilities, and relationships among users, listings, agreements, messages, payments, and reports.
Data Model	Entity Relationship Diagram (ERD)	Defines persistent entities, keys, and relationships required for relational database design.
Decision Model	Decision Table / Decision Tree	Clarifies business decisions such as listing approval, account verification, rent reminder status, and report escalation.

3.7 Domain Analysis

Domain analysis identifies the major real-world concepts that must be represented in the software. These domain objects later guide class diagram development, database schema design, API design, and test case preparation. The following table summarizes the principal domain concepts for Thikana.

Domain Concept	Description	Key Relationships
User	Represents any registered account on the platform, including tenant, buyer, owner, agency, or admin.	User may create listings, send messages, submit reports, save favorites, and request verification.

Domain Concept	Description	Key Relationships
Listing	Represents a property advertisement with location, price, type, availability, and descriptive details.	Listing belongs to one owner/agency and may have many photos, favorites, reviews, messages, and reports.
Property Location	Represents structured address and GPS coordinates used for map display and area intelligence.	Location belongs to a listing and interacts with map and places services.
Photo / Media	Represents listing photos, floor plans, and optional 360-degree panoramic media.	Many media records belong to one listing.
Favorite	Represents a saved listing selected by a tenant or buyer.	Favorite links one user with one listing.
Message	Represents an in-app conversation message between two users.	Message connects sender, receiver, and optionally a related listing.
Verification Request	Represents submitted NID/student ID or owner verification evidence awaiting admin decision.	Verification request belongs to one user and is reviewed by an administrator.
Report	Represents a complaint about a suspicious listing, fake account, or inappropriate behavior.	Report is filed by a user and targets either a listing or another user.
Agreement	Represents a digital rental agreement between owner and tenant for a specific listing.	Agreement links listing, owner, tenant, rent amount, duration, and status.
Payment Record	Represents rent payment history or future payment status for an agreement.	Payment record belongs to one agreement and may trigger reminders.
Review	Represents rating and written feedback submitted after an interaction.	Review links reviewer, owner/listing, rating value, and comment.

Domain Concept	Description	Key Relationships
Notification	Represents system-generated alerts for OTP, messages, rent due reminders, verification status, or listing changes.	Notification belongs to one or more users and may be delivered by email, SMS, FCM, or in-app channel.

3.8 Requirements Modeling Summary

The requirements modeling activities confirm that **Thikana** is not only a property listing website but a multi-role digital real estate ecosystem. The elicitation results emphasize trust, verification, location intelligence, user-friendly search, and direct communication as core success factors. The actor model defines the human and system participants; the business process model clarifies the major workflows; the functional decomposition organizes the platform into manageable modules; and the use case catalogue provides a direct foundation for detailed functional requirements, system diagrams, traceability, and testing.

The models produced in this section shall be used as the baseline for the next SRS sections. Any future change in feature scope, user role, data entity, or external integration shall be checked against this requirements model to maintain consistency across design, implementation, and validation activities.

4 SPECIFIC REQUIREMENTS

This section constitutes the core technical specification of Thikana — Smart Real Estate & Rental Management Platform. It provides a complete and traceable catalog of the functional requirements, external interface requirements, and non-functional requirements that the system shall satisfy. Each requirement is uniquely identified, assigned a priority level, linked to an originating source, and accompanied by measurable acceptance criteria so that the development team and quality assurance team can validate implementation objectively.

The requirements in this section are written using the normative keyword SHALL. This indicates mandatory behavior for the system unless the requirement is explicitly marked as Low priority or identified as a future enhancement. The initial academic prototype shall prioritize core capabilities such as user authentication, property listing, search and filtering, map-based discovery, communication, verification workflow, and basic administrative moderation.

4.1 Functional Requirements

Functional requirements define the behaviors, functions, and services that **Thikana** must provide to its users. Requirements are grouped by the main user classes and cross-cutting system functions identified in Section 3. The grouping improves readability and supports traceability from actor goals to system features.

Priority definitions used throughout this section are as follows:

Priority	Level	Definition
HIGH	Mandatory / MVP	Essential to the core MVP. Non-delivery would make the platform incomplete or unusable for the academic demonstration.
MEDIUM	Important	Important for operational completeness and user trust. Non-delivery would reduce usefulness but would not block the initial prototype.
LOW	Optional / Future	Optional enhancement or future increment. Non-delivery is acceptable for the initial release with documented deferral.

4.1.1 Tenant / Buyer Requirements (FR-TB)

The following requirements govern interactions between **Thikana** and Tenant / Buyer users. These users search for rental properties, bachelor housing, sublets, offices, or properties for purchase; compare listings; view area intelligence; communicate with owners; save favorites; and participate in verification, review, and roommate matching workflows.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-TB-001	The system SHALL allow tenants and buyers to register using name, phone number, email address, password, and selected user role.	High	Tenant / Buyer	Account is created; duplicate phone/email is rejected; user receives OTP within 60 seconds.
FR-TB-002	The system SHALL allow tenants and buyers to verify their phone number through OTP before accessing full platform features.	High	Tenant / Buyer	Correct OTP changes user status to phone-verified; expired or incorrect OTP is rejected with clear error message.
FR-TB-003	The system SHALL authenticate tenants and buyers using email/phone and password with JWT-based secure session handling.	High	Tenant / Buyer	User logs in successfully; JWT is issued; protected pages reject unauthenticated requests.
FR-TB-004	The system SHALL provide a personalized tenant/buyer dashboard showing saved listings, recent searches, messages, alerts, and verification status.	Medium	Tenant / Buyer	Dashboard loads within 3 seconds and displays current user-specific data.
FR-TB-005	The system SHALL allow tenants and buyers to search properties by keyword, location, property type, or area name.	High	Tenant / Buyer	Search returns matching active listings and displays a no-results message when no match exists.
FR-TB-006	The system SHALL allow filtering by price range, property type, bedrooms, bathrooms, furnished status, amenities, availability, and listing category.	High	Tenant / Buyer	Applied filters refine the result set correctly and show the active filter state to the user.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-TB-007	The system SHALL display property results in list view with thumbnail, title, price, location, key details, status, and owner/agency indicator.	High	Tenant / Buyer	Each listing card displays all required summary data and opens the detail page when selected.
FR-TB-008	The system SHALL provide map-based property browsing using Google Maps or Mapbox with property pins for active listings.	High	Tenant / Buyer	Map loads pins for matching listings; clicking a pin displays a preview without leaving the map.
FR-TB-009	The system SHALL allow users to sort property results by price, newest listing, relevance, and availability status.	Medium	Tenant / Buyer	Sorting changes the order of displayed results according to the selected sort criterion.
FR-TB-010	The system SHALL allow tenants and buyers to view full property details including description, address, price, photos, amenities, floor plan, owner information, and availability.	High	Tenant / Buyer	Property detail page displays all approved listing data accurately.
FR-TB-011	The system SHALL allow users to view uploaded property images and floor plans in a gallery interface.	High	Tenant / Buyer	Gallery opens images without broken links and supports navigation among all uploaded media.
FR-TB-012	The system SHALL allow users to view 360-degree virtual tours when panoramic media is available for a listing.	Medium	Tenant / Buyer	Virtual tour viewer loads available 360-degree media and displays a fallback message if no tour exists.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-TB-013	The system SHALL display nearby facilities such as schools, colleges, hospitals, pharmacies, markets, mosques, and transport stops for a selected property.	Medium	Tenant / Buyer	Nearby facility panel displays facility name, type, and approximate distance using map service data.
FR-TB-014	The system SHALL allow users to enter a destination address and receive an estimated commute time from the selected property.	Medium	Tenant / Buyer	Commute estimate is displayed using available directions data or a clear error message if unavailable.
FR-TB-015	The system SHALL allow authenticated users to save listings to a personal favorites list and remove them later.	Medium	Tenant / Buyer	Saved listing appears in favorites; removal updates the list immediately.
FR-TB-016	The system SHALL notify users when a saved property changes price, status, or availability.	Medium	Tenant / Buyer	Notification is delivered through in-app and push/email channel within 5 minutes of the change event.
FR-TB-017	The system SHALL allow tenants and buyers to send messages to property owners or agencies from a listing page.	High	Tenant / Buyer	Message is stored, delivered to recipient inbox, and sender can view the conversation.
FR-TB-018	The system SHALL allow users to submit NID or student ID images for manual verification.	Medium	Tenant / Buyer	Document upload is stored securely; verification status becomes Pending; admin is notified.
FR-TB-019	The system SHALL allow users to report fake listings, suspicious accounts, misleading information, or unsafe interactions.	High	Tenant / Buyer	Report is logged with target, reason, reporter ID, timestamp, and status Pending Review.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-TB-020	The system SHALL allow tenants to leave a 1–5 star rating and written review for an owner after a completed interaction.	Medium	Tenant / Buyer	Review is saved, associated with the owner/listing, and reflected in average rating after moderation rules are applied.
FR-TB-021	The system SHALL allow users to create roommate requests specifying area, budget, gender preference, lifestyle preferences, and contact preference.	Low	Tenant / Buyer	Roommate request is stored and visible in filtered roommate results.

4.1.2 Property Owner and Real Estate Agency Requirements (FR-OA)

The following requirements govern functions available to Property Owner and Real Estate Agency users. These users create and manage listings, upload media, respond to enquiries, initiate rental agreements, track rent activity, and view analytics. Agencies may manage multiple listings and perform owner-level operations at larger scale.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-OA-001	The system SHALL allow property owners and agencies to register and select the appropriate owner or agency role.	High	Owner / Agency	User account is created with selected role and receives access to the corresponding dashboard.
FR-OA-002	The system SHALL allow owners and agencies to create property listings with title, description, type, address, location, price/rent, contact rule, and availability status.	High	Owner / Agency	Listing is saved only when required fields are valid; published listing becomes searchable.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-OA-003	The system SHALL allow owners and agencies to upload at least 10 property photos per listing in JPEG, PNG, or WEBP format.	High	Owner / Agency	Accepted files upload successfully; unsupported formats and oversized files are rejected.
FR-OA-004	The system SHALL allow owners and agencies to upload a floor plan image or PDF for each listing.	Medium	Owner / Agency	Floor plan file is stored and displayed on the listing detail page.
FR-OA-005	The system SHALL allow owners and agencies to add or update listing coordinates through address geocoding or map pin selection.	High	Owner / Agency	Listing latitude and longitude are saved and used correctly in map search.
FR-OA-006	The system SHALL allow owners and agencies to set listing category including flat, house, bachelor, sublet, office, or buy/sell property.	High	Owner / Agency	Selected category is saved and can be used as a search filter.
FR-OA-007	The system SHALL allow owners and agencies to update listing availability as Available, Rented, Sold, Under Negotiation, Inactive, or Draft.	High	Owner / Agency	Status change is saved and reflected in search results within 2 minutes.
FR-OA-008	The system SHALL allow owners and agencies to edit listing content after publication while preserving change history for moderation.	High	Owner / Agency	Updated fields appear on the listing and old/new values are recorded in audit log.
FR-OA-009	The system SHALL allow owners and agencies to deactivate or archive a listing when it is no longer active.	Medium	Owner / Agency	Inactive listing is removed from public search and remains available in owner dashboard history.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-OA-010	The system SHALL support embedding 360-degree panoramic images for virtual tours of selected properties.	Medium	Owner / Agency	Panoramic media is attached to listing and visible through supported viewer.
FR-OA-011	The system SHALL display listing activity metrics including views, saves, enquiries, and message count to owners and agencies.	Medium	Owner / Agency	Analytics panel displays current metrics for each listing and updates after user interactions.
FR-OA-012	The system SHALL allow owners and agencies to respond to tenant/buyer messages from an inbox interface.	High	Owner / Agency	Reply is delivered to the sender and stored in conversation history.
FR-OA-013	The system SHALL allow owners to reveal contact number only to verified tenants or selected users according to privacy settings.	Medium	Owner	Phone number remains hidden unless reveal condition is met and owner setting permits disclosure.
FR-OA-014	The system SHALL allow owners to initiate a digital rental agreement for a selected verified tenant and property.	Medium	Owner	Agreement is generated with property, owner, tenant, rent amount, duration, and status Draft/Pending.
FR-OA-015	The system SHALL allow both owner and tenant to review, accept, or reject a digital rental agreement.	Medium	Owner / Tenant	Agreement status changes correctly and both parties receive status notification.
FR-OA-016	The system SHALL allow owners and tenants to record monthly rent payment status and view payment history.	Medium	Owner / Tenant	Payment record is saved with amount, due date, paid date, method, and status.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-OA-017	The system SHALL allow owners and agencies to receive alerts for new enquiries, reports, verification updates, and listing status changes.	Medium	Owner / Agency	Alerts are delivered through in-app notification and configured push/email channel.
FR-OA-018	The system SHALL allow agency users to manage multiple property listings under a single agency account.	Medium	Agency	Agency dashboard displays all agency listings with filter, edit, status, and analytics controls.
FR-OA-019	The system SHALL generate weekly and monthly trend reports for agency-level users.	Low	Agency	Report includes listing views, enquiries, saves, and status distribution for selected date range.

4.1.3 System Administrator Requirements (FR-AD)

The following requirements govern the administrative functions required to operate **Thikana** safely and reliably. Administrators verify users, moderate listings, handle user reports, manage disputes, monitor platform activity, and control system-wide data and policies.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-AD-001	The system SHALL allow administrators to log in through a secure admin interface with role-based permissions.	High	Admin	Only authorized admin accounts can access admin routes and dashboard.
FR-AD-002	The system SHALL provide an admin dashboard showing total users, active listings, pending verification requests, reports, messages, and transaction/payment records.	High	Admin	Dashboard displays current aggregate metrics and loads within 3 seconds for prototype data volume.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-AD-003	The system SHALL allow administrators to view, approve, reject, or request additional documents for verification submissions.	High	Admin	Verification decision updates user status and notifies the user.
FR-AD-004	The system SHALL allow administrators to review reported fake listings, suspicious users, and inappropriate behavior.	High	Admin	Report detail is visible with reporter, target, reason, timestamp, and current status.
FR-AD-005	The system SHALL allow administrators to flag, hide, restore, or permanently suspend listings based on moderation outcome.	High	Admin	Moderation action changes listing visibility and records an audit log entry.
FR-AD-006	The system SHALL allow administrators to suspend, ban, or reactivate user accounts for policy violation or security concerns.	High	Admin	Account status update prevents/permits login according to action taken.
FR-AD-007	The system SHALL allow administrators to manage property categories, amenity lists, and configurable listing status values.	Medium	Admin	Configuration changes appear in listing forms and filters without code modification.
FR-AD-008	The system SHALL allow administrators to view and export user activity reports including login history, listings viewed, favorites, and messages sent.	Medium	Admin	Report export is generated in PDF or Excel format and opens without errors.
FR-AD-009	The system SHALL allow administrators to manage rent reminder templates, verification notification templates, and system alert content.	Medium	Admin	Template changes are saved and used in subsequent notifications.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-AD-010	The system SHALL allow administrators to search and filter users by role, verification status, account status, phone number, and registration date.	Medium	Admin	Filtered admin list returns accurate matching records.
FR-AD-011	The system SHALL allow administrators to view platform-wide analytics including listing growth, user growth, reports, enquiries, and rental agreement counts.	Medium	Admin	Analytics dashboard displays selected period metrics and graphs/tables accurately.
FR-AD-012	The system SHALL maintain an audit log for all administrative actions including verification decisions, content moderation, account suspension, and configuration changes.	High	Admin / System	Audit log records actor ID, action, target, timestamp, IP address, and old/new values where applicable.

4.1.4 Common System Requirements (FR-CM)

The following requirements apply across all user roles and system modules. They define common behaviors related to authentication, authorization, communication, notifications, logging, search, reporting, and data protection.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-CM-001	The system SHALL enforce unique phone number and email address across all user accounts.	High	System	Registration with duplicate phone/email is rejected with an explanatory message.
FR-CM-002	The system SHALL lock an account temporarily after five consecutive failed login attempts.	High	Security	Account is locked for 15 minutes and further login attempts are denied during lockout.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-CM-003	The system SHALL support password reset through OTP or secure time-limited email link.	High	Security	User can reset password only with valid OTP/link; expired token is rejected.
FR-CM-004	The system SHALL enforce role-based access control for tenant, buyer, owner, agency, and admin roles.	High	Security	Unauthorized role cannot access restricted pages or API endpoints.
FR-CM-005	The system SHALL deliver real-time message notifications using Firebase Cloud Messaging or equivalent service.	High	System	New message generates notification within 5 seconds under normal network conditions.
FR-CM-006	The system SHALL store message history and allow users to access past conversations from the inbox.	Medium	System	Conversation history is retrievable, ordered by time, and associated with relevant listing where applicable.
FR-CM-007	The system SHALL support sharing property details, images, and documents within a conversation when enabled.	Low	System	Shared items appear in the conversation and remain accessible to both participants.
FR-CM-008	The system SHALL send automated rent due reminders seven days before the configured due date.	Medium	System / Scheduler	Reminder is sent through push/email and recorded in notification log.
FR-CM-009	The system SHALL allow authenticated users to update their profile information including name, email, phone, photo, and contact preferences.	Medium	All Users	Profile changes are saved and reflected across dashboards.
FR-CM-010	The system SHALL provide global search for properties, users, reports, and listings according to user role permissions.	Medium	System	Search results are restricted to authorized data scope.

Req. ID	Description	Priority	Source	Acceptance Criteria
FR-CM-011	The system SHALL generate downloadable reports in PDF or Excel format for authorized users.	Medium	System	Export files contain selected report data and open without formatting errors.
FR-CM-012	The system SHALL soft-delete critical business records such as listings, reports, agreements, and payment records before permanent disposal.	High	System	Deleted records are hidden from public view but retained for audit/dispute resolution period.

4.2 External Interface Requirements

External interface requirements define how **Thikana** shall interact with users, third-party systems, communication services, and hardware environments. These requirements ensure that the platform can operate as a responsive web application with reliable integrations for maps, storage, authentication, notifications, and payment-related workflows.

4.2.1 User Interface Requirements

Req. ID	Interface Requirement	Priority	Source	Acceptance Criteria
UI-001	The system SHALL provide a responsive web interface usable on desktop, laptop, tablet, and smartphone screens.	High	UI/UX	Layout adapts correctly from 360px mobile width to 1920px desktop width.
UI-002	The system SHALL use a consistent visual identity including teal primary color, golden accent color, and clean cream/white background.	Medium	UI/UX	All major screens follow the documented brand color palette.
UI-003	The system SHALL provide key screens including Home/Search, Property Detail, Owner Dashboard, Admin Panel, Messaging Inbox, User Profile, Favorites, and Verification Center.	High	UI/UX	Each key screen is reachable from navigation based on user role.

Req. ID	Interface Requirement	Priority	Source	Acceptance Criteria
UI-004	The system SHALL display inline validation errors near the related input field using plain language.	High	UI/UX	Invalid form submission shows field-specific error message without losing entered data.
UI-005	The system SHALL provide a persistent header containing search access, user menu, and notification icon after login.	Medium	UI/UX	Header is visible across authenticated pages and navigation links work correctly.
UI-006	The system SHALL provide minimum 44x44 pixel touch targets for interactive elements on mobile devices.	Medium	Accessibility	Buttons and tappable controls meet minimum touch target requirement.
UI-007	The system SHALL provide descriptive alt text for property images and functional icons where applicable.	Medium	Accessibility	Accessibility audit confirms alt text for all meaningful images.
UI-008	The system SHALL display loading states or skeleton placeholders during network data fetching.	Medium	UI/UX	Users see loading feedback instead of blank pages while data is loading.

4.2.2 Software Interface Requirements

Req. ID	Software Interface Requirement	Priority	Source	Acceptance Criteria
SI-001	The system SHALL integrate with Google Maps Platform or Mapbox for map display, geocoding, property pins, nearby facility lookup, and commute estimation.	High	External API	Map loads property pins and retrieves location-related data using configured API key.

Req. ID	Software Interface Requirement	Priority	Source	Acceptance Criteria
SI-002	The system SHALL integrate with Firebase Authentication or equivalent OTP service for phone verification and optional Google OAuth login.	High	External API	OTP verification and optional Google login complete through configured authentication service.
SI-003	The system SHALL integrate with Firebase Storage or equivalent cloud storage for property photos, floor plans, NID documents, and panoramic images.	High	External API	Uploaded files are stored securely and retrievable through authorized URLs.
SI-004	The system SHALL integrate with Firebase Realtime Database or equivalent service for chat messages, online status, and notification counters.	High	External API	Real-time message update is visible without full page refresh.
SI-005	The system SHALL support payment gateway sandbox integration with SSLCommerz, bKash, or Nagad for future BDT transaction flow.	Medium	External API	Sandbox payment request and response can be simulated and logged.
SI-006	The system SHALL use MySQL 8.0+ or compatible relational database for persistent platform records.	High	Database	Core entities are stored with foreign keys and data integrity constraints.
SI-007	The system SHALL expose secure REST API endpoints over HTTPS for frontend-backend communication.	High	Backend	API returns JSON responses and rejects unauthorized requests.
SI-008	The system SHALL support Three.js, Pannellum, or equivalent viewer for 360-degree panoramic property tours.	Medium	Frontend	Panoramic media renders in a browser-based viewer without plugin installation.

4.2.3 Communication Interface Requirements

Req. ID	Communication Requirement	Priority	Source	Acceptance Criteria
CI-001	The system SHALL use HTTPS with TLS 1.2 or higher for all client-server communication.	High	Security	HTTP requests redirect to HTTPS and TLS certificate is valid.
CI-002	The system SHALL exchange API data using JSON format over REST endpoints.	High	Backend	All API responses are valid JSON with appropriate HTTP status codes.
CI-003	The system SHALL support WebSocket or Firebase-managed real-time communication for chat and notification features.	High	System	Messages appear in recipient inbox with real-time update under normal conditions.
CI-004	The system SHALL send transactional emails through SMTP or a configured email delivery service.	Medium	External Service	Registration, password reset, agreement, and reminder emails are delivered and logged.
CI-005	The system SHALL send OTP messages through an SMS gateway or Firebase phone authentication service.	High	External Service	OTP message is delivered to valid Bangladeshi mobile number during registration/verification.
CI-006	The system SHALL send browser push notifications through Firebase Cloud Messaging or equivalent push service.	Medium	External Service	Push notification is received by subscribed user for configured events.

4.3 Non-Functional Requirements

Non-functional requirements define the quality attributes, operational constraints, and measurable standards that **Thikana** must satisfy. These requirements apply across the platform and are critical for usability, performance, security, reliability, scalability, maintainability, localization, and legal compliance.

4.3.1 Usability Requirements (NFR-US)

NFR ID	Description	Priority	Category	Acceptance Criteria
NFR-US-001	A first-time user SHOULD be able to search for a property and open a listing detail page within 3 minutes without training.	Medium	Usability	Observed prototype walkthrough confirms completion within 3 minutes by sample users.
NFR-US-002	The registration process SHALL require no more than five major steps including OTP verification.	High	Usability	Registration flow contains five or fewer user-facing steps.
NFR-US-003	All forms SHALL provide clear labels, placeholders, and field-level validation feedback.	High	Usability	Required fields and validation errors are visible and understandable.
NFR-US-004	The interface SHOULD conform to WCAG 2.1 Level AA for keyboard navigation, contrast, and screen-reader support.	Medium	Accessibility	Accessibility checklist shows no critical violations in core screens.
NFR-US-005	The interface SHALL support mobile-first responsive design for property search, listing detail, messaging, and dashboards.	High	Usability	Core screens render correctly on 360px mobile and desktop viewports.

4.3.2 Performance Requirements (NFR-PE)

NFR ID	Description	Priority	Category	Acceptance Criteria
NFR-PE-001	The search results page SHALL load within 2 seconds for queries returning up to 50 listings on a standard 4G connection.	High	Performance	Performance test shows average load time ≤ 2 seconds for 50 listings.
NFR-PE-002	Property detail pages including image gallery SHOULD load within 3 seconds for optimized media.	Medium	Performance	Page load test shows average load time ≤ 3 seconds for standard listing detail.
NFR-PE-003	The map view SHALL render up to 200 property pins without noticeable interaction lag.	Medium	Performance	Map remains interactive with 200 pins under prototype test data.
NFR-PE-004	In-app messages SHALL be delivered with maximum latency of 500 milliseconds under normal network conditions.	High	Performance	Chat test shows message delivery latency ≤ 500 ms for local/prototype conditions.
NFR-PE-005	Image uploads up to 5MB SHOULD be processed and stored within 5 seconds.	Medium	Performance	Upload test completes within 5 seconds for supported file size.

4.3.3 Security Requirements (NFR-SE)

NFR ID	Description	Priority	Category	Acceptance Criteria
NFR-SE-001	All passwords SHALL be hashed using bcrypt with minimum cost factor 12 before storage.	High	Security	Database inspection confirms no plaintext password is stored.
NFR-SE-002	All protected API endpoints SHALL require a valid JWT token except public search and public listing endpoints.	High	Security	Unauthorized request receives 401/403 response.
NFR-SE-003	The system SHALL enforce RBAC so tenants cannot access owner, agency, or admin-only endpoints.	High	Security	Role permission tests confirm restricted access is denied.
NFR-SE-004	All sensitive data in transit SHALL be encrypted using TLS 1.2 or higher.	High	Security	Security test confirms HTTPS/TLS enforcement.
NFR-SE-005	NID and student ID document images SHALL be stored in private, non-public cloud storage accessible only to authorized admins.	High	Security	Direct public URL access to identity documents is denied.
NFR-SE-006	Authentication endpoints SHALL implement rate limiting to prevent brute-force attacks.	High	Security	More than five failed attempts triggers lockout/throttling.
NFR-SE-007	SQL injection and XSS SHALL be prevented through parameterized queries, input validation, and output encoding.	High	Security	Security tests fail to execute common SQLi/XSS payloads.
NFR-SE-008	State-changing requests SHOULD implement CSRF protection where cookie-based sessions are used.	Medium	Security	CSRF test request without token is rejected for protected operations.

4.3.4 Reliability and Availability Requirements (NFR-RL)

NFR ID	Description	Priority	Category	Acceptance Criteria
NFR-RL-001	The production platform SHOULD target 99.5% uptime excluding planned maintenance windows.	Medium	Reliability	Monitoring report confirms target availability after deployment.
NFR-RL-002	The system SHALL display user-friendly error messages without exposing stack traces or sensitive internal details.	High	Reliability	Forced error test shows safe generic message to user.
NFR-RL-003	The database SHALL be backed up automatically at least once per day.	High	Reliability	Backup schedule exists and produces recoverable backup files.
NFR-RL-004	Recovery from backup SHOULD be achievable within 4 hours for production deployment.	Medium	Reliability	Recovery drill completes within 4 hours.
NFR-RL-005	The frontend SHALL provide loading states during asynchronous data fetching.	Medium	Reliability	No core page shows blank state for normal network delay.

4.3.5 Scalability and Maintainability Requirements (NFR-SM)

NFR ID	Description	Priority	Category	Acceptance Criteria
NFR-SM-001	The academic prototype SHALL support at least 500 concurrent users under controlled test conditions.	Medium	Scalability	Load test simulates 500 concurrent users without critical failure.

NFR ID	Description	Priority	Category	Acceptance Criteria
NFR-SM-002	The architecture SHOULD be designed to scale to 10,000 concurrent users in production through horizontal scaling.	Low	Scalability	Design documentation shows stateless API layer and scalable storage strategy.
NFR-SM-003	The backend SHALL follow separation of concerns using controller, service, and model layers.	High	Maintainability	Code review confirms module separation.
NFR-SM-004	Environment-specific configuration SHALL be stored in environment variables or .env files, not hardcoded.	High	Maintainability	Repository scan finds no hardcoded production secrets.
NFR-SM-005	The codebase SHALL be maintained in Git with meaningful commit messages and branch naming conventions.	Medium	Maintainability	Repository history shows consistent commits and branches.

4.3.6 Localization and Compliance Requirements (NFR-LC)

NFR ID	Description	Priority	Category	Acceptance Criteria
NFR-LC-001	The system SHALL display monetary values in Bangladeshi Taka (BDT/টাকা).	High	Localization	All price and rent fields show BDT currency formatting.
NFR-LC-002	The system SHALL validate Bangladeshi mobile numbers beginning with 01 and containing 11 digits.	High	Localization	Invalid phone number formats are rejected during registration and profile update.
NFR-LC-003	Dates SHOULD be displayed in DD-MM-YYYY format and Asia/Dhaka timezone.	Medium	Localization	Date/time values display in Bangladeshi convention.

NFR ID	Description	Priority	Category	Acceptance Criteria
NFR-LC-004	The system SHALL support UTF-8 encoding for English and Bengali text.	High	Localization	Bangla text can be saved and displayed without character corruption.
NFR-LC-005	User personal data SHALL not be shared with third parties without explicit user consent.	High	Compliance	Privacy review confirms consent requirement before external sharing.
NFR-LC-006	User-reported fake listings or suspicious accounts SHOULD be reviewed by an administrator within 48 hours.	Medium	Compliance	Report workflow records review decision within the target service time.

4.4 Requirements Summary

Section 4 defines the complete requirement baseline for the **Thikana** platform. In total, this section specifies 54 functional requirements across four groups (FR-TB, FR-OA, FR-AD, FR-CM), 14 external interface requirements across three categories (UI, Software Interface, Communication), and 31 non-functional requirements across six quality attribute categories. Each requirement is uniquely identified, prioritized, sourced, and accompanied by measurable acceptance criteria to support objective validation during testing and academic evaluation.

Requirement Group	Prefix	Total	High	Med/Low
Tenant / Buyer Functional Requirements	FR-TB-001 to FR-TB-021	21	10	11
Owner / Agency Functional Requirements	FR-OA-001 to FR-OA-019	19	8	11
System Administrator Functional Requirements	FR-AD-001 to FR-AD-012	12	7	5
Common System Functional Requirements	FR-CM-001 to FR-CM-012	12	6	6

Requirement Group	Prefix	Total	High	Med/Low
External Interface Requirements (UI + SI + CI)	UI-001–008, SI-001–008, CI-001–006	22	14	8
Non-Functional Requirements (US+PE+SE+RL+SM+LC)	NFR-US, PE, SE, RL, SM, LC	31	18	13
TOTAL	All Groups Combined	107	63	44

These requirements shall be used as the authoritative basis for design, implementation, test case preparation, validation, and later requirements traceability. Any change to the feature set or system scope should be assessed against this section to determine its impact on user workflows, data entities, external services, and acceptance criteria.

5 DATA REQUIREMENTS

This section defines the data requirements for **Thikana** — Smart Real Estate & Rental Management Platform. It describes the major data entities, their attributes, relationships, integrity rules, validation constraints, reports, retention policies, backup requirements, and disposal rules. The purpose of this section is to ensure that the database design supports all functional requirements described in Section 4 and provides a reliable foundation for property search, listing management, verification, communication, agreement handling, payments, analytics, and administrative moderation.

The **Thikana** platform shall store structured transactional data in a relational database such as MySQL, while media files such as property photos, floor plans, panoramic images, and identity verification documents may be stored in Firebase Storage or a compatible cloud object storage service. Real-time data such as chat notifications and message delivery status may use Firebase Realtime Database or an equivalent real-time service. All persistent data shall be protected through validation, access control, audit logging, and backup mechanisms.

5.1 Data Model Overview

The logical data model of **Thikana** is centered on users, property listings, property media, communication records, verification records, favorites, reviews, rental agreements, payments, reports, and system logs. Each entity has a clearly defined purpose and is connected to other entities through primary keys and foreign keys. The database shall enforce referential integrity so that orphaned property, message, payment, or report records cannot exist without their corresponding parent records.

Entity	Key Attributes	Purpose and Relationships
User	user_id, name, phone, email, password_hash, role, verification_status, created_at	Stores account and role information for tenants, buyers, owners, agencies, and admins. One user can post many listings, send many messages, save many favorites, submit reports, and participate in agreements.
Listing	listing_id, owner_id, title, property_type, listing_category, price, address, lat, lng, status, created_at	Stores property advertisements. One listing belongs to one owner or agency, has many photos, can receive reviews, can be saved by many users, and may be linked to one agreement.
Photo / Media	media_id, listing_id, file_url, media_type, is_primary, is_360, uploaded_at	Stores listing images, floor plans, and virtual tour media. Many media records belong to one listing.

Entity	Key Attributes	Purpose and Relationships
Favorite	favorite_id, user_id, listing_id, created_at	Represents saved properties. Many users can save many listings through the Favorite relationship.
Message	message_id, sender_id, receiver_id, listing_id, content, sent_at, read_status	Stores in-app chat messages between tenants/buyers and owners/agencies. Each message belongs to one sender and one receiver.
VerificationDocument	verification_id, user_id, document_type, document_url, status, reviewed_by, reviewed_at	Stores NID, student ID, or other verification document submission records for manual admin review.
Review	review_id, listing_id, reviewer_id, owner_id, rating, comment, status, created_at	Stores rating and feedback submitted by tenants or buyers after a completed interaction.
Agreement	agreement_id, listing_id, owner_id, tenant_id, rent_amount, start_date, end_date, status	Stores digital rental agreements linking one listing, one owner, and one tenant.
Payment	payment_id, agreement_id, amount, due_date, paid_date, method, status, transaction_ref	Stores rent payment history and payment tracking records under an agreement.
Report	report_id, reporter_id, target_type, target_id, reason, status, created_at	Stores reports submitted against fake listings, suspicious users, misleading content, or unsafe interactions.
AreaFacility	facility_id, listing_id, facility_type, name, distance, source	Stores or caches nearby facility data such as schools, hospitals, markets, mosques, and transport stops when needed.
Notification	notification_id, user_id, type, message, status, created_at	Stores notifications for messages, verification updates, saved-property alerts, rent reminders, and admin actions.

Entity	Key Attributes	Purpose and Relationships
AuditLog	audit_id, actor_id, action, entity_type, entity_id, old_value, new_value, timestamp	Stores security-relevant and business-critical user actions for accountability and troubleshooting.

5.2 Data Dictionary

The data dictionary defines the major data attributes that shall be stored by the Thikana system. The final physical schema may include additional technical fields such as created_by, updated_by, deleted_at, version, and indexing fields. However, the following attributes represent the minimum logical data required to satisfy the core system requirements.

5.2.1 User Entity

Field	Type	Length	Allowed Values / Constraint	Description
user_id	INT	11	Auto-increment	Primary key and unique identifier for each user account.
name	VARCHAR	100	Text	Full name of the registered user.
phone	VARCHAR	15	Bangladesh format: 01XXXXXXXXX	Mobile number used for login, OTP verification, and contact. Must be unique.
email	VARCHAR	120	Valid email format	User email address used for login, notification, and password recovery. Must be unique.
password_hash	VARCHAR	255	bcrypt hash	Encrypted password hash. Plaintext passwords shall never be stored.
role	ENUM	-	tenant, buyer, owner, agency, admin	Defines role-based permission and dashboard access.

Field	Type	Length	Allowed Values / Constraint	Description
phone_verified	BOOLEAN	1	0 or 1	Indicates whether the user has completed OTP verification.
verification_status	ENUM	-	unverified, pending, verified, rejected	Indicates identity document verification state.
status	ENUM	-	active, suspended, banned, deleted	Defines whether the account can access platform services.
created_at	DATETIME	-	System timestamp	Date and time when the account was created.

5.2.2 Listing Entity

Field	Type	Length	Allowed Values / Constraint	Description
listing_id	INT	11	Auto-increment	Primary key and unique identifier for each property listing.
owner_id	INT	11	FK -> User.user_id	Identifies the owner or agency who posted the listing.
title	VARCHAR	150	Required	Short title of the listing displayed in search results.
description	TEXT	-	Required	Detailed description of the property, facilities, and rental/sale conditions.
property_type	ENUM	-	flat, house, bachelor, sublet, office, land, commercial	Type/category of the property.
listing_category	ENUM	-	rent, buy, sell, sublet	Indicates the property transaction category.

Field	Type	Length	Allowed Values / Constraint	Description
price	DECIMAL	10,2	Positive BDT value	Monthly rent or sale price in Bangladeshi Taka.
address	VARCHAR	255	Required	Human-readable property address.
division	VARCHAR	60	Bangladesh division	Administrative division for location filtering.
district	VARCHAR	60	Bangladesh district	Administrative district for location filtering.
area	VARCHAR	100	Text	Local area or neighborhood name.
lat / lng	DECIMAL	10,7	Valid GPS coordinates	Latitude and longitude used for map-based search and distance calculation.
bedrooms	INT	2	0 or positive number	Number of bedrooms or usable rooms.
bathrooms	INT	2	0 or positive number	Number of bathrooms.
furnished_status	ENUM	-	furnished, semi_furnished, unfurnished	Indicates furnishing level of the property.
availability_status	ENUM	-	available, rented, sold, under_negotiation, inactive	Current public availability state of the listing.
created_at	DATETIME	-	System timestamp	Date and time when the listing was created.

5.2.3 Media, Favorite, and Area Intelligence Entities

Field	Type	Length	Allowed Values / Constraint	Description
media_id	INT	11	Auto-increment	Unique identifier for each uploaded listing media item.
listing_id	INT	11	FK -> Listing.listing_id	Identifies the listing to which the media belongs.
file_url	VARCHAR	500	Secure storage URL	Path or URL of the uploaded image, floor plan, or 360-degree media file.
media_type	ENUM	-	photo, floor_plan, panorama, document	Specifies the type of uploaded media.
is_primary	BOOLEAN	1	0 or 1	Indicates whether the media is the primary display image.
favorite_id	INT	11	Auto-increment	Unique identifier for a saved listing record.
favorite.user_id	INT	11	FK -> User.user_id	Identifies the user who saved the listing.
favorite.listing_id	INT	11	FK -> Listing.listing_id	Identifies the saved property listing.
facility_type	ENUM	-	school, hospital, market, mosque, transport, pharmacy, other	Classification of nearby facility shown in area intelligence.
distance	DECIMAL	8,2	Non-negative value	Approximate distance from the property in meters or kilometers.
source	VARCHAR	80	Google Maps / Manual / Cached	Indicates source of area intelligence data.

5.2.4 Communication, Review, and Report Entities

Field	Type	Length	Allowed Values / Constraint	Description
message_id	INT	11	Auto-increment	Unique identifier for each chat message.
sender_id	INT	11	FK -> User.user_id	User who sends the message.
receiver_id	INT	11	FK -> User.user_id	User who receives the message.
message.listing_id	INT	11	Nullable FK -> Listing.listing_id	Optional listing context for a conversation.
content	TEXT	-	Required	Text content of the message.
read_status	ENUM	-	sent, delivered, read	Indicates message delivery and read status.
review_id	INT	11	Auto-increment	Unique identifier for each review.
rating	DECIMAL	3,2	1.00 to 5.00	Star rating assigned by the reviewer.
comment	TEXT	-	Optional text	Written review comment.
report_id	INT	11	Auto-increment	Unique identifier for each report.
target_type	ENUM	-	listing, user, review, message	Type of content or user being reported.
reason	VARCHAR	255	Required	Reason for reporting fake, misleading, abusive, or unsafe content.
report.status	ENUM	-	pending, reviewing, resolved, rejected	Administrative processing status of the report.

5.2.5 Verification, Agreement, and Payment Entities

Field	Type	Length	Allowed Values / Constraint	Description
verification_id	INT	11	Auto-increment	Unique identifier for a verification request.
document_type	ENUM	-	nid, student_id, passport, trade_license	Type of document submitted for verification.
document_url	VARCHAR	500	Private storage URL	Secure storage path for the uploaded verification document.
reviewed_by	INT	11	FK -> User.user_id	Admin user who reviewed the verification request.
agreement_id	INT	11	Auto-increment	Unique identifier for a rental agreement.
agreement.listing_id	INT	11	FK -> Listing.listing_id	Listing associated with the agreement.
owner_id	INT	11	FK -> User.user_id	Owner participating in the agreement.
tenant_id	INT	11	FK -> User.user_id	Tenant participating in the agreement.
rent_amount	DECIMAL	10,2	Positive BDT value	Monthly rent amount recorded in the agreement.
start_date / end_date	DATE	-	Valid date range	Agreement validity period. End date must be later than start date.
agreement.status	ENUM	-	draft, pending, accepted, rejected, expired, terminated	Current lifecycle state of the rental agreement.

Field	Type	Length	Allowed Values / Constraint	Description
payment_id	INT	11	Auto-increment	Unique identifier for payment record.
amount	DECIMAL	10,2	Positive BDT value	Rent or utility amount paid or due.
payment.status	ENUM	-	pending, paid, failed, overdue, cancelled	Payment record status.
transaction_ref	VARCHAR	100	Unique when available	External gateway or manual transaction reference number.

5.3 Entity Relationship Summary

The following relationship rules summarize the logical cardinalities that shall be maintained in the **Thikana** database. These relationships shall be implemented using foreign key constraints where appropriate and shall be reflected in the ERD in the system models section.

Source Entity	Target Entity	Relationship	Description
User	Listing	One-to-Many	One owner or agency user may create many listings; each listing belongs to exactly one owner or agency.
Listing	Photo / Media	One-to-Many	One listing may contain multiple photos, floor plans, and panoramic images.
User	Favorite	One-to-Many	One user may save many listings. Duplicate saves for the same user and listing shall be prevented.
Listing	Favorite	One-to-Many	One listing may be saved by many users through Favorite records.
User	Message	One-to-Many	One user may send and receive many messages.

Source Entity	Target Entity	Relationship	Description
Listing	Message	Zero/One-to-Many	A message may optionally be associated with a specific listing.
User	VerificationDocument	One-to-Many	One user may submit multiple verification documents or resubmissions.
Listing	Review	One-to-Many	One listing may receive multiple reviews from verified users.
User	Review	One-to-Many	One user may write multiple reviews, but duplicate reviews for the same completed interaction shall be restricted.
Listing	Agreement	One-to-Zero/One	A rented listing may be linked to one active agreement at a time.
Agreement	Payment	One-to-Many	One agreement may generate multiple monthly rent payment records.
User	Report	One-to-Many	One user may file many reports against listings, users, reviews, or messages.
User	AuditLog	One-to-Many	One user may create many audit log entries through important actions.

5.4 Database Constraints and Integrity Rules

The database shall enforce integrity rules to protect system consistency, prevent invalid data entry, and support trustworthy property transactions. These rules are mandatory for the production system and should be simulated or enforced at application level in the academic prototype when full database constraints are not feasible.

Rule ID	Constraint Type	Constraint Description
DC-001	Primary Key	Every persistent entity SHALL have a unique primary key.

Rule ID	Constraint Type	Constraint Description
DC-002	Unique User Identity	User phone numbers and email addresses SHALL be unique across active accounts.
DC-003	Role Constraint	User role SHALL be restricted to tenant, buyer, owner, agency, or admin.
DC-004	Foreign Key Integrity	Listings, messages, reviews, payments, reports, and favorites SHALL reference valid parent records.
DC-005	Listing Status Constraint	Listing status SHALL be restricted to available, rented, sold, under_negotiation, inactive, or deleted.
DC-006	Price Constraint	Property price, rent amount, payment amount, and utility amount SHALL be stored as DECIMAL and SHALL NOT be negative.
DC-007	GPS Constraint	Latitude and longitude SHALL contain valid geographic coordinate values before a listing appears in map view.
DC-008	Media Constraint	Only approved file types such as JPEG, PNG, WEBP, PDF, and supported panoramic formats SHALL be accepted.
DC-009	Verification Privacy	Identity document URLs SHALL be stored as private references and SHALL NOT be exposed publicly.
DC-010	Soft Delete Rule	Deleted listings SHALL be soft-deleted first and retained for dispute resolution before permanent deletion.
DC-011	Audit Rule	Critical write operations SHALL create audit log records with actor, entity, timestamp, and action details.
DC-012	No Orphan Agreement	A rental agreement SHALL NOT exist without a valid listing, owner, and tenant.
DC-013	Payment Reference Rule	A completed payment SHOULD store a unique transaction reference when processed through a payment gateway.
DC-014	Review Validity	A review SHOULD only be accepted after a completed interaction or agreement-related event.

5.5 Reports and Data Outputs

Thikana shall generate standard reports for tenants, owners, agencies, and administrators. Reports shall be available through role-based access control and may be exported in PDF or Excel format where required. Report generation shall use authorized data only and shall not expose private verification documents or sensitive internal system fields to unauthorized users.

Report Name	Description	Accessible By
Payment History Report	Monthly payment log for a specific agreement showing due amount, paid amount, due date, paid date, method, and status.	Tenant, Owner, Admin
Listing Analytics Report	Summary of views, saves, enquiries, message count, status changes, and listing performance over a selected date range.	Owner, Agency, Admin
Platform Summary Report	Platform-wide summary showing total users, active listings, pending reports, verification requests, agreements, and transaction counts.	Admin
Tenant Verification Log	List of verification requests grouped by pending, approved, rejected, or resubmission-required status.	Admin
User Activity Report	Login history, messages sent, listings viewed, favorites saved, reports filed, and key actions by a user.	Admin
Reported Content Report	Administrative report of fake listings, suspicious users, unsafe interactions, and moderation outcomes.	Admin
Saved Property Alert Log	Record of notifications triggered when saved listings changed price, availability, or status.	Tenant / Buyer, Admin
Agency Portfolio Report	Summary of all listings managed by an agency, including active, rented, sold, inactive, and under-negotiation properties.	Agency, Admin
Area Demand Report	Aggregated demand by area based on search frequency, listing views, saved listings, and enquiries.	Admin, Agency

5.6 Data Acquisition, Validation, Retention, Disposal, and Backup

This subsection defines how **Thikana** acquires, validates, stores, retains, disposes, and backs up data. Because the platform handles personally identifiable information, property data, communication records, identity documents, and payment-related records, data governance must be applied consistently across all modules.

Category	Requirement Description
Data Acquisition	User profile data shall be collected during registration. Property data shall be submitted by owners and agencies. Area intelligence data shall be acquired through map and places services or cached from approved sources. Verification documents shall be uploaded directly by users and reviewed by admins.
Validation	The system shall validate phone numbers, email addresses, required fields, file types, file sizes, price values, GPS coordinates, role values, listing status, and date ranges before storing data. Invalid inputs shall be rejected with clear error messages.
Data Integrity	Foreign key constraints, unique constraints, ENUM constraints, audit logs, transaction controls, and application-level validation shall be used to maintain consistency. Financial values shall use DECIMAL types to avoid rounding errors.
Data Retention	Active user accounts, listings, messages, agreements, and reviews shall remain available while the account is active. Payment records shall be retained for at least 5 years for financial audit purposes. Verification documents shall be retained only as long as required for platform trust and dispute handling.
Data Disposal	Deleted listings shall first be soft-deleted and retained for 90 days before permanent removal. User account deletion shall anonymize personal data while preserving non-personal transaction records required for system integrity and audit.
Backup	The MySQL database shall be backed up automatically at least once per day. Backup files shall be stored separately from the primary server. Firebase or object storage media shall follow provider backup policy and additional export procedures where available.
Recovery	The system should support recovery from the most recent valid backup within 4 hours in production. Backup restoration shall be tested periodically to verify data usability.

Category	Requirement Description
Privacy Protection	NID, student ID, phone number, email address, and private messages shall not be disclosed to unauthorized users. Access shall be controlled through RBAC and admin audit logs.

5.7 Data Volume and Growth Assumptions

The academic prototype will use manually seeded test data and a limited number of user-submitted records. However, the database design should be prepared for gradual production growth across major cities and divisions of Bangladesh. The following planning assumptions are used for initial capacity estimation.

Data Category	Growth Assumption
Users	500 prototype users; scalable design target 100,000+ production users.
Listings	200 prototype listings; scalable design target 500,000+ historical listings.
Media Files	Up to 10 photos per listing in prototype; production should support photos, floor plans, and 360-degree media with storage optimization.
Messages	Prototype message history for demonstration; production should support millions of chat records with pagination and archival.
Reports and Logs	All moderation reports and security events should be retained according to administrative and legal policy.
Search Data	Frequently searched locations and filters may be indexed to improve performance and analytics.

5.8 Data Security Classification

All data stored by **Thikana** shall be classified according to sensitivity so that appropriate access controls, storage rules, and retention policies can be applied. Sensitive records shall be protected through encryption in transit, restricted storage access, and audit logging.

Classification	Example Data	Access Rule
Public Data	Published listing title, price, area, public photos, property type, and visible amenities.	Visible to all users unless listing is inactive or removed.
Authenticated User Data	Saved favorites, message inbox, profile details, and personal dashboard data.	Visible only to the owning user and authorized system functions.
Confidential Data	Phone number, email, identity documents, agreement details, payment records, and private messages.	Accessible only to authorized users and admins with logged access.
Administrative Data	Audit logs, report decisions, moderation notes, admin actions, and system configuration.	Accessible only to authorized administrators.
Derived Analytics Data	Aggregated search trends, listing views, area demand, and engagement metrics.	May be shown in dashboards without exposing individual private records.

Together, these data requirements provide the foundation for a secure, organized, and scalable property management ecosystem. They shall be used during database schema design, API design, data migration planning, test case preparation, and future enhancement of the **Thikana** platform.

6 SYSTEM MODELS AND ANALYSIS MODELS

This section presents the major system models and analysis models of Thikana - Smart Real Estate & Rental Management Platform. These models provide visual and structured representations of the system boundary, external actors, use cases, workflows, object structure, database relationships, data flow, and decision logic. The models are intended to help stakeholders, developers, testers, database designers, and academic evaluators understand how the platform shall operate before implementation.

All diagrams in this section are provided as PlantUML source code. The code blocks can be copied directly into the PlantUML online server or any PlantUML-compatible rendering tool to generate diagrams. The diagrams are based on the Thikana functional scope, including property search, listing management, tenant verification, direct communication, digital agreements, payment tracking, area intelligence, analytics, and administrative moderation.

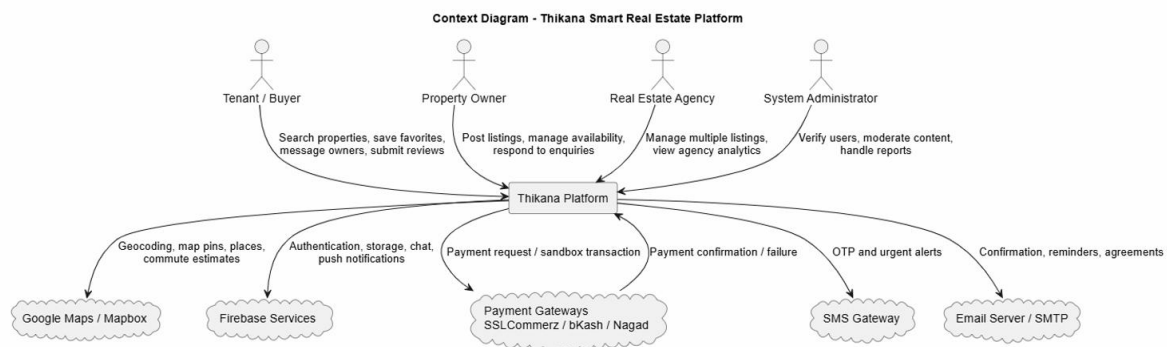
6.1 Context Diagram

Purpose

The context diagram shows Thikana as a single system and illustrates its interaction with human actors and external services.

Explanation

Thikana communicates with tenants/buyers, property owners, real estate agencies, and system administrators through the web interface. It also interacts with Google Maps or Mapbox for maps and area intelligence, Firebase for authentication, storage, real-time chat and notifications, payment gateways for sandbox or future live payments, SMS gateways for OTP delivery, and email servers for transactional messages.



6.2 Use Case Diagram

Purpose

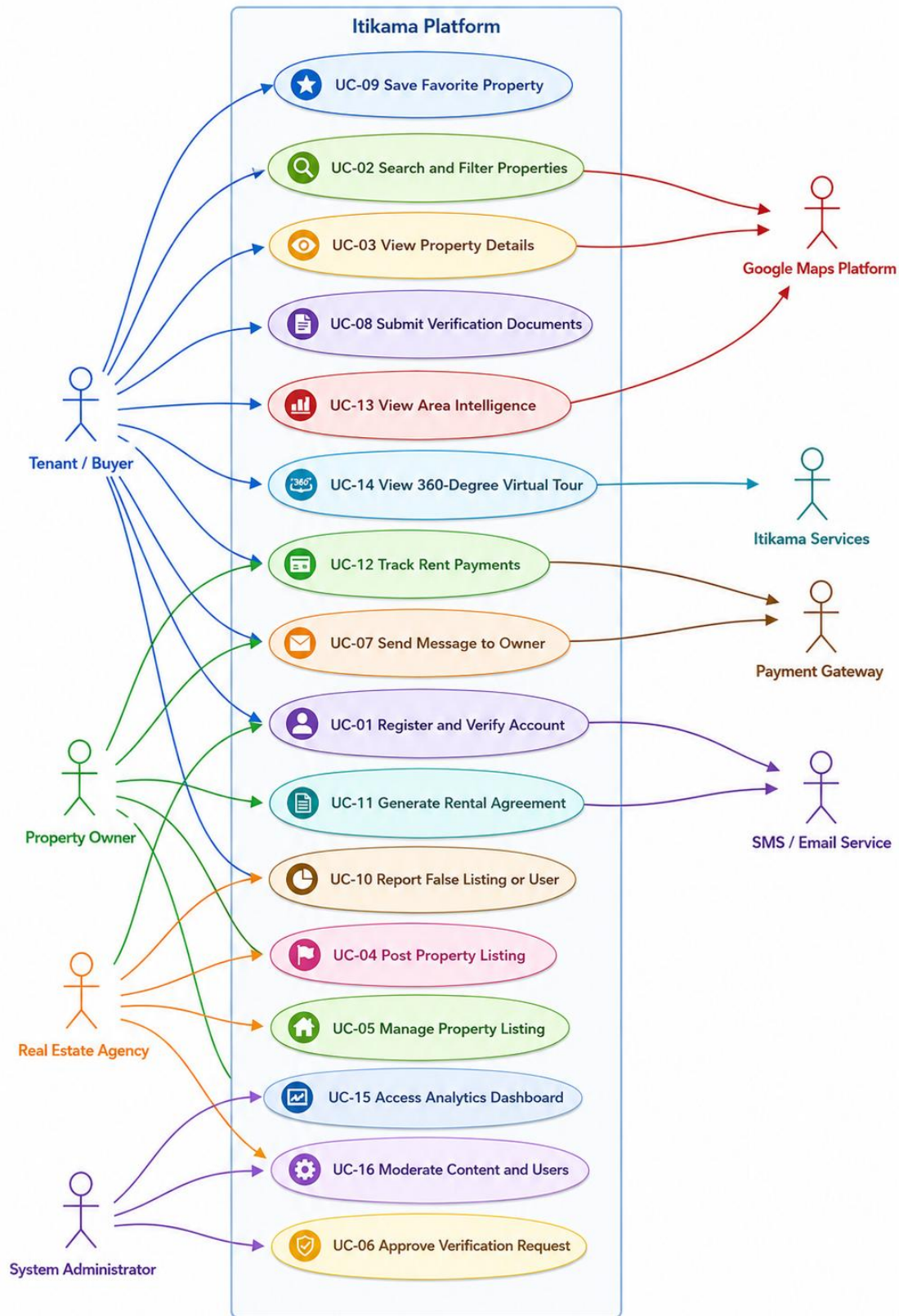
The use case diagram identifies the major actors of **Thikana** and the system functions that each actor can access.

Explanation

Tenants and buyers use the system for searching properties, viewing details, saving favorites, messaging owners, submitting verification documents, viewing area intelligence, and reviewing virtual tours.

Property owners and agencies post and manage listings, communicate with users, generate agreements, track payments, and view analytics. Administrators verify users, moderate content, handle reports, and monitor platform-level analytics.

Use Case Diagram - Itikama Smart Real Estate Platform



6.3 Detailed Use Case Specifications

The following use case specifications describe the priority interactions between the actors and the **Thikana** platform. These descriptions provide a narrative view of the major system behaviors that are later represented through activity, sequence, state transition, class, ERD, and data flow models.

Use Case ID	Use Case Name	Primary Actor(s)	Main Flow Summary	Postcondition
UC-01	Register and Verify Account	Tenant, Buyer, Owner, Agency	User provides account information; system sends OTP; user verifies phone; role-based dashboard opens.	User has an active, phone-verified account.
UC-02	Search and Filter Properties	Tenant, Buyer	User enters keyword/location and filters; system retrieves active listings and displays results in list and map views.	Matching properties are displayed or no-result guidance is shown.
UC-03	View Property Details	Tenant, Buyer	User opens a listing; system displays description, price, photos, amenities, owner information, and availability status.	User can decide whether to save, message, report, or view area intelligence.
UC-04	Post Property Listing	Owner, Agency	Owner enters listing details, uploads media, sets price and availability; system validates and publishes or queues listing.	Listing is created and becomes visible according to status and moderation rules.

Use Case ID	Use Case Name	Primary Actor(s)	Main Flow Summary	Postcondition
UC-05	Manage Property Listing	Owner, Agency, Admin	Authorized user edits property details, status, photos, availability, or moderation state.	Listing data and public visibility are updated.
UC-06	Save Favorite Property	Tenant, Buyer	User saves a listing to favorites and can later remove it.	Favorite list is updated and alert eligibility is created.
UC-07	Send Message to Owner	Tenant, Buyer, Owner, Agency	User opens chat from listing detail and sends message; owner receives real-time notification and may reply.	Conversation history is stored and notification is delivered.
UC-08	Submit Verification Documents	Tenant	Tenant uploads NID or student ID; system stores document privately and marks status pending.	Admin receives verification request for review.
UC-09	Approve Verification Request	Admin	Admin reviews submitted document and approves, rejects, or requests more information.	User verification status and badge are updated.
UC-10	Report Fake Listing or User	Any User	User submits report with reason; system creates moderation case for admin review.	Report status becomes submitted or under review.
UC-11	Generate Rental Agreement	Owner	Owner selects tenant and listing; system prepares pre-filled agreement; tenant accepts or rejects.	Agreement becomes active or rejected.

Use Case ID	Use Case Name	Primary Actor(s)	Main Flow Summary	Postcondition
UC-12	Track Rent Payments	Tenant, Owner	System creates due schedule; tenant logs payment; owner/admin can view payment history.	Payment record is updated and receipt/history is available.
UC-13	View Area Intelligence	Tenant, Buyer	User opens area panel; system loads nearby facilities and commute information.	Nearby schools, hospitals, markets, mosques, transport and commute estimates are displayed.
UC-14	View 360-Degree Virtual Tour	Tenant, Buyer	User opens virtual tour; system loads panoramic media in a viewer.	User can remotely explore the property interior.
UC-15	Access Analytics Dashboard	Owner, Agency, Admin	Authorized user opens analytics dashboard; system displays views, saves, enquiries and trend data.	Dashboard metrics and reports are available.
UC-16	Moderate Content and Users	Admin	Admin reviews reports, suspicious listings, users, reviews, and verification records.	Admin action is recorded and affected content or user status is updated.

6.4 Activity Diagrams

Activity diagrams describe the flow of actions and decisions inside major business processes. The following activity diagrams model core **Thikana** workflows that are critical for the MVP and future expansion.

6.4.1 User Registration and Verification Activity Diagram

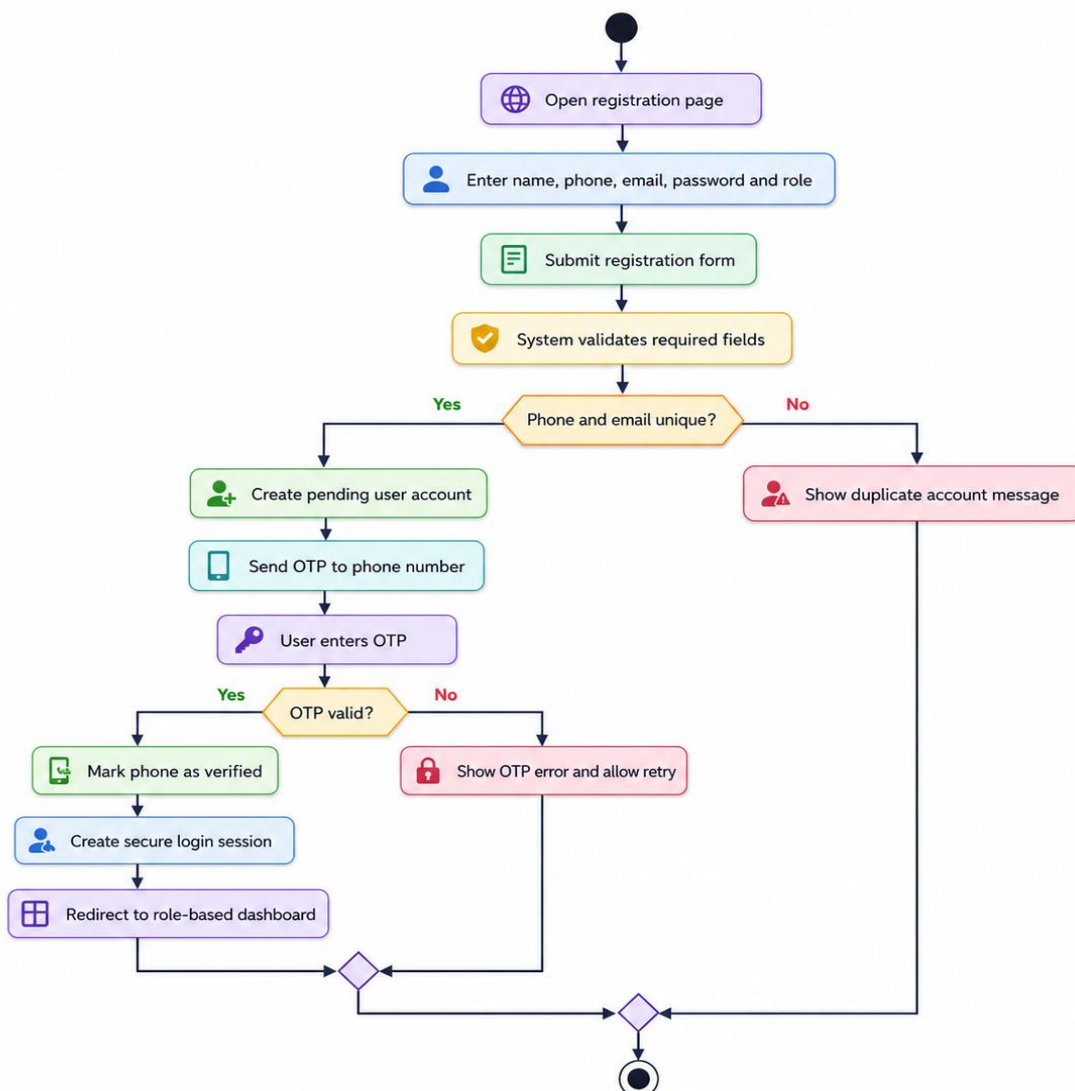
Purpose

This diagram models how a user registers, verifies OTP, and reaches the correct dashboard.

Explanation

The registration workflow validates required inputs, enforces unique phone and email values, sends OTP, verifies the OTP, and creates a verified session for the user. Duplicate accounts and invalid OTPs result in corrective messages.

Activity Diagram - User Registration and Verification



6.4.2 Smart Property Search and Filtering Activity Diagram

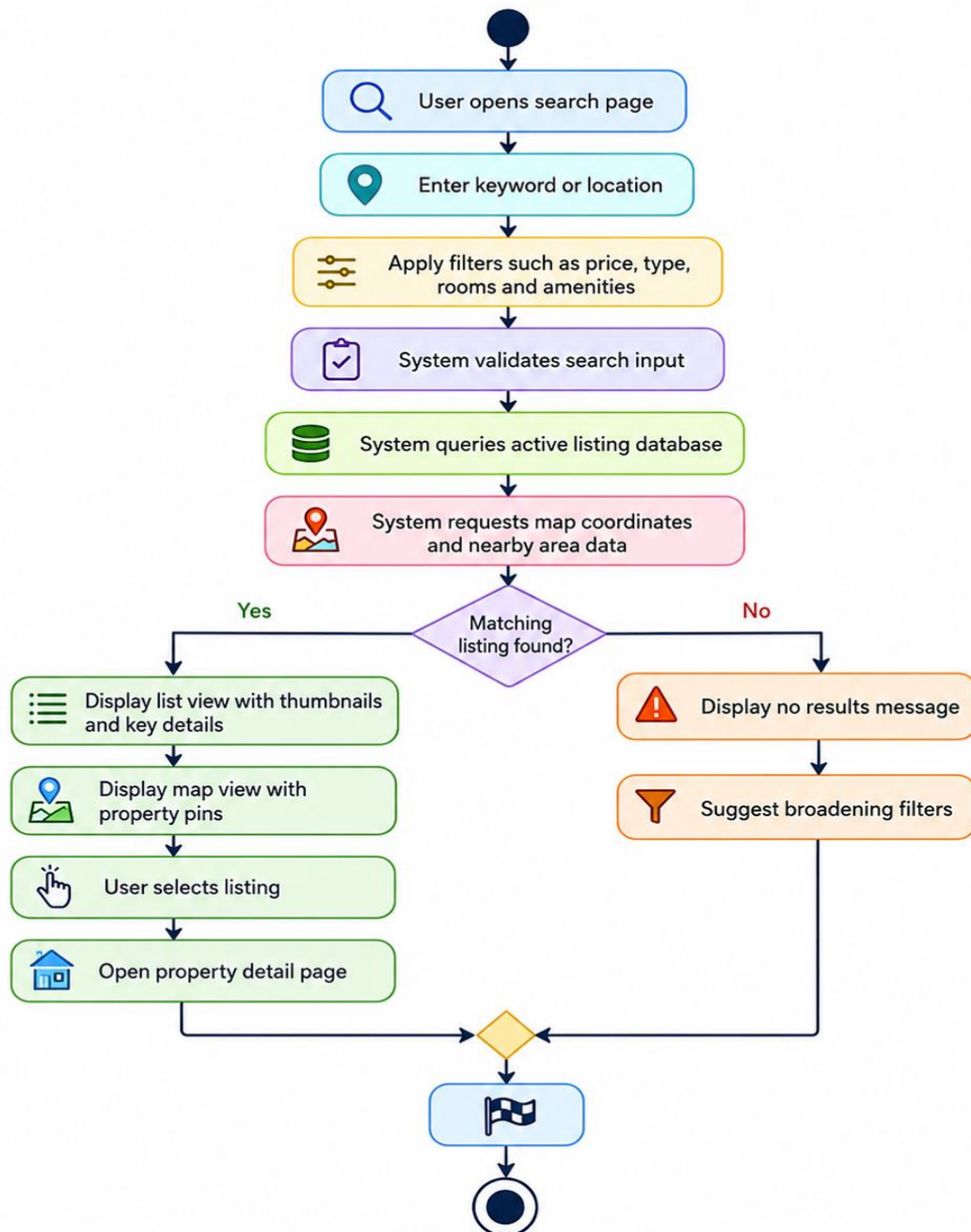
Purpose

This diagram models how a tenant or buyer searches and filters property listings.

Explanation

The search process accepts keywords, locations, and filters; retrieves active listings; enriches the results with map data; and displays either matching properties or no-result guidance.

Activity Diagram - Smart Property Search and Filtering



6.4.3 Property Listing Creation Activity Diagram

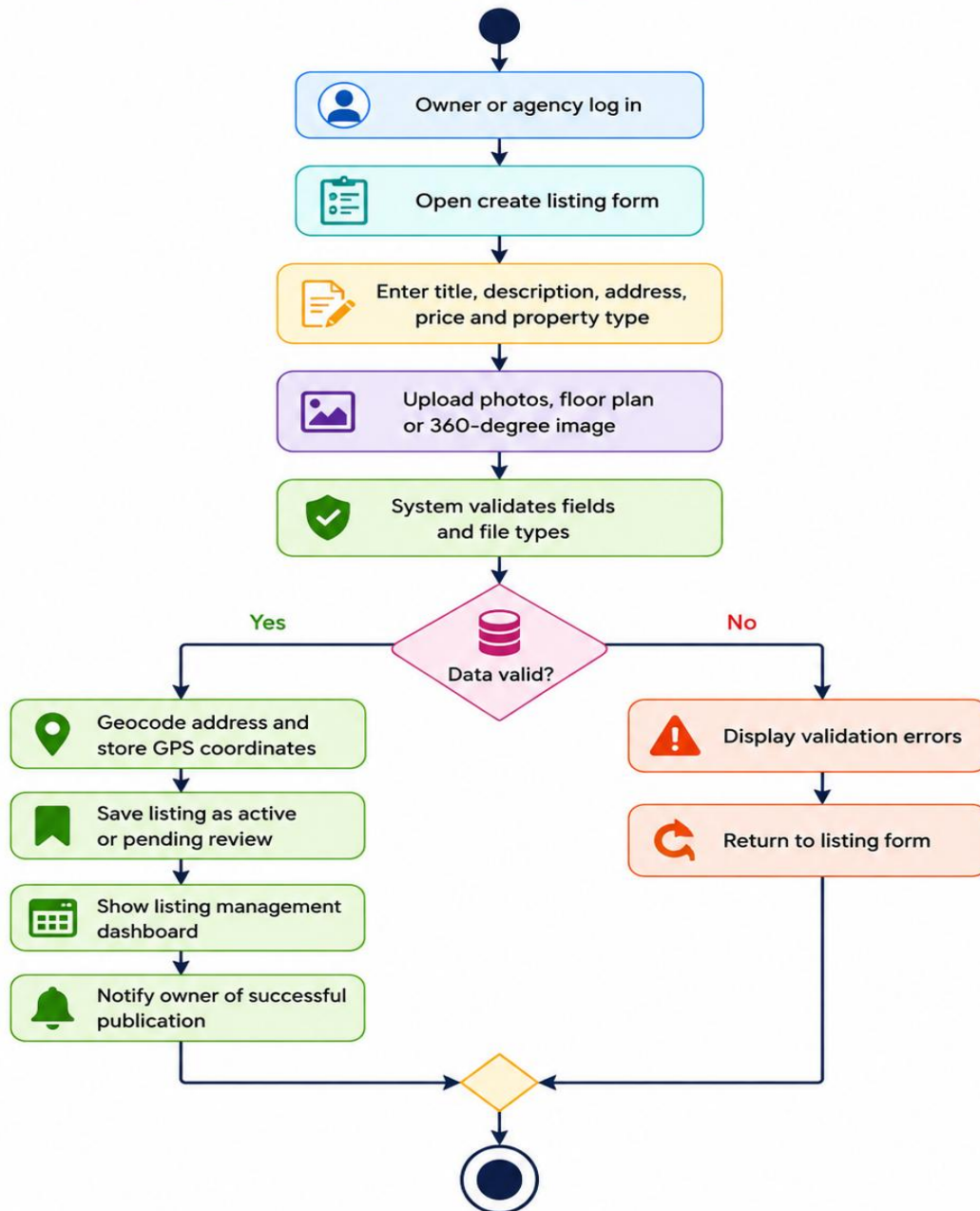
Purpose

This diagram models how owners and agencies create and publish property listings.

Explanation

The listing workflow validates the form, uploads media, geocodes the address, stores listing records, and publishes the listing or returns validation errors.

Activity Diagram - Property Listing Creation and Publication



6.4.4 Tenant Verification Activity Diagram

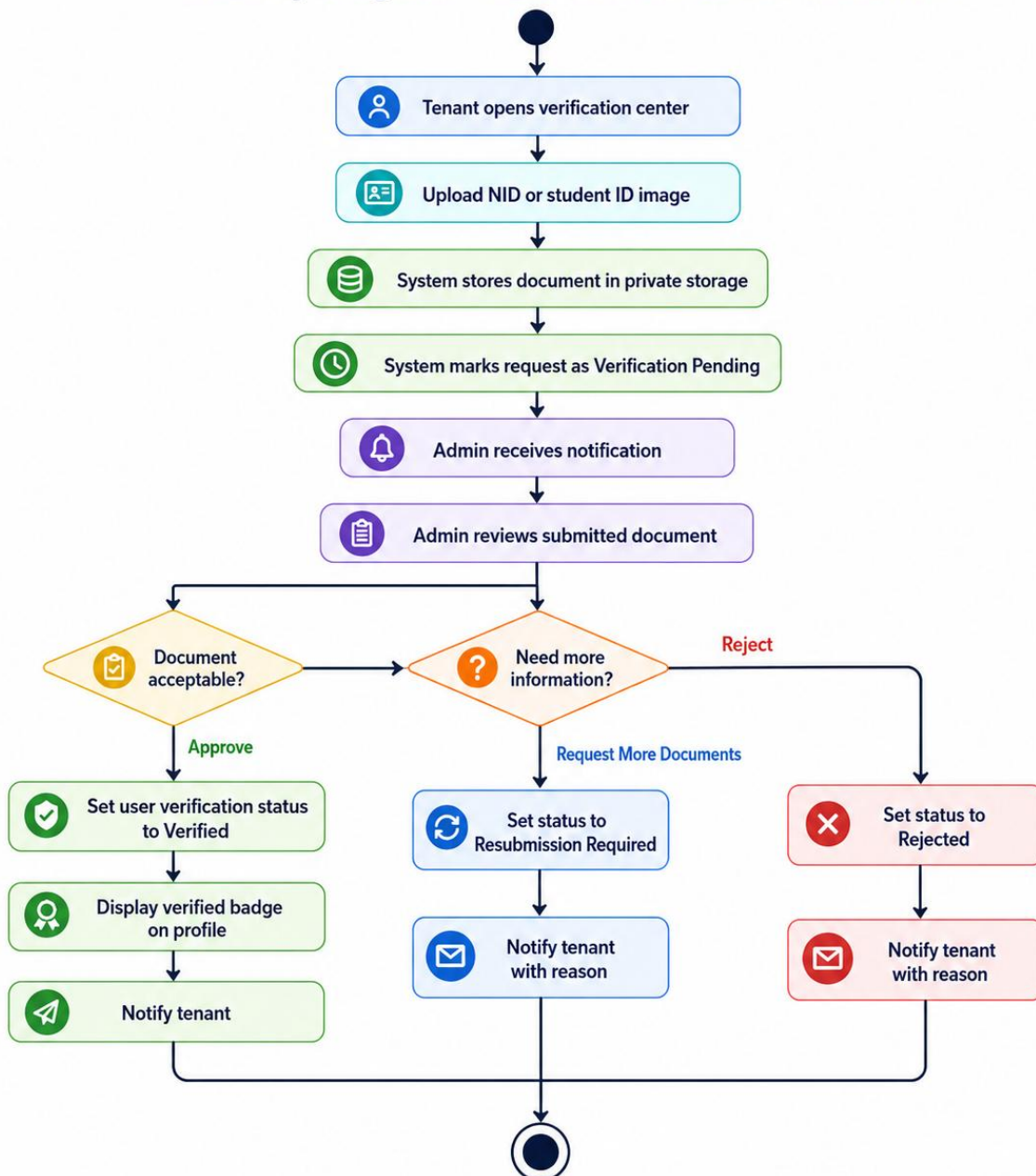
Purpose

This diagram models how a tenant submits identity documents and how the administrator reviews the request.

Explanation

The workflow includes document upload, private storage, pending status, admin review, approval, rejection, or request for further documents, followed by notification to the tenant.

Activity Diagram - Tenant Verification Workflow



6.4.5 Rental Agreement and Rent Tracking Activity Diagram

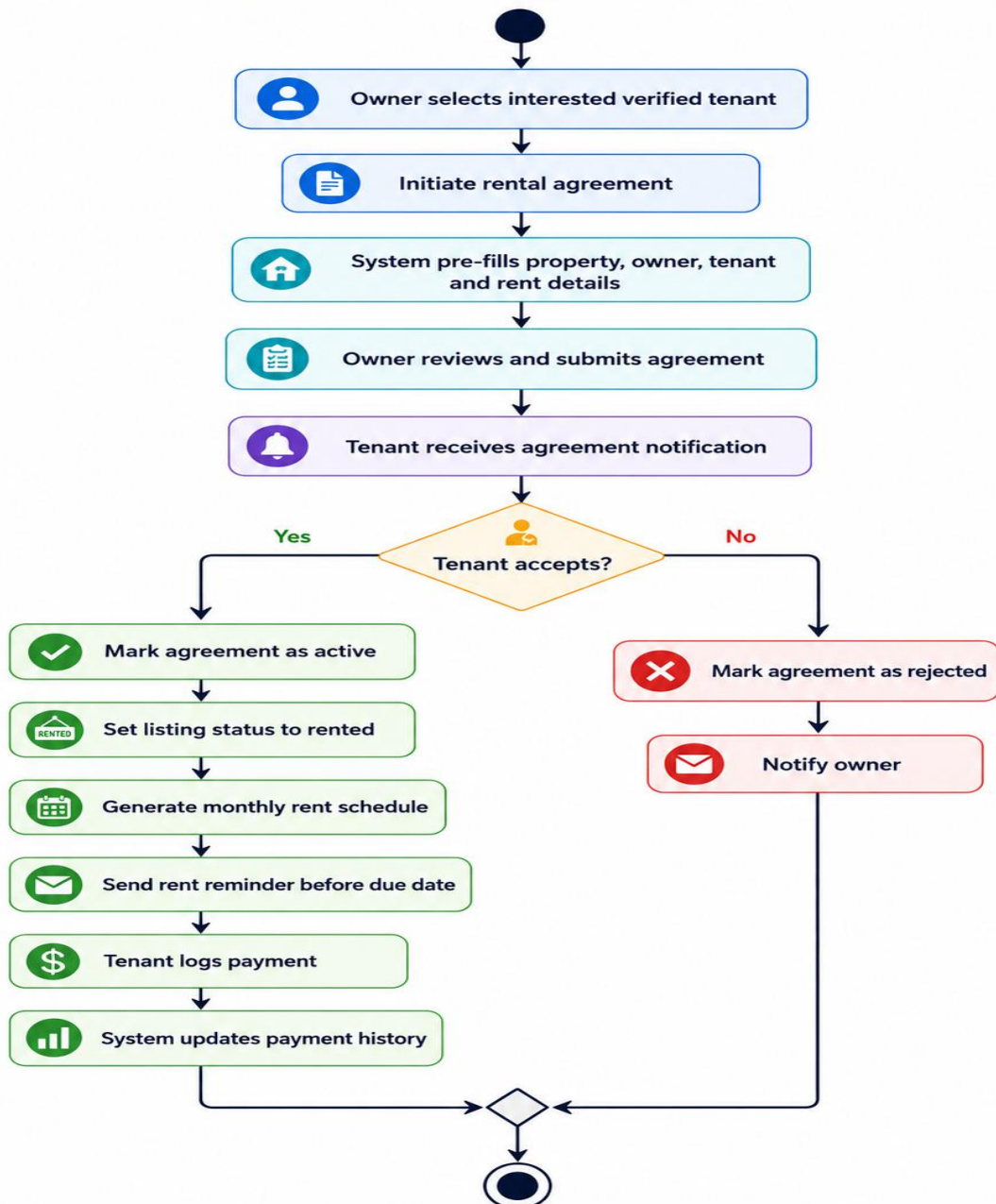
Purpose

This diagram models the rental agreement and payment tracking process.

Explanation

The process starts when an owner selects a verified tenant, generates a pre-filled agreement, waits for tenant acceptance, activates the agreement, updates the listing status, and manages rent reminders and payment records.

Activity Diagram - Rental Agreement and Rent Tracking



6.5 Swimlane Diagrams

Swimlane diagrams show how responsibility moves between actors and the system during multi-party workflows. They are especially useful for **Thikana** because major platform actions involve tenants, owners, administrators, external map services, Firebase, and payment services.

6.5.1 Property Search and Owner Inquiry Swimlane Diagram

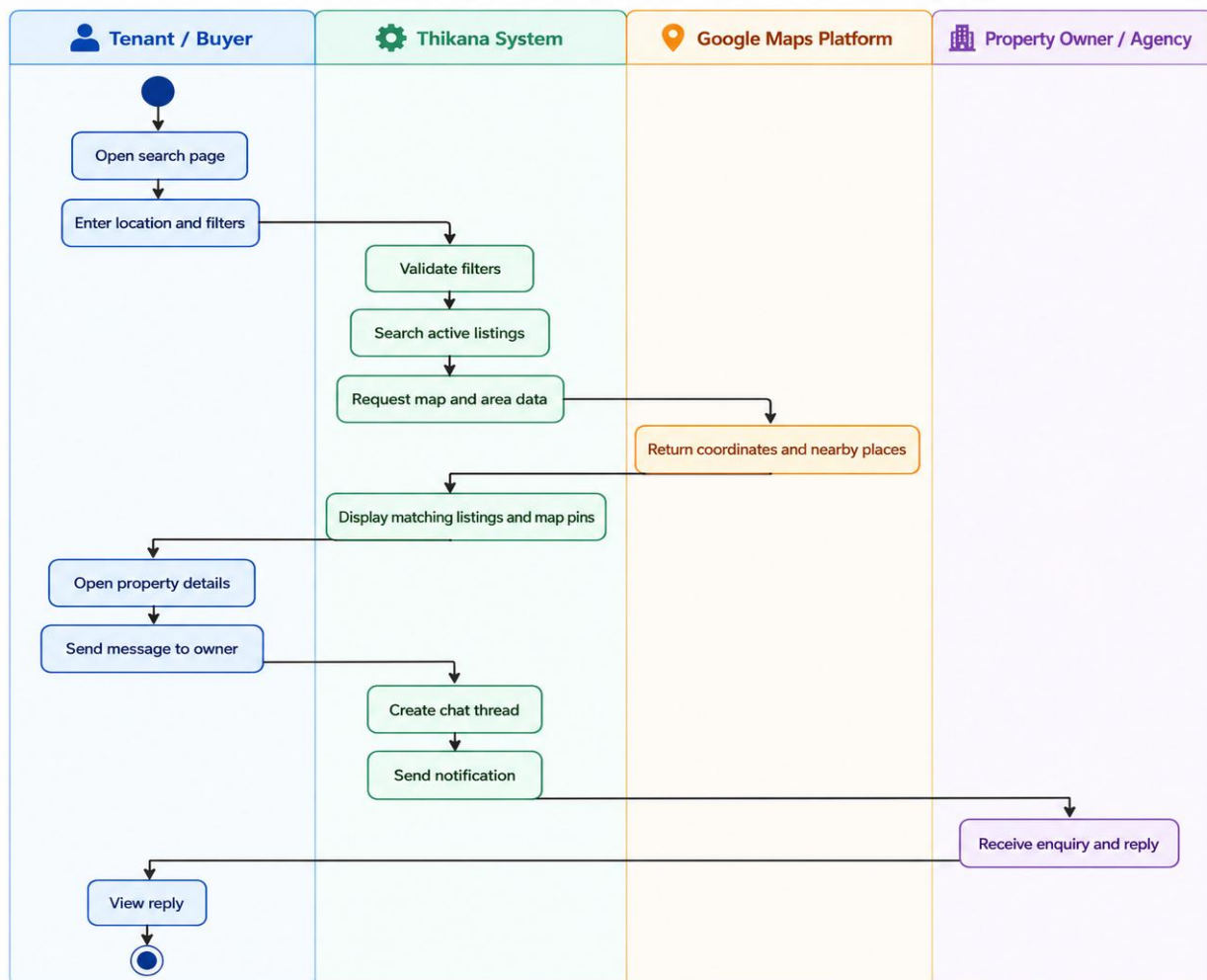
Purpose

This swimlane diagram shows how a tenant/buyer discovers a property and starts communication with an owner or agency.

Explanation

The workflow passes from the tenant/buyer to the Thikana system, then to Google Maps for map and facility data, and finally to the property owner through the messaging and notification service.

Swimlane Diagram - Property Search and Owner Inquiry



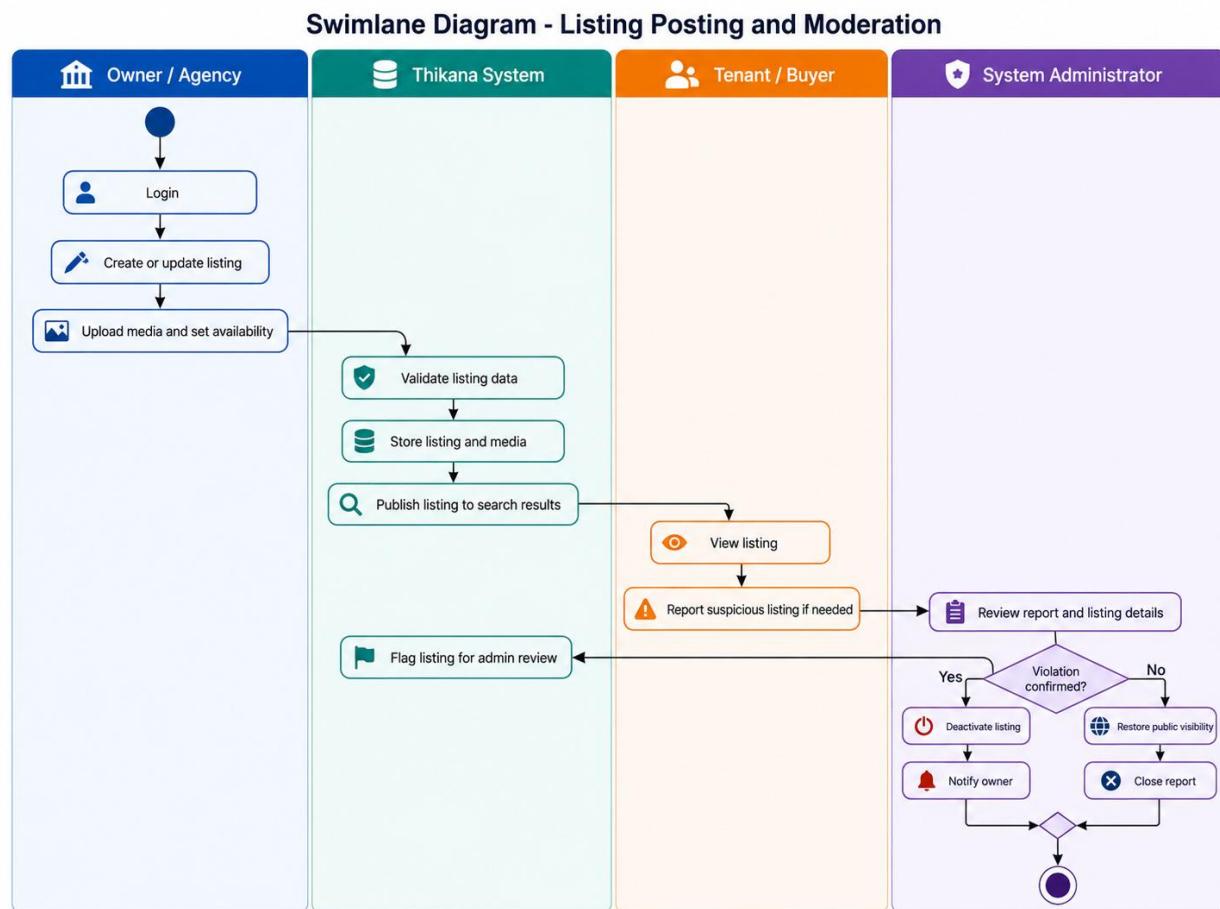
6.5.2 Listing Posting and Moderation Swimlane Diagram

Purpose

This swimlane diagram shows how listings are created and how reported listings are reviewed by administrators.

Explanation

Owners and agencies create listings, the system validates and publishes them, users may report suspicious content, and the administrator either deactivates or restores the listing.



6.5.3 Rental Agreement and Payment Tracking Swimlane Diagram

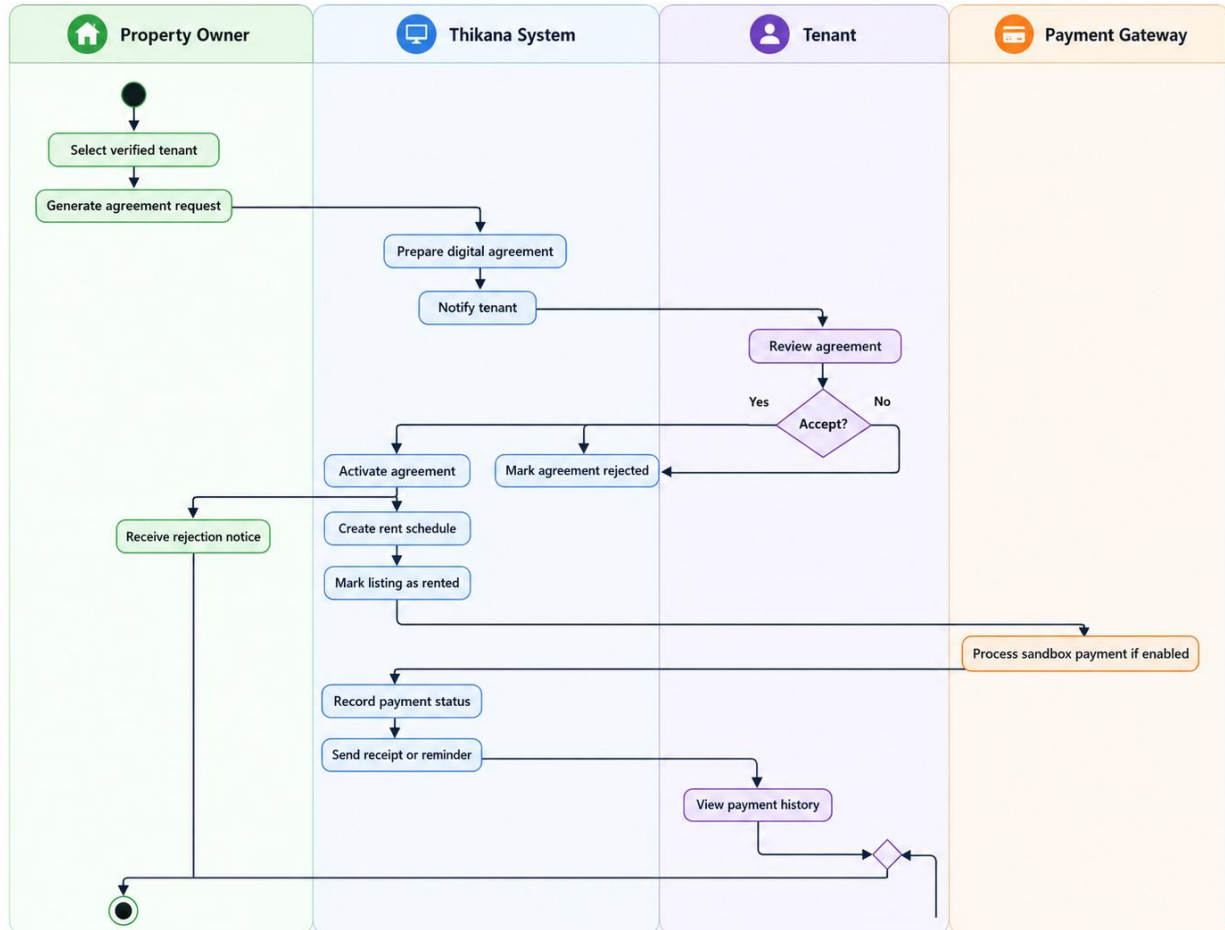
Purpose

This swimlane diagram shows the collaboration between owner, tenant, Thikana system, and payment gateway during rental agreement handling.

Explanation

The owner initiates the agreement, the system prepares and sends it to the tenant, the tenant accepts or rejects, and the system creates payment schedules and records payment status.

Swimlane Diagram - Rental Agreement and Payment Tracking



6.6 Sequence Diagrams

Sequence diagrams describe object-to-object or component-to-component communication over time. The following sequence diagrams show how the frontend, controllers, services, databases, and external providers collaborate to complete major Thikana functions.

6.6.1 Registration, Login and OTP Verification Sequence Diagram

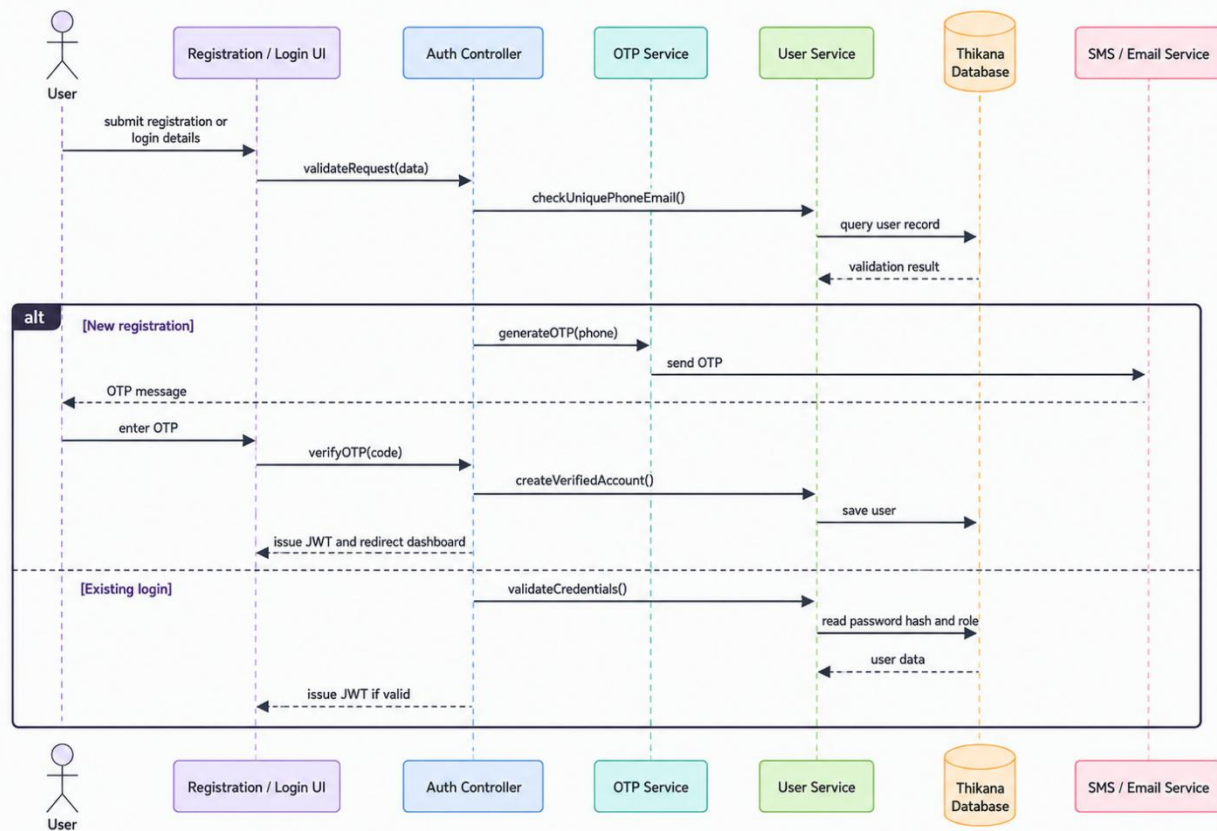
Purpose

This sequence diagram shows how user registration and authentication requests are processed.

Explanation

The flow includes UI submission, authentication controller validation, user service lookup, OTP generation, SMS/email delivery, account creation, and JWT issuance.

Sequence Diagram - Registration, Login and OTP Verification



6.6.2 Smart Property Search Sequence Diagram

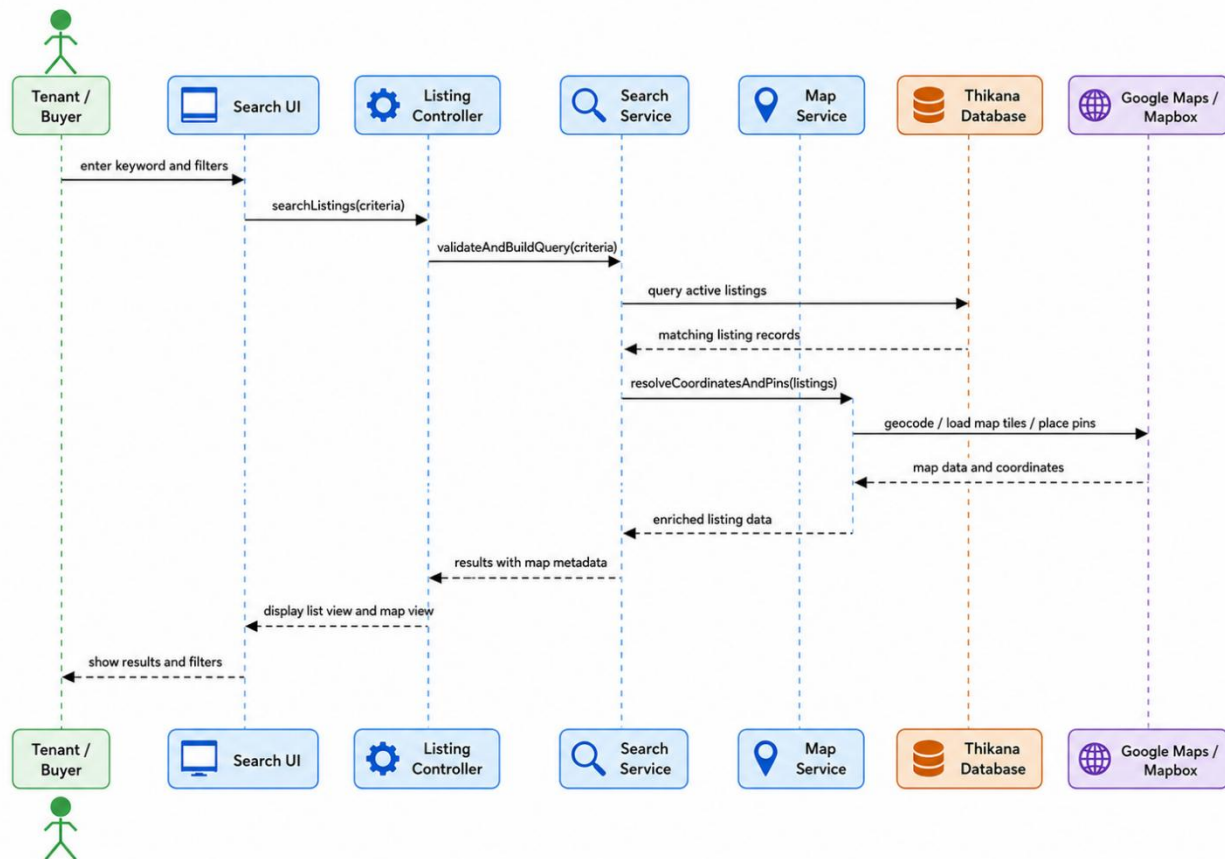
Purpose

This sequence diagram shows how search requests move through the listing controller, search service, database, map service, and UI.

Explanation

The system validates criteria, searches active listings, enriches them with coordinates and map metadata, and returns a combined list and map view to the user.

Sequence Diagram - Smart Property Search



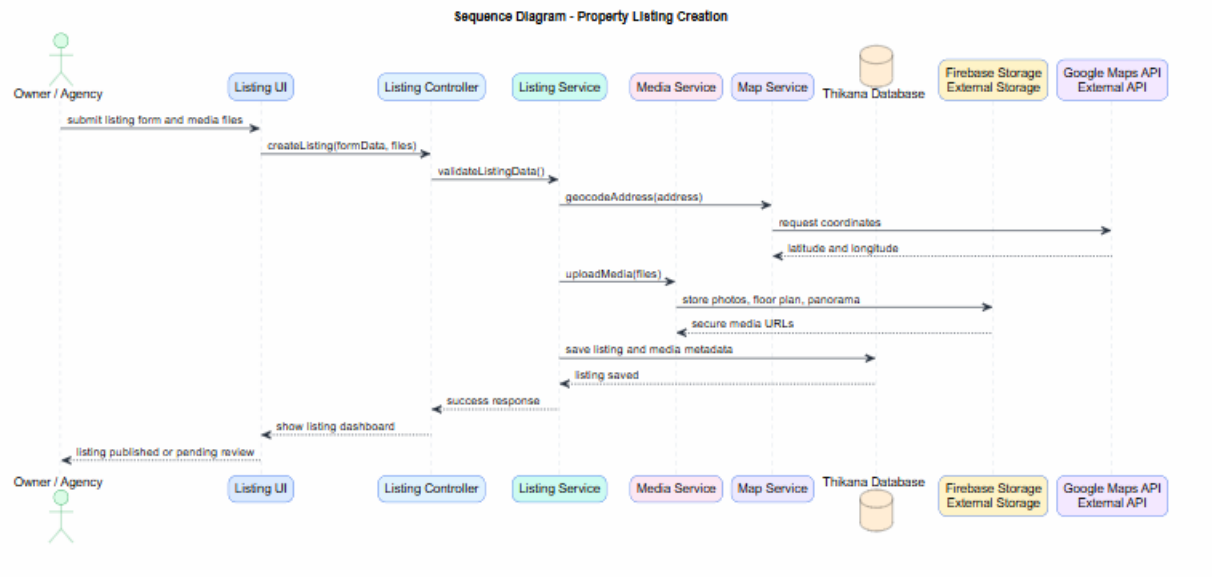
6.6.3 Property Listing Creation Sequence Diagram

Purpose

This sequence diagram shows how owners and agencies create listings with property details and media files.

Explanation

The listing controller validates data, the map service geocodes the address, the media service stores uploaded files, and the listing service persists metadata in the database.



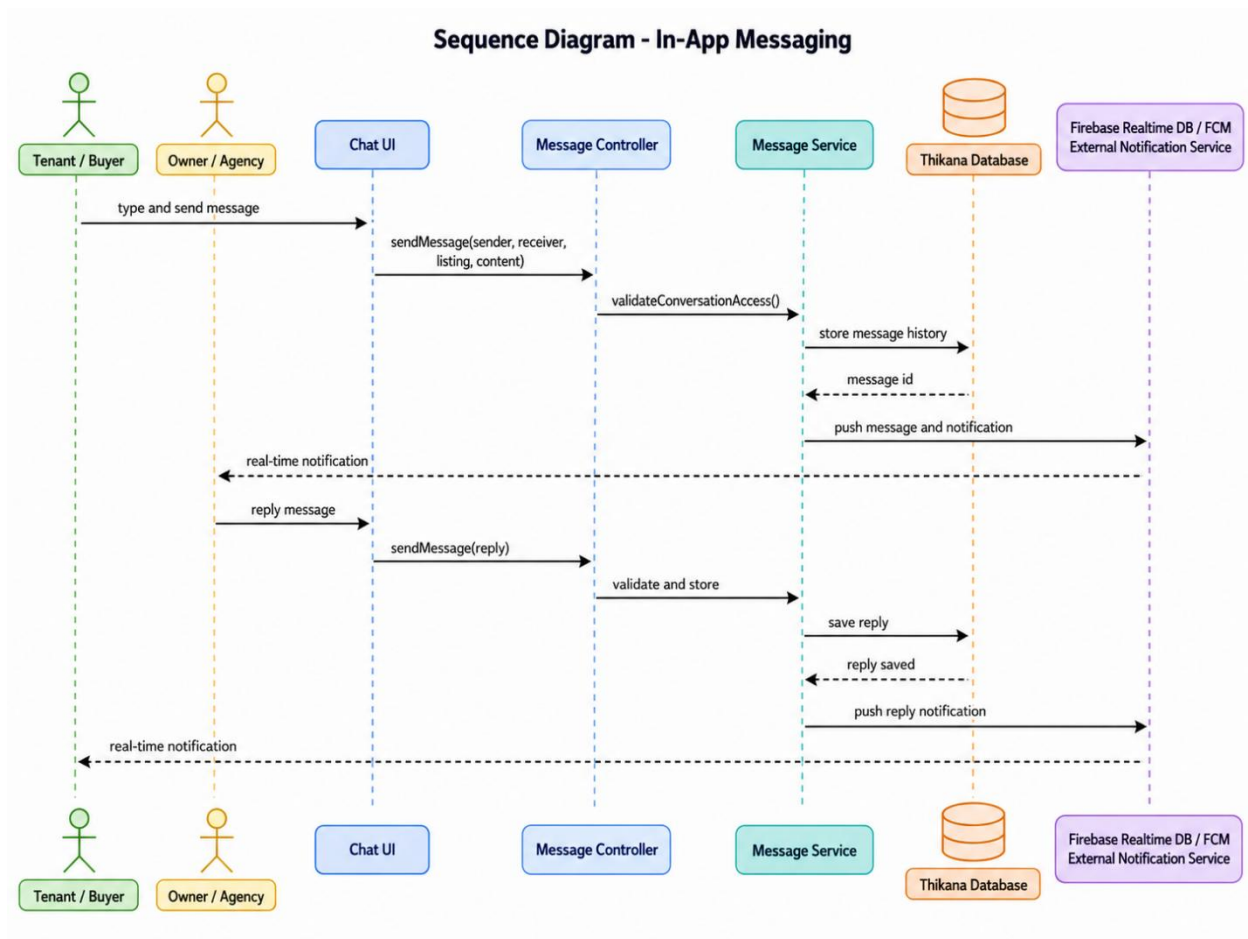
6.6.4 In-App Messaging Sequence Diagram

Purpose

This sequence diagram shows how messages are stored and delivered in real time.

Explanation

The message controller validates access, stores conversation history, and uses Firebase Realtime Database or Cloud Messaging to push notifications between tenants/buyers and owners/agencies.



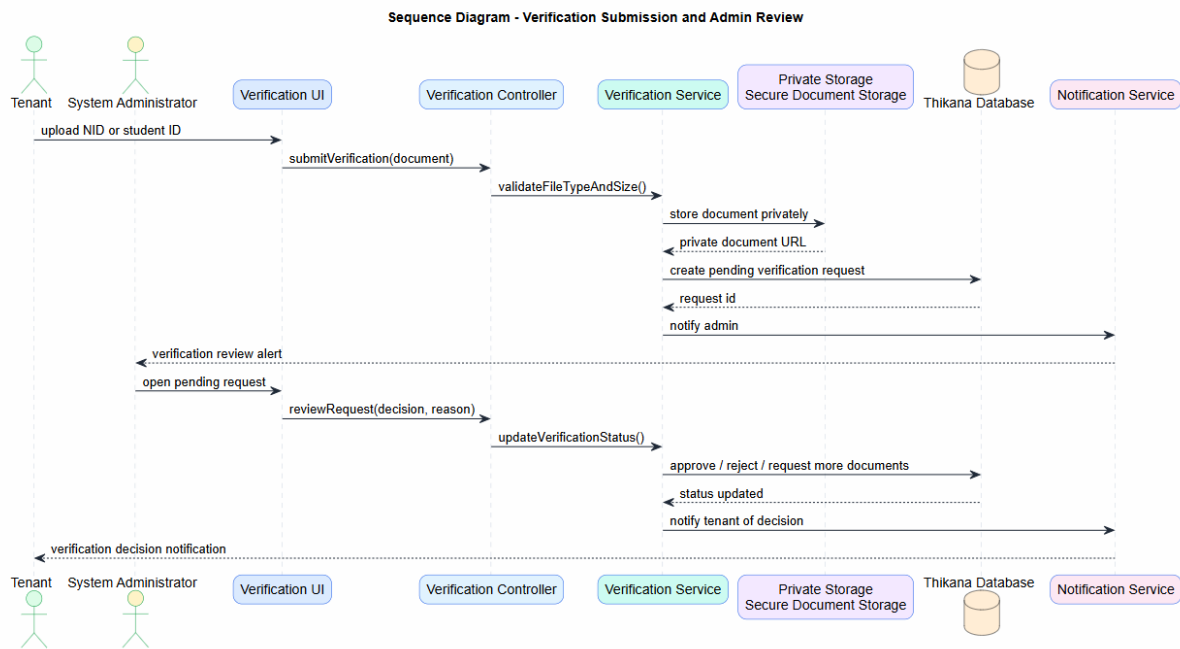
6.6.5 Verification Submission and Admin Review Sequence Diagram

Purpose

This sequence diagram shows how identity verification documents are submitted, stored privately, reviewed, and resolved.

Explanation

The verification controller validates uploaded files, stores them in private storage, creates pending requests, notifies admins, and updates user verification status based on admin decisions.



6.7 State Transition Diagrams

State transition diagrams describe how major business objects change status over time. Thikana contains several status-driven objects, including user accounts, property listings, rental agreements, and moderation reports.

6.7.1 User Account Verification Lifecycle

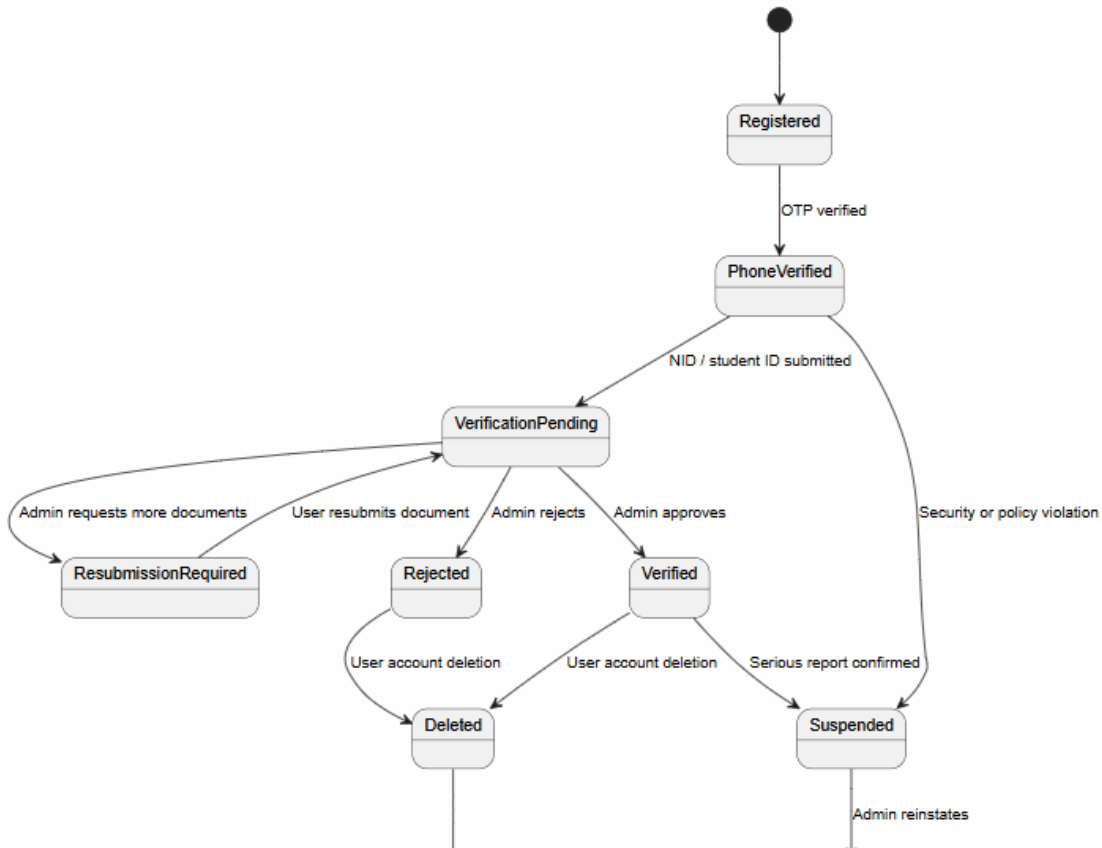
Purpose

This diagram describes the major states of a user account from registration to verification, suspension, deletion, or rejection.

Explanation

A user account starts as registered, becomes phone verified after OTP, may enter verification pending status after document submission, and can become verified, rejected, suspended, or deleted depending on system and admin actions.

State Transition Diagram - User Account Verification Lifecycle



6.7.2 Property Listing Lifecycle

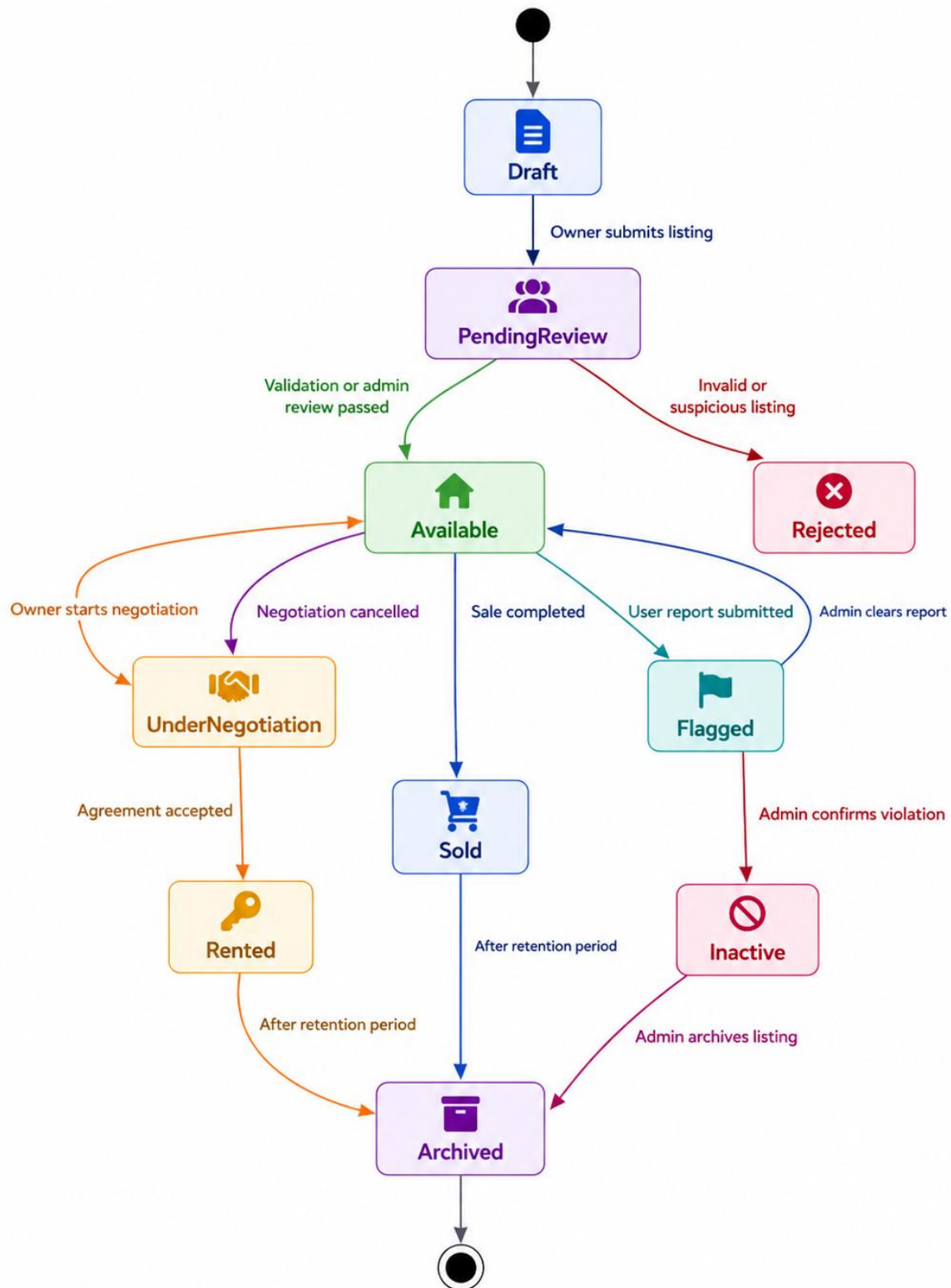
Purpose

This diagram describes the status changes of a property listing.

Explanation

A listing can move from draft to pending review, available, under negotiation, rented, sold, flagged, inactive, and archived. Reported listings require administrative decision before being restored or removed.

State Transition Diagram - Property Listing Lifecycle



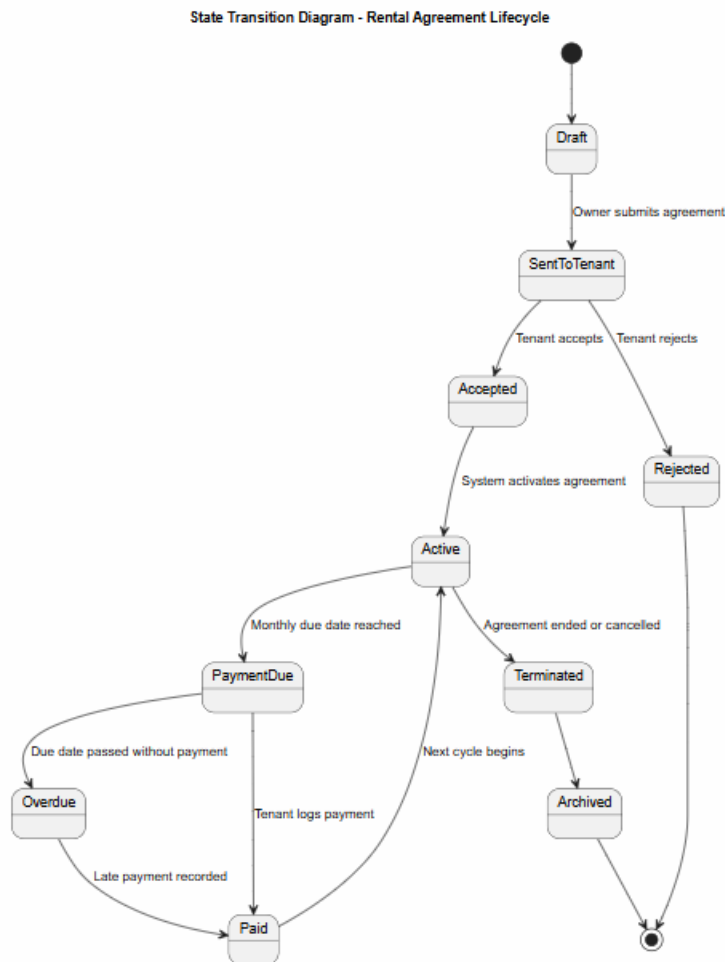
6.7.3 Rental Agreement Lifecycle

Purpose

This diagram describes the status changes of a rental agreement and its rent payment cycle.

Explanation

An agreement starts as draft, is sent to the tenant, becomes accepted or rejected, and after activation it cycles through payment due, paid, overdue, and finally terminated or archived states.



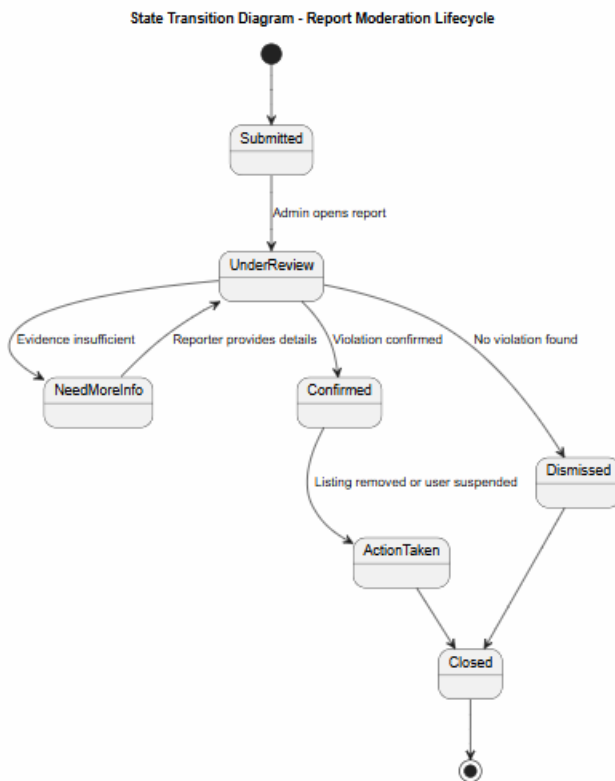
6.7.4 Report Moderation Lifecycle

Purpose

This diagram describes how fake listing, suspicious user, and content reports are processed.

Explanation

A report moves from submitted to under review, may require more information, and is either confirmed or dismissed. Confirmed reports lead to action taken and then closure.



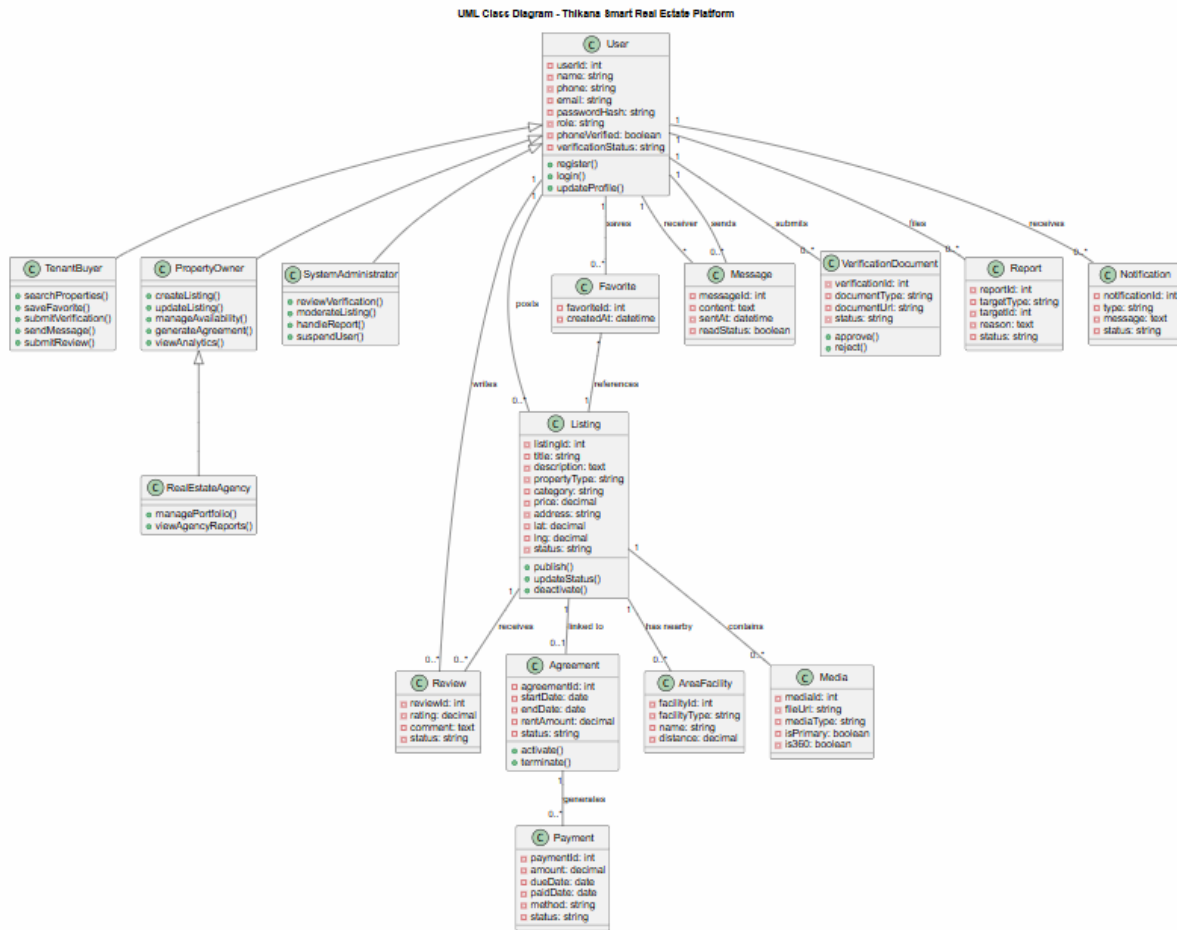
6.8 Class Diagram

Purpose

The class diagram represents the object-oriented structure of the Thikana platform.

Explanation

The User class is the parent abstraction for tenants/buyers, property owners, real estate agencies, and administrators. Listing, Media, Favorite, Message, VerificationDocument, Review, Agreement, Payment, Report, Notification, and AreaFacility represent the main domain objects. Their relationships support property discovery, listing management, communication, verification, agreements, payments, analytics, and moderation.



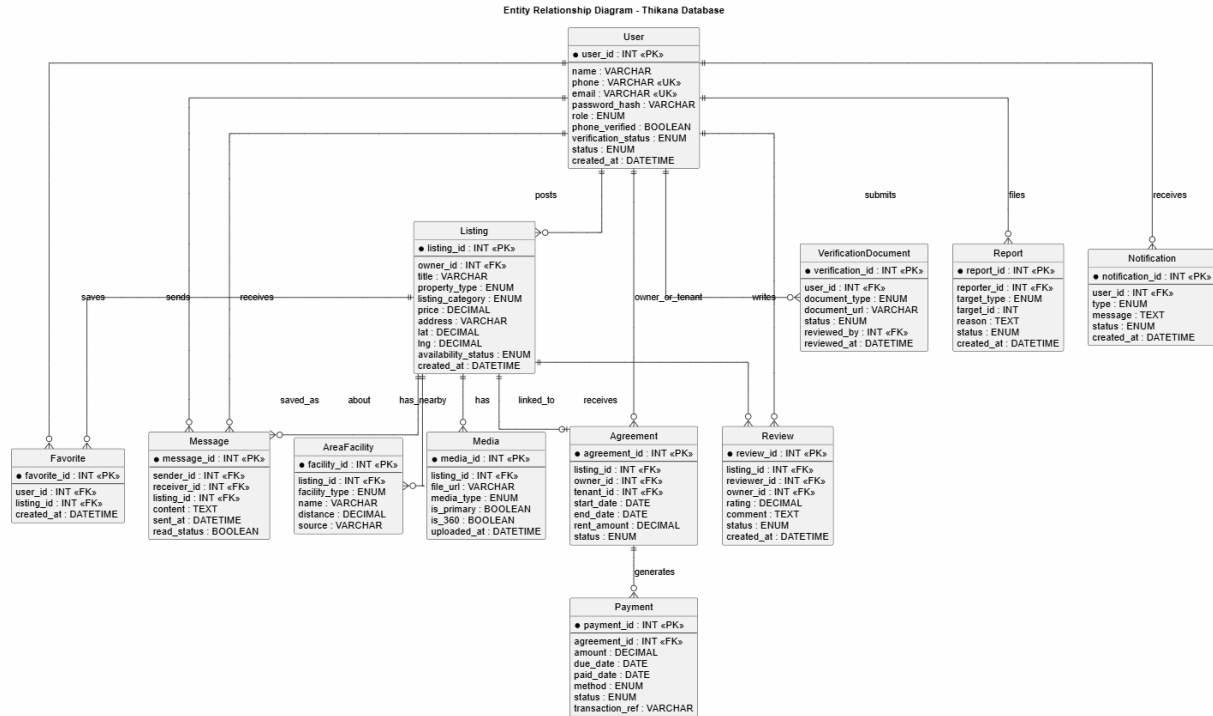
6.9 Entity Relationship Diagram (ERD)

Purpose

The ERD models the database entities, primary keys, foreign keys, and major relationships required by Thikana.

Explanation

The database structure is centered on users and listings. Users can post listings, save favorites, send messages, submit verification documents, write reviews, file reports, receive notifications, and participate in agreements. Listings can contain media, receive reviews, become part of rental agreements, and store nearby area facility data.



6.10 Data Flow Diagrams

Data flow diagrams describe how information moves between external actors, Thikana processes, external services, and internal data stores. They help clarify the input, processing, storage, and output behavior of the platform.

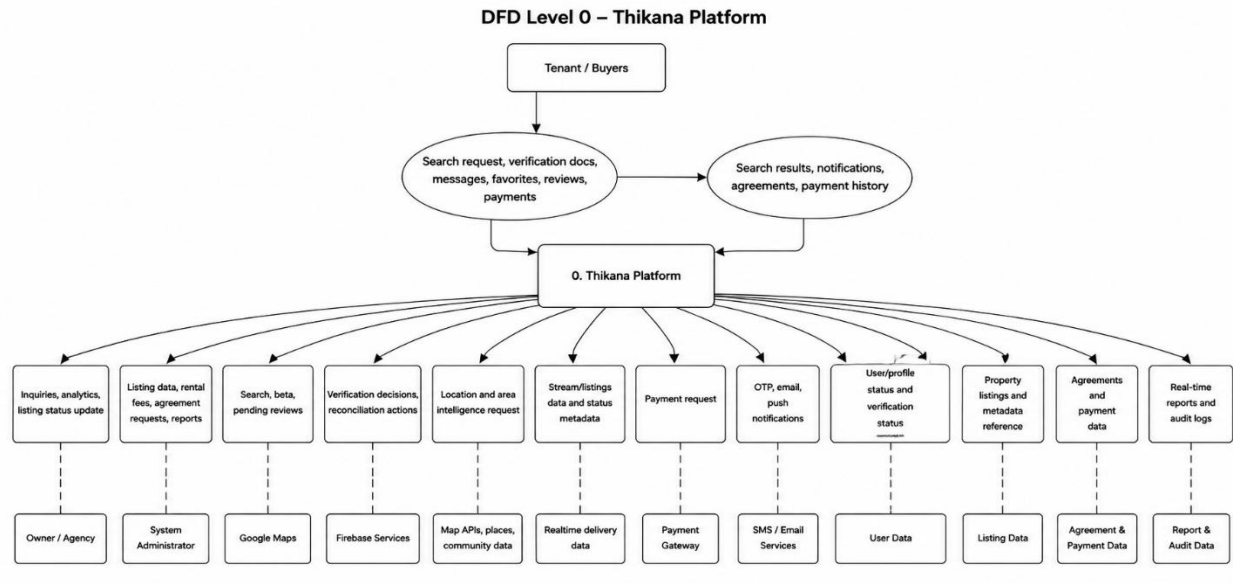
6.10.1 DFD Level 0 - Thikana Platform

Purpose

This diagram presents the Thikana platform as a single process and shows its data exchange with users, admins, external systems, and data stores.

Explanation

Tenants, buyers, owners, agencies, and administrators send requests and receive responses through Thikana. The system exchanges map, Firebase, payment, SMS, and email data with external services and stores users, listings, agreements, payments, reports, and audit data internally.



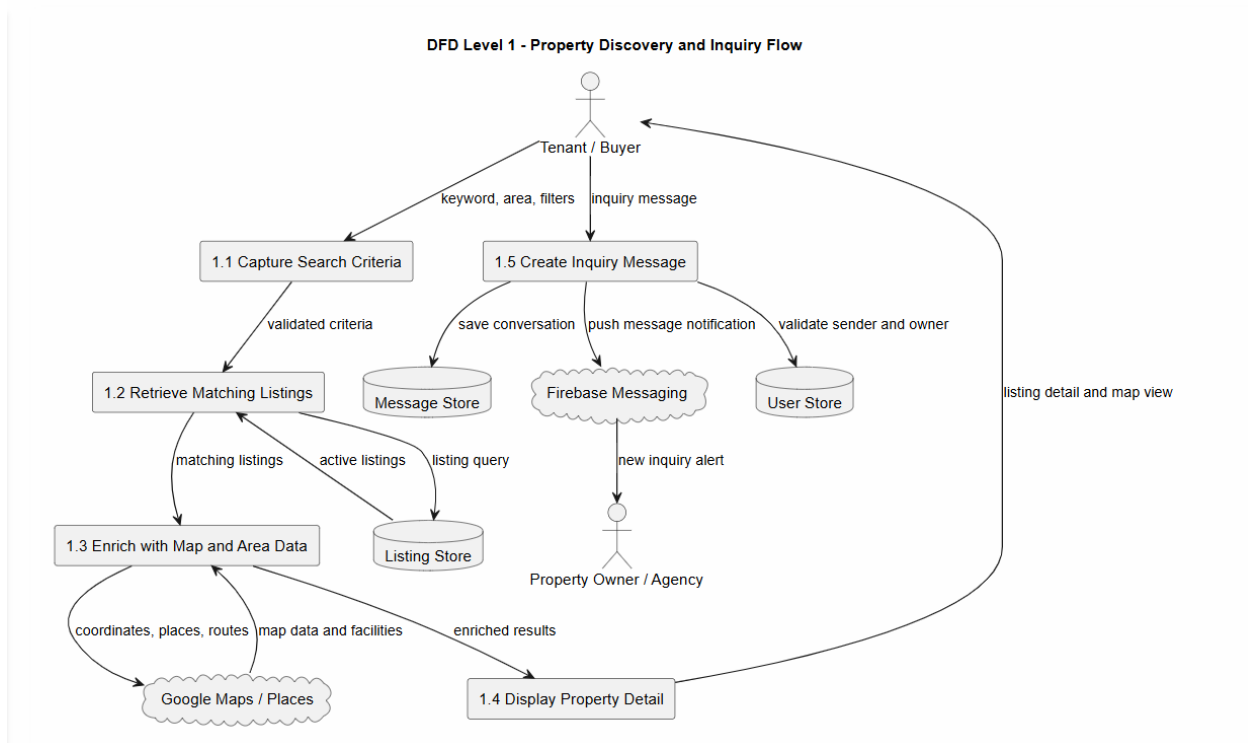
6.10.2 DFD Level 1 - Property Discovery and Inquiry Flow

Purpose

This diagram decomposes the property discovery process into smaller data-processing steps.

Explanation

The tenant or buyer submits search criteria, the system retrieves matching listings, enriches them with map and places data, displays listing details, and creates an inquiry message to the owner or agency through the messaging service.



6.11 Decision Tables

Decision tables define business logic in a compact and testable form. The following decision tables clarify how **Thikana** determines listing visibility and user verification outcomes.

6.11.1 Property Listing Visibility Decision Table

Rule	Listing Status	Owner Status	Reported?	Admin Review Status	System Action
R1	Available	Active	No	Not required	Show listing in public search and map view
R2	Under Negotiation	Active	No	Not required	Show listing with under-negotiation badge
R3	Rented	Active	No	Not required	Hide from normal search; allow owner/admin history view

Rule	Listing Status	Owner Status	Reported?	Admin Review Status	System Action
R4	Sold	Active	No	Not required	Hide from normal search; allow owner/admin history view
R5	Available	Suspended/Banned	No	Not required	Hide listing until owner account is restored
R6	Any Status	Any Status	Yes	Pending Review	Remove from public search and send to admin queue
R7	Inactive/Deleted	Any Status	No	Not required	Hide from all public views

6.11.2 Tenant Verification Decision Table

Rule	Phone Status	Document Status	Admin Finding	System Action
R1	Phone OTP verified	Valid document submitted	Document clear and matches profile	Approve verification and show Verified Tenant badge
R2	Phone OTP verified	Valid document submitted	Document unclear or incomplete	Request resubmission with explanation
R3	Phone OTP verified	Invalid document type	Not applicable	Reject or request valid NID/student ID
R4	Phone not verified	Any document status	Not applicable	Block document review until phone is verified
R5	Phone OTP verified	No document submitted	Not applicable	Keep user unverified and allow basic browsing only

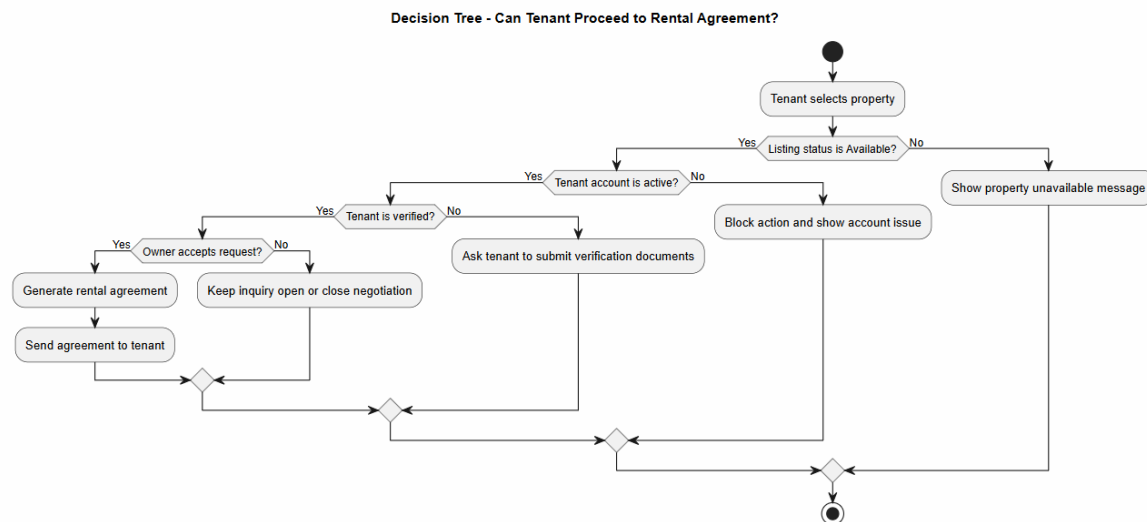
6.12 Decision Tree

Purpose

The decision tree models whether a tenant can proceed from property interest to rental agreement generation.

Explanation

The system checks listing availability, tenant account status, tenant verification status, and owner acceptance before generating a rental agreement. If any condition fails, the system displays the appropriate corrective action.



6.13 Model Coverage Summary

The following table summarizes the purpose of each analysis model and the system area it supports. This summary helps developers, testers, and evaluators trace the models back to major Thikana workflows.

Model	Purpose	Supported System Area
Context Diagram	System boundary and external services	All user roles and integrations
Use Case Diagram	Actor-to-function access mapping	Functional scope and role-based capability
Activity Diagrams	Step-by-step workflow logic	Registration, search, listing, verification, agreements
Swimlane Diagrams	Responsibility distribution among actors	Search inquiry, listing moderation, agreement/payment
Sequence Diagrams	Component interaction over time	Auth, search, listing, messaging, verification
State Transition Diagrams	Lifecycle of status-driven objects	User accounts, listings, agreements, reports
Class Diagram	Object-oriented domain structure	Core backend design and service responsibility
ERD	Database entities and relationships	Persistent data model and relational schema
DFDs	Data movement among users, processes and stores	Input, processing, storage and output analysis
Decision Tables / Tree	Business rule clarification	Listing visibility, verification, agreement eligibility

7. REQUIREMENTS TRACEABILITY MATRIX

This section provides the Requirements Traceability Matrix (RTM) for Thikana, a smart real estate and rental management platform for Bangladesh. The matrix links each functional and non-functional requirement to the use cases that realize it, the design entities or service classes responsible for implementation, and the test cases that verify correct behavior. It supports completeness checking, change impact analysis, auditability, and test coverage management.

The RTM is derived from the functional requirements FR-01 to FR-55 in Section 3, the data entities in Section 4, the quality attributes and other constraints in Sections 6 to 8, and the use case summary in Appendix B. Because the current SRS does not include a full UML class diagram, the Design Class column uses the primary data entities and implementation service classes implied by the logical data model and system features.

7.1 Purpose and Methodology

The purpose of this Requirements Traceability Matrix is to ensure that every stated requirement has a clear implementation and verification path. Forward traceability maps each requirement to use cases, design classes, and test cases. Backward traceability maps each design class to the requirements that justify its existence. This supports requirements validation, change control, test planning, and academic evaluation.

- Completeness verification: every requirement is linked to at least one use case, design class, and test case.
- Change impact analysis: affected use cases, design elements, and tests can be identified quickly when a requirement changes.
- Consistency assurance: design and testing artifacts remain aligned with the formal SRS requirements.
- Audit trail support: reviewers can verify how requirements are implemented and tested.
- Test coverage validation: QA activities can confirm that all requirements have planned verification cases.

7.2 RTM Key and Status Definitions

Column / Code	Definition
Req. ID / NFR ID	Unique identifier assigned to a functional or non-functional requirement.
Requirement Description	Concise summary of the requirement; full descriptions are in the relevant SRS section.
Priority	High = essential for MVP/core operation; Medium = important but can be staged; Low = optional or future enhancement.
Mapped Use Case(s)	Use case identifier(s) from Appendix B that exercise or realize the requirement. UC-17 is proposed for roommate matching because Appendix B currently omits that use case.
Design Class(es)	Primary data entities, service classes, or interface components responsible for implementing the requirement.
Test Case ID	Unique test case identifier assigned for verification. Functional tests use TC-001 to TC-055; non-functional tests use TC-056 to TC-108.
Verified	Requirement has a clear acceptance path and is feasible for implementation/testing in the current SRS scope.
Pending	Requirement is mapped but requires future increment work, external setup, security/accessibility audit, production validation, or further implementation detail.
Open	Requirement lacks enough specification for meaningful implementation or test design. No requirement in this RTM is marked Open; however, UC-17 is recommended for Appendix B completeness.

7.3 Forward Traceability - Functional Requirements

The following subsections map all fifty-five functional requirements from Section 3 of the Thikana SRS to the use cases, design classes, and test cases responsible for satisfying them.

7.3.1 User Registration and Authentication Requirements (FR-01 to FR-08)

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-01	Allow users to register with name, phone number, email, and password	High	UC-01	User, AuthService	TC-001	Verified
FR-02	Send one-time password (OTP) to registered phone number for verification	High	UC-01	User, OTPService, NotificationService	TC-002	Verified
FR-03	Enforce unique phone numbers and email addresses per account	High	UC-01	User, AuthService	TC-003	Verified
FR-04	Support role selection during registration: Tenant/Buyer, Property Owner, or Agency	High	UC-01	User, Role	TC-004	Verified
FR-05	Authenticate users using JWT with configurable expiry	High	UC-01	AuthService, User	TC-005	Verified
FR-06	Support password reset via OTP or email link	High	UC-01	AuthService, OTPService, NotificationService	TC-006	Verified
FR-07	Display a Verified badge on accounts that complete phone verification	Medium	UC-01, UC-08, UC-09	User, VerificationRequest	TC-007	Verified
FR-08	Support OAuth 2.0 login via Google as optional authentication	Low	UC-01	AuthService, OAuthService, User	TC-008	Pending

7.3.2 Smart Property Search and Filtering Requirements (FR-09 to FR-16)

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-09	Provide keyword search accepting property type, location, or area name	High	UC-02	SearchService, Listing	TC-009	Verified

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-10	Support filters by price, type, bedrooms, bathrooms, furnished status, and amenities	High	UC-02	SearchService, Listing	TC-010	Verified
FR-11	Display search results with thumbnail, title, price, location, and key details	High	UC-02	Listing, Photo, UserInterface	TC-011	Verified
FR-12	Provide Google Maps powered map view showing property pins for active listings	High	UC-02	MapService, Listing	TC-012	Verified
FR-13	Allow users to click map pins to preview listing details without leaving map view	Medium	UC-02, UC-03	MapService, Listing, UserInterface	TC-013	Verified
FR-14	Support sorting by price, date posted, and relevance	Medium	UC-02	SearchService, Listing	TC-014	Verified
FR-15	Display total count of matching results and current filter state	Medium	UC-02	SearchService, UserInterface	TC-015	Verified
FR-16	Provide AI-powered recommendation panel based on history and preferences	Low	UC-02, UC-15	RecommendationService, User, Listing	TC-016	Pending

7.3.3 Property Listing and Management Requirements (FR-17 to FR-24)

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-17	Allow owners to create listings with title, description, type, location, price, and contact details	High	UC-04	Listing, User	TC-017	Verified

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-18	Support uploading a minimum of 10 photos per listing in JPEG, PNG, or WEBP	High	UC-04	Photo, MediaStorage, Listing	TC-018	Verified
FR-19	Allow floor plan image or PDF upload for each listing	Medium	UC-04	MediaAsset, Listing, MediaStorage	TC-019	Verified
FR-20	Allow owners to set and update availability status: Available, Rented, Sold, Under Negotiation	High	UC-05	Listing	TC-020	Verified
FR-21	Support embedding a 360-degree panoramic view for each listing	Medium	UC-14	Photo, VirtualTour, Listing	TC-021	Verified
FR-22	Allow owners to edit all listing fields after publication without admin approval	High	UC-05	Listing, AuditLog	TC-022	Verified
FR-23	Display listing activity summary: views, saves, and enquiry count	Medium	UC-15	AnalyticsService, Listing, Favorite, Message	TC-023	Verified
FR-24	Automatically deactivate rented or sold listings after 30 days	Low	UC-05	Listing, SchedulerService	TC-024	Pending

7.3.4 Area Intelligence Requirements (FR-25 to FR-29)

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-25	Display nearby schools, colleges, hospitals, pharmacies, markets, mosques, and transport stops	Medium	UC-13	AreaIntelligence, MapService, Listing	TC-025	Verified
FR-26	Show distances to nearby facilities using Google Maps Places API	Medium	UC-13	AreaIntelligence, MapService	TC-026	Verified

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-27	Allow users to enter a destination and receive commute time estimate	Medium	UC-13	DirectionsService, MapService	TC-027	Verified
FR-28	Display area safety rating based on aggregated owner and tenant reviews	Low	UC-13	AreaIntelligence, Review	TC-028	Pending
FR-29	Indicate approximate internet availability based on operator coverage data	Low	UC-13	AreaIntelligence, CoverageDataService	TC-029	Pending

7.3.5 Tenant Verification Requirements (FR-30 to FR-34)

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-30	Allow tenants to submit NID or student ID photo for manual admin verification	Medium	UC-08	VerificationRequest, User, MediaStorage	TC-030	Verified
FR-31	Display Verified Tenant badge on admin-approved profiles	Medium	UC-08, UC-09	User, VerificationRequest	TC-031	Verified
FR-32	Allow users to report fraudulent listings or suspicious users	High	UC-10	Report, Listing, User	TC-032	Verified
FR-33	Notify admin when a new verification request or report is submitted	Medium	UC-09, UC-10	NotificationService, Report, VerificationRequest	TC-033	Verified
FR-34	Allow admins to approve, reject, or request further documents	Medium	UC-09	VerificationRequest, AdminPanel, User	TC-034	Verified

7.3.6 Direct Communication Requirements (FR-35 to FR-39)

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-35	Provide in-app messaging between tenants and owners	High	UC-07	Message, User	TC-035	Verified
FR-36	Deliver real-time message notifications using Firebase Cloud Messaging	High	UC-07	Message, NotificationService	TC-036	Verified
FR-37	Store message history and allow access to past conversations from inbox	Medium	UC-07	Message, User	TC-037	Verified
FR-38	Allow sharing property details, images, and documents within conversation	Low	UC-07	Message, MediaAsset, Listing	TC-038	Pending
FR-39	Allow owners to reveal phone number to verified tenants only	Medium	UC-07, UC-08	User, Message, VerificationRequest	TC-039	Verified

7.3.7 Property Management and Agreement Requirements (FR-40 to FR-44)

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-40	Generate digital rental agreement pre-filled with property, parties, rent, and duration	Medium	UC-11	Agreement, Listing, User	TC-040	Verified
FR-41	Allow both parties to review and digitally accept or reject agreement	Medium	UC-11	Agreement, User, AuditLog	TC-041	Verified
FR-42	Send automated rent due reminders via push notification and email	Medium	UC-12	Payment, Agreement, NotificationService	TC-042	Verified

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-43	Allow tenants to log monthly rent payments and generate payment history	Medium	UC-12	Payment, Agreement	TC-043	Verified
FR-44	Support utility expense tracking for tenants with monthly summaries	Low	UC-12	UtilityExpense, Payment	TC-044	Pending

7.3.8 Save Favorites and Alerts Requirements (FR-45 to FR-47)

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-45	Allow authenticated users to save listings to a personal favorites list	Medium	UC-06	Favorite, User, Listing	TC-045	Verified
FR-46	Notify users when a saved property is rented, sold, or price changes	Medium	UC-06	Favorite, NotificationService, Listing	TC-046	Verified
FR-47	Allow users to remove listings from favorites at any time	Medium	UC-06	Favorite	TC-047	Verified

7.3.9 Roommate Matching Requirements (FR-48 to FR-50)

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-48	Allow users to post roommate requests with area, budget, lifestyle, and gender preference	Low	UC-17 Proposed	RoommateRequest, User	TC-048	Pending
FR-49	Display roommate requests filtered by location and preferences	Low	UC-17 Proposed	RoommateRequest, SearchService	TC-049	Pending

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-50	Allow users to contact each other through messaging from roommate listing	Low	UC-17 Proposed, UC-07	RoommateRequest, Message, User	TC-050	Pending

7.3.10 Owner Reputation and Analytics Requirements (FR-51 to FR-55)

Req. ID	Requirement Description	Priority	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
FR-51	Allow tenants to leave star rating and written review after completed interaction	Medium	UC-03, UC-15	Review, User, Listing	TC-051	Verified
FR-52	Display owner's average rating and total review count on profile and listings	Medium	UC-03, UC-15	Review, User, Listing	TC-052	Verified
FR-53	Provide owners with analytics dashboard showing views, enquiries, and saves over time	Medium	UC-15	AnalyticsService, Listing, Favorite, Message	TC-053	Verified
FR-54	Provide admin dashboard showing total users, listings, reports, and transaction counts	Medium	UC-15, UC-16	AdminDashboard, AnalyticsService, User, Listing, Report, Payment	TC-054	Verified
FR-55	Generate weekly and monthly trend reports for agency-level users	Low	UC-15	AnalyticsService, ReportService	TC-055	Pending

7.4 Forward Traceability - Non-Functional Requirements

The table below maps the non-functional requirements derived from quality attributes, localization requirements, legal and regulatory requirements, deployment requirements, logging requirements, and academic project requirements. These requirements are systemic and often apply across multiple use cases.

NFR ID	NFR Description (Summary)	Category	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
NFR-US-001	New users complete property search and view a listing within 3 minutes	Usability	UC-02, UC-03	UserInterface, SearchService, Listing	TC-056	Verified
NFR-US-002	Registration process requires no more than 5 steps with OTP verification	Usability	UC-01	UserInterface, AuthService, OTPService	TC-057	Verified
NFR-US-003	All form fields display clear placeholder text and inline validation feedback	Usability	All Use Cases	UserInterface	TC-058	Verified
NFR-US-004	Conform to WCAG 2.1 Level AA including keyboard and screen reader support	Usability / Accessibility	All Use Cases	UserInterface	TC-059	Pending
NFR-US-005	All error messages use plain language and corrective action	Usability	All Use Cases	UserInterface, ErrorHandler	TC-060	Verified
NFR-US-006	Interface supports English now and Bengali language labels in future versions	Usability / Localization	All Use Cases	UserInterface, I18nService	TC-061	Pending
NFR-PF-001	Search results load within 2 seconds for up to 50 listings on 4G	Performance	UC-02	SearchService, Listing	TC-062	Verified
NFR-PF-002	Property detail pages with image gallery load within 3 seconds	Performance	UC-03, UC-14	Listing, Photo, MediaStorage	TC-063	Verified

NFR ID	NFR Description (Summary)	Category	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
NFR-PF-003	Interactive map renders pins for up to 200 listings without noticeable lag	Performance	UC-02, UC-13	MapService, Listing	TC-064	Verified
NFR-PF-004	In-app messages delivered with max latency of 500 milliseconds	Performance	UC-07	Message, NotificationService	TC-065	Verified
NFR-PF-005	Support 500 concurrent prototype users and scalable architecture for 10,000 production users	Performance / Scalability	All Use Cases	All Services	TC-066	Pending
NFR-PF-006	Image uploads processed and stored within 5 seconds for files up to 5 MB	Performance	UC-04	MediaStorage, Photo	TC-067	Verified
NFR-SE-001	Hash all passwords using bcrypt with minimum cost factor 12	Security	UC-01	User, AuthService	TC-068	Verified
NFR-SE-002	Require valid JWT token for all API endpoints except public search/listing endpoints	Security	All Protected Use Cases	AuthService, User, Role	TC-069	Verified
NFR-SE-003	Enforce role-based access control so tenants cannot access owner/admin endpoints	Security	All Use Cases	Role, AuthService, User	TC-070	Verified
NFR-SE-004	Encrypt all client-server data transmission using TLS 1.2 or higher	Security	All Use Cases	NetworkConfig, AuthService	TC-071	Verified
NFR-SE-005	Store NID document images in private cloud bucket accessible only to admins	Security	UC-08, UC-09	MediaStorage, VerificationRequest, AdminPanel	TC-072	Verified
NFR-SE-006	Rate limit auth endpoints: max 5 failed attempts before 15-minute lockout	Security	UC-01	AuthService, AuditLog	TC-073	Verified

NFR ID	NFR Description (Summary)	Category	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
NFR-SE-007	Prevent SQL injection and XSS through parameterized queries and output encoding	Security	All Use Cases	BackendAPI, UserInterface	TC-074	Pending
NFR-SE-008	Implement CSRF protection on all state-changing API requests	Security	All Write Use Cases	BackendAPI, AuthService	TC-075	Pending
NFR-SF-001	Warn users about unverified listings and advise in-person verification before payment	Safety	UC-03	Listing, UserInterface	TC-076	Verified
NFR-SF-002	Process no real payments during academic prototype; use sandbox mode	Safety	UC-12	Payment, PaymentGateway Adapter	TC-077	Verified
NFR-SF-003	Admin reviews user-reported fake listings or suspicious accounts within 48 hours	Safety	UC-10, UC-16	Report, AdminPanel, NotificationService	TC-078	Pending
NFR-SF-004	Do not store or display full NID numbers publicly; documents admin-only	Safety / Privacy	UC-08, UC-09	VerificationRequest, MediaStorage, UserInterface	TC-079	Verified
NFR-RL-001	Production system targets 99.5% uptime excluding planned maintenance	Reliability / Availability	All Use Cases	DeploymentConfig, BackendAPI	TC-080	Pending
NFR-RL-002	Unexpected server errors show user-friendly messages without stack traces	Reliability	All Use Cases	ErrorHandler, UserInterface	TC-081	Verified
NFR-RL-003	Automated daily backups; recovery from backup within 4 hours	Reliability / Backup	All Use Cases	BackupService, Database	TC-082	Verified

NFR ID	NFR Description (Summary)	Category	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
NFR-RL-004	Frontend implements loading states and skeleton screens during data fetching	Reliability / Usability	All Use Cases	UserInterface	TC-083	Verified
NFR-MT-001	Backend follows RESTful API design with controller, service, model separation	Maintainability	All Use Cases	BackendAPI, Controllers, Services, Models	TC-084	Verified
NFR-MT-002	Database schema supports adding property types and user roles without structural redesign	Maintainability / Scalability	UC-01, UC-04, UC-05	Database, User, Listing, Role	TC-085	Verified
NFR-MT-003	Environment-specific API keys and credentials managed through environment variables	Maintainability / Security	All Use Cases	DeploymentConfig	TC-086	Verified
NFR-MT-004	Codebase maintained in Git with clear branching strategy and commit conventions	Maintainability	All Use Cases	Repository, DevOpsProcess	TC-087	Pending
NFR-LOC-001	Primary interface language English; Bengali toggle in subsequent increment	Localization	All Use Cases	I18nService, UserInterface	TC-088	Pending
NFR-LOC-002	Display all monetary values in BDT / ₳ only in prototype	Localization	UC-02, UC-03, UC-11, UC-12	Listing, Payment, UserInterface	TC-089	Verified
NFR-LOC-003	Display dates as DD-MM-YYYY and times as 12-hour AM/PM in Asia/Dhaka timezone	Localization	All Use Cases	DateTimeService, UserInterface	TC-090	Verified
NFR-LOC-004	Validate Bangladesh mobile numbers starting with 01 and 11 digits total	Localization	UC-01	User, AuthService, ValidationService	TC-091	Verified

NFR ID	NFR Description (Summary)	Category	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
NFR-LOC-005	Address structure follows Division, District, Upazila, Union, Area, Road/House	Localization	UC-02, UC-04, UC-13	Listing, Address, MapService	TC-092	Verified
NFR-LOC-006	Default map center Dhaka and map services support Bangla place names where available	Localization	UC-02, UC-13	MapService, UserInterface	TC-093	Pending
NFR-LOC-007	Use UTF-8 encoding to support English and Bengali characters	Localization	All Use Cases	Database, UserInterface	TC-094	Verified
NFR-CL-001	Comply with Bangladesh Digital Security Act 2018 for user data protection and content moderation	Compliance	All Use Cases	CompliancePolicy, User, Report	TC-095	Pending
NFR-CL-002	Do not share user personal data with third parties without explicit consent	Compliance / Privacy	All Use Cases	User, ConsentPolicy, BackendAPI	TC-096	Verified
NFR-CL-003	Property listings involving transactions comply with Bangladesh real estate regulations	Compliance	UC-04, UC-05, UC-11, UC-12	Listing, Agreement, Payment	TC-097	Pending
NFR-DP-001	System deployable on Ubuntu 20.04 VPS using documented installation scripts	Deployment	All Use Cases	DeploymentConfig, DevOpsProcess	TC-098	Pending
NFR-DP-002	Maintain README and setup guide in GitHub repository	Deployment / Academic	All Use Cases	Repository, DevOpsProcess	TC-099	Pending
NFR-DP-003	Manage database credentials, API keys, and JWT secret through .env file	Deployment / Security	All Use Cases	DeploymentConfig	TC-100	Verified
NFR-LG-001	Log all API requests with method, endpoint, status code, and response time	Logging	All Use Cases	BackendAPI, AuditLog	TC-101	Verified

NFR ID	NFR Description (Summary)	Category	Mapped Use Case(s)	Design Class(es)	Test Case ID	Status
NFR-LG-002	Record authentication events in a separate security audit log	Logging / Security	UC-01	AuditLog, AuthService	TC-102	Verified
NFR-LG-003	Log application errors with stack traces server-side for debugging	Logging	All Use Cases	ErrorHandler, AuditLog	TC-103	Verified
NFR-LG-004	Integrate production error monitoring using Sentry or custom dashboard	Logging / Monitoring	All Use Cases	MonitoringService, AdminDashboard	TC-104	Pending
NFR-ACD-001	Project follows Incremental Development Model with documented increments	Academic / Process	All Use Cases	DevOpsProcess, Repository	TC-105	Verified
NFR-ACD-002	All source code maintained in GitHub with meaningful commit messages	Academic / Process	All Use Cases	Repository	TC-106	Pending
NFR-ACD-003	Prototype demonstrates user registration, property listing, search filters, and map view	Academic / Process	UC-01, UC-02, UC-04	AuthService, Listing, SearchService, MapService	TC-107	Verified
NFR-ACD-004	Final demonstration includes working prototype deployed to publicly accessible URL	Academic / Deployment	All Use Cases	DeploymentConfig, DevOpsProcess	TC-108	Pending

7.5 Backward Traceability - Design Class to Requirements

The backward traceability table maps the major design entities and service classes to the requirements that justify them. This view confirms that each class exists for a documented business, functional, or quality reason.

Design Class	Requirements Traced to This Class	Design Justification
User	FR-01-FR-08, FR-30-FR-32, FR-35, FR-39, FR-40-FR-41, FR-45, FR-48, FR-51-FR-52, NFR-SE-001, NFR-CL-002	Represents tenants, buyers, owners, agencies, and admins; central identity object for authentication, profile, roles, verification, and interactions.
Role	FR-04, NFR-SE-002, NFR-SE-003, NFR-MT-002	Defines role-specific permissions for tenant/buyer, owner, agency, and admin workflows.
AuthService	FR-01-FR-08, NFR-SE-001-NFR-SE-008, NFR-LOC-004	Implements registration, OTP, JWT, OAuth, password reset, rate limiting, and access control.
Listing	FR-09-FR-29, FR-32, FR-40, FR-45-FR-47, FR-51-FR-54, NFR-PF-001-NFR-PF-003, NFR-LOC-002, NFR-LOC-005	Core property entity containing location, type, price, status, media, owner relationship, and public search visibility.
Photo / MediaAsset	FR-18, FR-19, FR-21, FR-30, FR-38, NFR-PF-002, NFR-PF-006, NFR-SE-005	Handles property photos, floor plans, 360-degree media, shared attachments, and verification document uploads.
SearchService	FR-09-FR-16, FR-49, NFR-PF-001	Executes keyword search, filters, sorting, result counts, and recommendation data access.
MapService / AreaIntelligence	FR-12-FR-13, FR-25-FR-29, NFR-PF-003, NFR-LOC-005, NFR-LOC-006	Integrates maps, pins, places, distances, commute estimates, and location-contextual intelligence.
VerificationRequest	FR-07, FR-30-FR-34, FR-39, NFR-SE-005, NFR-SF-004	Stores and manages verification documents, status, admin decision, and verified profile badges.
Report	FR-32-FR-34, FR-54, NFR-SF-003, NFR-CL-001	Supports user reporting of fake listings or suspicious accounts and admin moderation workflows.
Message	FR-35-FR-39, FR-50, NFR-PF-004	Represents in-app conversations, message history, attachments, and communication between users.
Agreement	FR-40-FR-43, NFR-CL-003	Represents digital rental agreements, acceptance status, parties, listing, rent amount, and duration.

Design Class	Requirements Traced to This Class	Design Justification
Payment	FR-42-FR-44, FR-54, NFR-SF-002, NFR-LOC-002	Tracks rent payments, payment history, utility expenses, and transaction-related dashboard counts.
Favorite	FR-23, FR-45-FR-47, FR-53	Links users to saved listings and triggers saved-property alerts and analytics counts.
Review	FR-28, FR-51-FR-52	Stores rating and written review data supporting safety indicators and owner reputation.
AnalyticsService / AdminDashboard	FR-23, FR-53-FR-55, NFR-LG-004	Aggregates views, saves, enquiries, reports, users, listings, transactions, and trend reports.
NotificationService	FR-02, FR-06, FR-33, FR-36, FR-42, FR-46, NFR-SF-003	Delivers OTPs, password reset messages, admin alerts, chat notifications, rent reminders, and favorite alerts.
AuditLog / LoggingService	FR-22, FR-41, NFR-SE-006, NFR-LG-001-NFR-LG-003	Records authentication events, write operations, API requests, and server-side errors.
DeploymentConfig / DevOpsProcesses	NFR-MT-003, NFR-DP-001-NFR-DP-003, NFR-ACD-001-NFR-ACD-004	Controls deployment, environment configuration, repository management, public demo, and academic process evidence.
RoommateRequest	FR-48-FR-50	Proposed class for roommate matching; should be added to analysis models if the feature is retained.

7.6 Use Case - Class - Test Case Cross-Reference Matrix

This cross-reference matrix provides a use-case-centric view of system coverage. It allows the testing team to identify the classes and test cases that must pass before each use case can be considered verified.

Use Case ID	Use Case Name	Primary Actor(s)	Design Classes Involved	Sample Requirements	Test Cases
UC-01	Register and Verify Account	Tenant, Owner, Agency	User, Role, AuthService, OTPService, NotificationService	FR-01-FR-08; NFR-US-002; NFR-SE-001; NFR-LOC-004	TC-001-TC-008, TC-057, TC-068, TC-091
UC-02	Search and Filter Properties	Tenant, Buyer	SearchService, Listing, MapService, UserInterface	FR-09-FR-16; NFR-PF-001; NFR-PF-003	TC-009-TC-016, TC-062, TC-064
UC-03	View Property Details	Tenant, Buyer	Listing, Photo, Review, AreaIntelligence	FR-13, FR-51-FR-52; NFR-PF-002; NFR-SF-001	TC-013, TC-051-TC-052, TC-063, TC-076
UC-04	Post a Property Listing	Owner, Agency	Listing, Photo, MediaStorage, Address	FR-17-FR-19; NFR-PF-006; NFR-ACD-003	TC-017-TC-019, TC-067, TC-107
UC-05	Manage Property Listing	Owner, Agency, Admin	Listing, SchedulerService, AuditLog	FR-20, FR-22, FR-24; NFR-CL-003	TC-020, TC-022, TC-024, TC-097
UC-06	Save Favorite Property	Tenant, Buyer	Favorite, Listing, NotificationService	FR-45-FR-47	TC-045-TC-047
UC-07	Send Message to Owner	Tenant, Buyer	Message, User, NotificationService	FR-35-FR-39, FR-50; NFR-PF-004	TC-035-TC-039, TC-050, TC-065
UC-08	Submit Verification Documents	Tenant	VerificationRequest, User, MediaStorage	FR-07, FR-30-FR-31, FR-39; NFR-SE-005; NFR-SF-004	TC-007, TC-030-TC-031, TC-039, TC-072, TC-079
UC-09	Approve Verification Request	Admin	VerificationRequest, AdminPanel, NotificationService	FR-07, FR-31, FR-33-FR-34	TC-007, TC-031, TC-033-TC-034

Use Case ID	Use Case Name	Primary Actor(s)	Design Classes Involved	Sample Requirements	Test Cases
UC-10	Report Fake Listing or User	Any User	Report, Listing, User, NotificationService	FR-32-FR-33; NFR-SF-003	TC-032-TC-033, TC-078
UC-11	Generate Rental Agreement	Owner	Agreement, Listing, User, AuditLog	FR-40-FR-41; NFR-CL-003	TC-040-TC-041, TC-097
UC-12	Track Rent Payments	Tenant, Owner	Payment, Agreement, NotificationService	FR-42-FR-44; NFR-SF-002	TC-042-TC-044, TC-077
UC-13	View Area Intelligence	Tenant, Buyer	AreaIntelligence, MapService, DirectionsService, Review	FR-25-FR-29; NFR-LOC-005-NFR-LOC-006	TC-025-TC-029, TC-092-TC-093
UC-14	View 360-Degree Virtual Tour	Tenant, Buyer	VirtualTour, Photo, Listing	FR-21; NFR-PF-002	TC-021, TC-063
UC-15	Access Analytics Dashboard	Owner, Agency, Admin	AnalyticsService, AdminDashboard, Listing, Review	FR-16, FR-23, FR-51-FR-55	TC-016, TC-023, TC-051-TC-055
UC-16	Moderate Content and Users	Admin	AdminPanel, Report, Listing, User	FR-32, FR-34, FR-54; NFR-CL-001	TC-032, TC-034, TC-054, TC-095
UC-17 Proposed	Roommate Matching	Tenant, Buyer	RoommateRequest, SearchService, Message	FR-48-FR-50	TC-048-TC-050

7.7 Requirements Coverage Summary

The coverage summary consolidates the forward and backward traceability information. Verification rate is calculated from the number of requirements currently marked Verified. Pending items remain traceable but require completion before release or final academic demonstration.

Requirement Category	Total	Mapped Use Cases	Mapped Classes	Test Cases	Status Summary	Verification Rate
FR-01 to FR-08: Registration and Authentication	8	UC-01	User, Role, AuthService, OTPService	TC-001-TC-008	7 Verified, 1 Pending	87.5%
FR-09 to FR-16: Search and Filtering	8	UC-02, UC-03, UC-15	SearchService, Listing, MapService	TC-009-TC-016	7 Verified, 1 Pending	87.5%
FR-17 to FR-24: Listing Management	8	UC-04, UC-05, UC-14, UC-15	Listing, Photo, MediaStorage, SchedulerService	TC-017-TC-024	7 Verified, 1 Pending	87.5%
FR-25 to FR-29: Area Intelligence	5	UC-13	AreaIntelligence, MapService, Review	TC-025-TC-029	3 Verified, 2 Pending	60.0%
FR-30 to FR-34: Tenant Verification	5	UC-08, UC-09, UC-10	VerificationRequest, Report, AdminPanel	TC-030-TC-034	5 Verified	100%
FR-35 to FR-39: Direct Communication	5	UC-07, UC-08	Message, NotificationService, User	TC-035-TC-039	4 Verified, 1 Pending	80.0%
FR-40 to FR-44: Agreements and Payments	5	UC-11, UC-12	Agreement, Payment, NotificationService	TC-040-TC-044	4 Verified, 1 Pending	80.0%
FR-45 to FR-47: Favorites and Alerts	3	UC-06	Favorite, Listing, NotificationService	TC-045-TC-047	3 Verified	100%

Requirement Category	Total	Mapped Use Cases	Mapped Classes	Test Cases	Status Summary	Verification Rate
FR-48 to FR-50: Roommate Matching	3	UC-17 Proposed, UC-07	RoommateRequest, SearchService, Message	TC-048-TC-050	3 Pending	0%
FR-51 to FR-55: Reputation and Analytics	5	UC-03, UC-15, UC-16	Review, AnalyticsService, AdminDashboard	TC-051-TC-055	4 Verified, 1 Pending	80.0%
NFR-001 to NFR-053: Quality, Localization, Compliance, Deployment	53	All Use Cases	All Classes / Services	TC-056-TC-108	39 Verified, 14 Pending	73.6%
TOTAL	108	UC-01 to UC-17 Proposed	19+ design entities/services	TC-001-TC-108	90 Verified, 18 Pending	83.3%

7.8 RTM Analysis and Observations

7.8.1 High-Coverage Areas

The strongest traceability coverage appears in core MVP workflows: user registration and verification, property search, property listing, map view, favorites, reporting, tenant verification, direct messaging, and basic analytics. These areas directly support the prototype scope described in the SRS: user registration, property listing, search with filters, and map view. The main entities with the widest requirement coverage are User, Listing, AuthService, SearchService, MapService, Message, and NotificationService.

7.8.2 Pending Verification Items

- The following items are mapped but require additional design, implementation, external service setup, audit, or future-increment validation before they can be fully verified:
- FR-08 OAuth login: optional third-party authentication depends on OAuth configuration and credential setup.
- FR-16 AI recommendation panel: future enhancement requiring recommendation logic and user-history data.

- FR-24 automatic listing deactivation: requires scheduler/cron implementation and aging rule testing.
- FR-28 and FR-29 area safety/internet indicators: depend on reliable aggregated review data and external operator coverage data.
- FR-38 document sharing in chat: requires attachment security and storage rules beyond basic messaging.
- FR-44 utility expense tracking: low-priority feature not essential for the MVP.
- FR-48 to FR-50 roommate matching: functional requirements exist, but Appendix B currently lacks a corresponding use case. UC-17 should be added.
- FR-55 trend reports: requires reporting-period aggregation and agency-level dashboard validation.
- Accessibility, security hardening, production availability, legal compliance, deployment, monitoring, and final demo NFRs require acceptance testing before release.

7.8.3 Change Impact Guidance

When a requirement changes, the project team should use this RTM as the primary impact analysis tool.

Step 1 - Locate the affected requirement in Tables 7.3 or 7.4.

Step 2 - Review the mapped use cases and update Appendix B if the user interaction changes.

Step 3 - Review mapped classes and update the data model, service design, or interface design as needed.

Step 4 - Update or invalidate the assigned test cases and re-run the relevant verification activities.

Step 5 - Update backward traceability, the coverage summary, and revision history after the change is approved.

7.8.4 Orphan Detection and Coverage Gaps

No functional requirement is left without a design class or test case mapping. However, the current Appendix B use case list stops at UC-16 and does not include a roommate matching use case even though FR-48 to FR-50 define roommate matching requirements. To remove this traceability gap, the project should add UC-17: Roommate Matching to Appendix B or defer FR-48 to FR-50 formally to a future SRS version.

All 55 functional requirements and 53 non-functional requirements have planned test case identifiers. The current RTM therefore provides 100% traceability coverage, while full verification remains pending for future-increment, compliance, accessibility, monitoring, and production-readiness items.

8. VALIDATION AND VERIFICATION

This section defines the strategy, checklists, rules, and acceptance criteria used to validate and verify the requirements specified in Sections 4 through 6 of this document. The objective of validation is to confirm that the requirements accurately represent stakeholder needs as established in Sections 1 and 2, while verification confirms that the requirements are internally consistent, complete, unambiguous, and traceable through the model artifacts defined in Sections 5, 6, and 7.

8.1 Requirement Review Strategy

The Requirement Review Strategy establishes a structured, repeatable process for examining each increment of the THIKANA requirements baseline before it is approved for design and implementation. The strategy is aligned with the Incremental Development Model described in Section 1, with reviews conducted at the end of each increment's requirements activity and prior to the start of design activities for that increment.

8.1.1 Review Levels

- Level 1 – Self-Review: The requirements author reviews their own work against the Inspection Checklist (Section 8.2) before submission.
- Level 2 – Peer Review: A team member not involved in authoring the section conducts an independent review using the same checklist and records defects in a Review Log.
- Level 3 – Formal Walkthrough: The full project team conducts a walkthrough of the reviewed section, focusing on consistency with Sections 1–2 and cross-references to Sections 5–7.
- Level 4 – Supervisor/Stakeholder Review: The course instructor or designated stakeholder reviews the consolidated document for academic and business suitability prior to baseline sign-off.

8.1.2 Review Roles

- Author: Prepares the requirement section and resolves identified defects.
- Reviewer: Independently inspects the section against the checklist and validation rules and logs defects.
- Moderator: Facilitates the walkthrough meeting, ensures all defects are discussed, and tracks resolution status.
- Recorder: Documents defects, decisions, and action items during review meetings.

8.1.3 Review Process

1. The Author completes a draft of the assigned requirements section and performs a Level 1 self-review using the Inspection Checklist.

2. The Author submits the draft to an assigned Reviewer for a Level 2 peer review.
3. The Reviewer logs all identified defects, classified as Major (blocks approval) or Minor (does not block approval but must be tracked).
4. The Author resolves all Major defects and re-submits the section for a Level 3 walkthrough with the full team.
5. The Moderator records the outcome of the walkthrough as Approved, Approved with Minor Comments, or Rejected.
6. Rejected sections return to Step 1; Approved or Approved-with-Comments sections proceed to Level 4 review.
7. The Supervisor/Stakeholder reviews the section for completeness against the project context and academic scope defined in Sections 1–2 and provides final sign-off.
8. Approved sections are added to the requirements baseline and become subject to formal change control.

8.1.4 Entry and Exit Criteria

- Entry Criteria: The section to be reviewed is complete in structure (all subsections present), references the correct module and actor names from Sections 1–2, and the author has completed the Level 1 self-review.
- Exit Criteria: All Major defects identified during Levels 2 and 3 have been resolved and verified; the section has been checked against the Validation Rules in Section 8.3; and the Supervisor/Stakeholder has granted sign-off.

8.1.5 Review Frequency

A formal review cycle (Levels 1–4) shall be conducted at the conclusion of the requirements activity for each increment of the Incremental Development Model. In addition, any change request approved through the project's change control process shall trigger a focused Level 2 and Level 3 re-review of the affected sections only.

8.2 Inspection Checklist

The following checklist shall be applied during Level 1 and Level 2 reviews described in Section 8.1. Each inspection item is associated with a defined pass/fail criterion. A section fails inspection if any item marked as a Major defect category (Completeness, Correctness, Consistency, Traceability, Security) does not meet its pass criterion.

Category	Inspection Item	Pass/Fail Criteria
Completeness	Does every functional requirement specify a unique ID, description, priority, source, and acceptance criteria?	Fail if any field is missing for any FR/NFR entry.

Category	Inspection Item	Pass/Fail Criteria
Completeness	Are all primary user classes (Tenant/Buyer, Property Owner, Real Estate Agency, System Administrator) represented by at least one requirement?	Fail if any actor has zero associated requirements.
Correctness	Does each requirement accurately reflect the approved Sections 1 and 2 baseline (scope, stakeholders, constraints)?	Fail if a requirement contradicts baseline scope or constraints.
Correctness	Are SHALL statements used consistently for mandatory requirements and SHOULD/MAY avoided unless intentional?	Fail if mandatory requirements use non-binding language.
Consistency	Are actor names, module names, and entity names used identically across Sections 3–6?	Fail if naming differs across sections (e.g., 'Agency' vs 'Real Estate Agency').
Consistency	Do class diagram attributes match the Data Dictionary (Section 5.2) in name, type, and constraints?	Fail if any attribute mismatch is found between Section 5 and Section 6.8.
Unambiguity	Is each requirement statement free of vague terms (e.g., 'fast', 'user-friendly', 'appropriate') without quantification?	Fail if vague terms are present without an associated measurable metric.
Verifiability	Can each requirement be verified through inspection, demonstration, analysis, or test?	Fail if no verification method can be identified for a requirement.
Traceability	Is each functional requirement mapped to at least one use case, one class, and one test case in the RTM (Section 7)?	Fail if any FR has a blank mapping in the RTM.
Traceability	Does each use case reference at least one functional requirement and one class from Section 6.8?	Fail if a use case has no linked requirement or class.

Category	Inspection Item	Pass/Fail Criteria
Feasibility	Is each requirement implementable within the defined technology stack and academic timeline?	Fail if a requirement assumes infrastructure or tools outside the defined stack without justification.
Prioritization	Has each requirement been assigned a priority (High/Medium/Low) consistent with its business value?	Fail if priority is missing or inconsistent with stated business rules.
Diagram Accuracy	Do PlantUML diagrams compile without syntax errors and reflect the textual description provided?	Fail if diagram code does not render or contradicts its explanation.
Security	Are authentication, authorization, and data protection requirements present for all sensitive operations (payments, verification, messaging)?	Fail if any sensitive operation lacks an associated security requirement.
Compliance	Do data retention, audit logging, and privacy-related requirements align with the NFR compliance category (Section 4.3)?	Fail if compliance NFRs are not reflected in corresponding functional or data requirements.

8.3 Validation Rules

The Validation Rules listed below shall be applied programmatically (where tooling permits) and manually during Level 2 and Level 3 reviews to confirm that the requirements specification conforms to the structural and semantic conventions established for the THIKANA SRS. A requirement, diagram, or table that violates any applicable rule shall be logged as a defect during review.

Rule ID	Validation Rule	Applies To
VR-001	Every functional requirement ID shall follow the format FR-XX-NNN, where XX identifies the owning actor/module (TB, PO, RA, SA, CS) and NNN is a zero-padded sequential number.	Section 4.1

Rule ID	Validation Rule	Applies To
VR-002	Every functional requirement shall contain a single, testable SHALL statement; compound requirements containing multiple SHALL statements shall be decomposed into separate requirements.	Section 4.1
VR-003	Every functional requirement shall declare exactly one Priority value from the set {High, Medium, Low}.	Section 4.1
VR-004	Every functional requirement shall declare a Source consistent with Sections 1–2 (e.g., Stakeholder Interview, Use Case, Business Rule).	Section 4.1
VR-005	Every functional requirement shall include at least one Acceptance Criterion expressed in a verifiable, observable form (e.g., 'the system displays...', 'the system rejects...').	Section 4.1
VR-006	Every non-functional requirement shall specify a quantifiable Metric with a unit of measure (e.g., seconds, percentage, requests/second).	Section 4.3
VR-007	Every non-functional requirement shall map to exactly one of the categories defined in Section 4.3 (Performance, Security, Reliability, Availability, Usability, Maintainability, Scalability, Portability, Accessibility, Auditability, Disaster Recovery, Logging, Compliance).	Section 4.3
VR-008	Every entity listed in the Data Dictionary (Section 5.2) shall appear as a class in the Class Diagram (Section 6.8) with matching attribute names and data types.	Sections 5, 6.8
VR-009	Every foreign key relationship declared in Section 5.3 shall correspond to an association, aggregation, or composition relationship in the Class Diagram and ERD.	Sections 5.3, 6.8, 6.9
VR-010	Every use case listed in Section 6.3 shall have a unique identifier in the format UC-NN and shall be referenced in the Use Case Diagram (Section 6.2).	Section 6.2, 6.3
VR-011	Every Main Flow step in a use case that results in a state change shall correspond to a transition defined in the relevant State Transition Diagram (Section 6.7).	Sections 6.3, 6.7

Rule ID	Validation Rule	Applies To
VR-012	Every decision table (Section 6.11) shall enumerate all logically possible condition combinations and shall not contain contradictory action outcomes for identical condition sets.	Section 6.11
VR-013	Every requirement (functional or non-functional) shall appear in the Requirements Traceability Matrix (Section 7) with a non-empty mapping to at least one Use Case/Class and one Test Case.	Section 7
VR-014	Every test case referenced in the RTM shall specify a precondition, test steps, expected result, and pass/fail status field.	Sections 7, 10
VR-015	All monetary values, dates, and identifiers referenced across sections shall use consistent formats (e.g., BDT for currency, ISO 8601 for dates, UUID or auto-increment integers for primary keys as declared in Section 5.3).	Sections 5, 6
VR-016	No requirement shall reference a technology, library, or third-party service outside the technology stack defined in Sections 1–2 (React.js, Node.js/Express, MySQL, Firebase services, Google Maps/Mapbox, SSLCommerz/bKash/Nagad Sandbox) unless explicitly listed as a Future Enhancement (Appendix C).	All Sections
VR-017	Glossary terms (Appendix A) shall cover all domain-specific terms used in Sections 3–10 at least once.	Appendix A
VR-018	Every open issue listed in Appendix B shall reference the section(s) of the document it affects.	Appendix B

8.4 Acceptance Criteria

Acceptance criteria define the conditions under which the THIKANA platform, or an increment thereof, is considered ready for stakeholder acceptance. Acceptance is assessed at two levels: (1) requirement-level acceptance criteria, as documented individually for each functional requirement in Section 4.1, and (2) system-level acceptance criteria, summarized per functional area in the table below, which represent the minimum demonstrable behavior expected at the conclusion of the increment in which that functional area is delivered.

Functional Area	System-Level Acceptance Criterion
User Registration & Authentication	A new user shall be able to register using a valid email/phone and password, receive a verification prompt via Firebase Authentication, and successfully log in within 3 attempts under normal network conditions.
User Profile Management	A registered user shall be able to view, update, and save profile details (including profile photo via Firebase Storage), with changes persisted and reflected immediately upon reload.
Smart Property Search & Filtering	A user shall be able to apply at least five filter criteria (e.g., location, price range, property type, number of bedrooms, availability) simultaneously and receive matching results within the defined performance NFR threshold.
Map-Based Property Discovery	A user shall be able to view property listings as markers on an interactive map (Google Maps/Mapbox) and select a marker to view a summary card without leaving the map view.
Property Listing Management	A Property Owner or Real Estate Agency shall be able to create, edit, publish, unpublish, and delete a property listing, with status changes reflected in the listing's lifecycle state (Section 6.7).
Media & Virtual Tour Management	A Property Owner shall be able to upload at least five images and an optional virtual tour link per listing, with images stored via Firebase Storage and displayed in a gallery on the listing detail page.
Area Intelligence System	A user viewing a property listing shall be able to view area-level statistics (e.g., average rent, nearby amenities) sourced from aggregated AnalyticsRecord data for the corresponding PropertyLocation.
Tenant & Owner Verification	A user who submits a verification request with required documents shall receive a status update (Pending, Approved, or Rejected) from a System Administrator, with the result reflected in their profile verification badge.

Functional Area	System-Level Acceptance Criterion
Direct Communication System	Two registered users shall be able to exchange text messages within a Conversation thread, with message delivery status and timestamps visible to both parties.
Rental Agreement Management	A Property Owner and Tenant/Buyer shall be able to generate, review, and digitally accept a rental agreement linked to a specific PropertyListing, with the agreement status transitioning through its defined lifecycle (Section 6.7).
Payment & Rent Tracking	A Tenant/Buyer shall be able to submit a rent payment through the integrated sandbox payment gateway (SSLCommerz/bKash/Nagad) and receive a confirmation, with the payment record visible in their payment history.
Favorites & Alerts	A user shall be able to mark a listing as a favorite and receive a notification (via Firebase Cloud Messaging) when a price or availability change occurs on a favorited listing.
Review & Reputation Management	A Tenant/Buyer who has completed a rental agreement shall be able to submit a rating and review for the associated Property Owner or listing, visible on the relevant profile or listing page.
Owner & Agency Analytics	A Property Owner or Real Estate Agency shall be able to view a dashboard summarizing listing views, inquiries, and conversion metrics for their active listings.
Administration & Moderation	A System Administrator shall be able to view, approve, reject, or remove reported listings, users, or reviews, with all moderation actions recorded in the AuditLog.
Reporting & Monitoring	A System Administrator shall be able to generate a system activity report for a specified date range, including user registrations, listings created, and transactions processed.

8.4.1 Overall System Acceptance Criteria

- All functional requirements classified as High priority in Section 4.1 shall be implemented and demonstrable through the user interface described in Section 4.2.1.
- All Non-Functional Requirements classified as High priority in Section 4.3 shall be measured and shall meet or exceed their specified metric thresholds under test conditions defined in Section 10.
- All use cases listed in Section 6.3 shall be executable end-to-end without unhandled errors for their defined Main Flow.
- All entities defined in the Data Dictionary (Section 5.2) shall be implemented in the MySQL database with constraints matching Section 5.3.
- The Requirements Traceability Matrix (Section 7) shall show 100% coverage, with no requirement lacking an associated use case, class, and test case.
- No open issue listed in Appendix B shall be classified as Critical at the time of final project submission.

8.4.2 Sign-Off Criteria

Final acceptance of the THIKANA requirements baseline, and of each subsequent increment delivered against it, shall require written or recorded sign-off from the Supervisor/Stakeholder confirming that: (a) the Inspection Checklist (Section 8.2) has been applied with no outstanding Major defects; (b) the Validation Rules (Section 8.3) have been satisfied; and (c) the applicable rows of the System-Level Acceptance Criteria table (Section 8.4) have been demonstrated through a working build or prototype.

9. RISK ANALYSIS

This section identifies risks that may affect the successful delivery of the THIKANA platform within the academic project timeline and technology stack defined in Sections 1–2. Each risk is classified by category, assessed for Probability and Impact (each rated High, Medium, or Low), assigned an overall Risk Level, and accompanied by a corresponding mitigation strategy. Risk Levels are derived using the Risk Matrix presented in Section 9.6, where Risk Level = $f(\text{Probability}, \text{Impact})$.

9.1 Technical Risks

Technical risks arise from the complexity of integrating the technologies specified in Sections 1–2 (React.js, Node.js/Express, MySQL, Firebase services, Google Maps Platform/Mapbox, and payment sandboxes) and from the evolving database schema required to support the entities defined in Section 5.

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
RT-01	Integration mismatches between Firebase Authentication tokens and the Node.js/Express backend session/JWT handling may cause login or authorization failures.	Medium	High	High	Define and document a clear token-verification contract early; implement a shared middleware module for token validation and write integration tests for FR-CS-001–FR-CS-005 authentication flows.
RT-02	Real-time messaging (Direct Communication System) may experience latency or message-delivery failures under concurrent load when	Medium	Medium	Medium	Implement message persistence in MySQL as the source of truth, use FCM only for push notifications, and load-test the

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
	relying on Firebase Cloud Messaging.				messaging module against the performance NFRs in Section 4.3.
RT-03	Rendering a large number of property markers on the map (Map-Based Property Discovery) may degrade frontend performance on lower-end devices.	Medium	Medium	Medium	Apply marker clustering and viewport-based lazy loading in the Google Maps/Mapbox integration; cap initial marker count and paginate results per Section 4.1 search requirements.
RT-04	Frequent schema changes to MySQL tables during incremental development may cause migration conflicts or data loss across increments.	Medium	High	High	Use a version-controlled migration tool (e.g., Sequelize/Knex migrations), enforce backward-compatible schema changes where possible, and require schema review as part of the Level 2 peer review (Section 8.1).
RT-05	Sandbox payment gateways (SSLCommerz/bKash/Nagad) may change API contracts or experience downtime,	Low	Medium	Medium	Abstract payment integration behind a service interface so the gateway implementation can

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
	blocking testing of Payment and Rent Tracking features.				be swapped without affecting FR-TB and FR-PO payment requirements; mock the gateway for unit testing.
RT-06	Inconsistent data formats (dates, currency, coordinates) across modules may cause integration defects between frontend, backend, and database layers.	Medium	Medium	Medium	Enforce the data format conventions defined in Validation Rule VR-015 (Section 8.3) through shared utility libraries and code review checklists.

9.2 Security Risks

Security risks relate to the protection of user data, authentication mechanisms, role-based access control, and sensitive documents handled by the Tenant and Owner Verification module, as well as the general resilience of the system against common web application vulnerabilities.

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
RS-01	Improper validation of Firebase Authentication tokens on the backend could allow unauthorized access to protected API endpoints.	Low	High	High	Implement centralized token-verification middleware tested against all FR-CS authentication requirements; include negative test cases (expired/forged

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
					tokens) in Section 10 security testing.
RS-02	Unsanitized user input in search filters, messaging, or listing forms could expose the system to SQL injection or stored Cross-Site Scripting (XSS).	Medium	High	High	Use parameterized queries/ORM for all MySQL access, apply input sanitization and output encoding on the React frontend, and include injection/XSS test cases in Section 10 Security Testing.
RS-03	Sensitive verification documents (e.g., NID images) stored in Firebase Storage may be exposed if storage access rules are misconfigured.	Medium	High	High	Configure Firebase Storage security rules to restrict document access to the uploading user and System Administrators only; audit access rules during each increment's security review.
RS-04	Weak session management (e.g., long-lived tokens, missing logout invalidation) could enable session hijacking.	Low	Medium	Medium	Apply short-lived access tokens with refresh-token rotation, implement explicit logout/token revocation, and verify against the Security NFRs defined in Section 4.3.

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
RS-05	Role-based access control gaps could allow a Tenant/Buyer or Property Owner to access System Administrator functions (e.g., Moderate Reports, Manage Users).	Medium	High	High	Implement and unit-test role-based authorization middleware for every endpoint, with explicit test cases for UC-12 and UC-14 covering unauthorized access attempts.

9.3 Operational Risks

Operational risks arise from team coordination, documentation consistency, version control practices, and environment setup activities that support the development process described in Section 8.1.

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
RO-01	Inconsistent use of actor names, module names, or requirement IDs across team members may cause documentation and traceability errors (violating VR-001, VR-016).	Medium	Medium	Medium	Maintain a shared glossary (Appendix A) and naming-convention reference document; enforce naming checks during Level 1/2 reviews described in Section 8.1.
RO-02	Lack of a shared development/staging environment may cause integration issues to surface late, close to demonstration deadlines.	Medium	Medium	Medium	Set up a shared staging environment early in the project using the defined technology stack; integrate

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
					continuously at the end of each increment rather than at project end.
RO-03	Poor version control discipline (e.g., direct commits to main branch, merge conflicts) may cause loss of work or unstable builds.	Medium	Low	Medium	Adopt a branching strategy (feature branches with pull requests), require at least one peer review before merge, and run automated build checks on each pull request.
RO-04	Miscommunication regarding which increment owns which functional area may result in duplicated or conflicting implementation effort.	Low	Medium	Low	Maintain an increment-to-module allocation table derived from Section 4.1 requirement IDs, reviewed at the start of each increment.

9.4 Schedule Risks

Schedule risks relate to the academic project timeline, the Incremental Development Model adopted for THIKANA, and the relative complexity of individual functional areas.

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
RSC-01	Effort for complex modules such as Rental Agreement Management and Payment & Rent Tracking may be	Medium	High	High	Allocate additional buffer time for these modules in the increment plan, prioritize their High-

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
	underestimated, causing schedule slippage.				priority requirements (Section 4.1) earliest within their increment, and track progress weekly.
RSC-02	Academic commitments (examinations, coursework from other courses) may reduce available development time during critical increments.	High	Medium	High	Plan the increment schedule around the known academic calendar, front-loading high-risk and high-priority modules into periods with fewer competing commitments.
RSC-03	Delays in obtaining sandbox credentials or approvals for third-party services (Firebase, Maps Platform, payment sandboxes) may delay integration testing.	Medium	Medium	Medium	Request all sandbox/developer accounts at project initiation, before the increment that requires them, and use mocked services as a fallback if approval is delayed.
RSC-04	Scope creep through informal addition of features not present in the approved Sections 1–2 baseline may extend the schedule beyond the academic timeline.	Medium	High	High	Apply formal change control: any new feature request must be evaluated against Section 1–2 scope and approved before being added to Section 4.1, per the

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
					review process in Section 8.1.

9.5 Resource Risks

Resource risks concern the availability of team members, access to third-party service quotas, hardware/devices for testing, and team familiarity with the required technology stack.

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
RR-01	Unavailability of a team member due to illness or withdrawal may reduce development capacity for an increment.	Medium	High	High	Cross-train team members on at least two functional areas, maintain shared documentation (this SRS and design artifacts) so any member can continue another's work.
RR-02	Free-tier quotas for Firebase services, Google Maps Platform/Mapbox, or SMS/notification services may be exhausted during testing, blocking development.	Medium	Medium	Medium	Monitor quota usage during development, use mock data/services for repetitive automated tests, and reserve live-service calls for manual and final acceptance testing.
RR-03	Limited access to varied devices/browsers may restrict cross-platform UI testing required by Section 4.2.1 and	Low	Medium	Low	Use browser developer tools for responsive-design emulation, and

Risk ID	Risk Description	Probability	Impact	Risk Level	Mitigation Strategy
	the Usability NFRs in Section 4.3.				schedule a focused device-testing session using available team members' personal devices before final submission.
RR-04	Gaps in team familiarity with React.js, Node.js/Express, PlantUML, or MySQL may slow initial development.	Medium	Medium	Medium	Allocate a short technology onboarding period at project start, pair less-experienced members with more experienced ones, and use the early increments for lower-complexity modules.

9.6 Risk Matrix

The Risk Matrix below classifies each identified risk by its Probability (columns) and Impact (rows), assigning an overall Risk Level of Low, Medium, or High. This matrix was used to derive the Risk Level values shown in the tables in Sections 9.1–9.5.

Impact \ Probability	Low	Medium	High
High	Medium RT-05	High RT-01, RT-04, RS-01–RS-03, RS-05, RSC-01	High RSC-02, RSC-04, RR-01

Impact \ Probability	Low	Medium	High
Medium	Low RR-03	Medium RT-02, RT-03, RT-06, RO-01, RO-02, RSC-03, RR-02, RR-04	High (none currently identified)
Low	Low RO-04	Low RO-03	Medium (none currently identified)

9.6.1 Risk Level Interpretation

- High: Requires immediate mitigation action and ongoing monitoring; mitigation strategy shall be implemented before the affected increment begins.
- Medium: Requires a documented mitigation strategy and periodic monitoring during the affected increment.
- Low: Monitored as part of routine project tracking; mitigation action is optional unless the risk escalates.

9.7 Mitigation Plan Summary

Mitigation strategies for each individual risk are documented in the Mitigation Strategy column of the tables in Sections 9.1–9.5. In addition to these item-level strategies, the following project-wide mitigation practices shall be adopted across all risk categories:

- All High Risk Level items shall be reviewed at the start of the increment in which the affected functional area is developed, and a designated team member shall be responsible for monitoring each High risk through to closure.
- Risks shall be re-assessed at the end of each increment as part of the Level 3 walkthrough described in Section 8.1, with Risk Levels updated based on new information.
- Any risk that materializes into an issue shall be logged in Appendix B (Open Issues List) with a reference to the originating Risk ID.
- Security-related risks (Section 9.2) shall be re-validated against the Security Testing activities defined in Section 10 prior to final project submission.

10. TESTING CONSIDERATIONS

This section defines the testing approach for the THIKANA platform, covering the levels of testing to be performed, their objectives, and representative sample test cases. Each test case references the corresponding functional requirement (Section 4.1), use case (Section 6.3), non-functional requirement (Section 4.3), or risk item (Section 9) it verifies, and is intended to be entered into the Requirements Traceability Matrix (Section 7) under the Test Case column. All test cases shall be executed against the technology stack and academic scope defined in Sections 1–2, and any test case that fails shall be logged as a defect and tracked through the review process described in Section 8.1.

10.1 Testing Strategy Overview

Testing for THIKANA follows a layered strategy aligned with the Incremental Development Model: each increment is subject to Functional Testing of its newly delivered requirements, Integration Testing of interfaces introduced or modified in that increment, and a partial System Testing pass covering affected end-to-end scenarios. Security Testing and Performance Testing are conducted on a recurring basis, with full passes performed prior to major milestones and a final pass prior to project submission. User Acceptance Testing is conducted near the end of the project once the core functional areas listed in Section 1 are implemented.

- Functional Testing verifies that individual functional requirements (Section 4.1) behave as specified, in isolation from other modules.
- Integration Testing verifies that modules and external services (Firebase, Google Maps Platform/Mapbox, payment sandboxes, MySQL) interact correctly.
- System Testing verifies complete end-to-end scenarios corresponding to the use cases defined in Section 6.3.
- Security Testing verifies that the security risks identified in Section 9.2 and the Security NFRs in Section 4.3 are mitigated.
- Performance Testing verifies that the Performance, Scalability, and Availability NFRs in Section 4.3 are met under realistic academic-scale load.
- User Acceptance Testing verifies that representative users from each actor class (Tenant/Buyer, Property Owner, Real Estate Agency, System Administrator) can complete their primary workflows successfully.

10.2 Functional Testing

Functional Testing exercises individual functional requirements from Section 4.1 against their documented Acceptance Criteria, typically through API-level or component-level tests. The sample test cases below illustrate representative coverage across all five requirement groups (Tenant/Buyer, Property Owner, Real Estate Agency, System Administrator, and Common System).

Test Case ID	Reference	Test Description & Steps	Expected Result
TC-FT-001	FR-CS-001	Preconditions: User is on the Registration page with no existing account. Steps: (1) Enter a valid email, phone number, and password meeting complexity rules; (2) Select role (Tenant/Buyer, Property Owner, or Real Estate Agency); (3) Submit the form.	The system creates a new User record via Firebase Authentication, stores the selected Role, and displays a confirmation message prompting the user to verify their email/phone.
TC-FT-002	FR-CS-003	Preconditions: A registered, verified user account exists. Steps: (1) Navigate to the Login page; (2) Enter correct email and password; (3) Submit.	The system authenticates the user, issues a valid session token, and redirects the user to their role-specific dashboard.
TC-FT-003	FR-TB-005	Preconditions: At least 10 published PropertyListing records exist across multiple locations and price ranges. Steps: (1) Open Smart Property Search; (2) Set filters for location, price range, property type, and number of bedrooms; (3) Apply filters.	The system returns only listings matching all selected filter criteria, ordered as defined by the default sort order, within the response-time NFR threshold.
TC-FT-004	FR-PO-002	Preconditions: A Property Owner is logged in. Steps: (1) Navigate to 'Create Listing'; (2) Enter required property details (title, description, price, location, type); (3) Upload at least one image; (4) Submit as Draft.	The system creates a new PropertyListing record with status 'Draft', associates the uploaded image with the listing via Firebase Storage, and displays the listing in the owner's 'My Listings' page.

Test Case ID	Reference	Test Description & Steps	Expected Result
TC-FT-005	FR-PO-004	Preconditions: A PropertyListing exists with status 'Draft' and meets publication eligibility rules (Section 6.11). Steps: (1) Open the listing; (2) Select 'Publish'.	The system transitions the listing's status from 'Draft' to 'Published' per the Property Listing Lifecycle (Section 6.7) and the listing becomes visible in search results.
TC-FT-006	FR-CS-010 / UC-08	Preconditions: Two registered users (a Tenant/Buyer and a Property Owner) exist, with no prior conversation. Steps: (1) Tenant/Buyer opens a published listing; (2) Selects 'Message Owner'; (3) Enters and sends a text message.	The system creates a new Conversation record linking both users, stores the Message with a timestamp, and the Property Owner receives a notification of the new message.
TC-FT-007	FR-TB-012 / UC-09	Preconditions: A Tenant/Buyer and Property Owner have agreed on rental terms for a published listing. Steps: (1) Property Owner initiates 'Generate Rental Agreement'; (2) System pre-fills agreement details from the PropertyListing; (3) Both parties review and accept.	The system creates a RentalAgreement record in 'Pending' status, then transitions to 'Active' once both parties accept, per the Rental Agreement Lifecycle (Section 6.7).
TC-FT-008	FR-TB-015 / UC-10	Preconditions: An Active RentalAgreement exists with a defined monthly rent amount. Steps: (1) Tenant/Buyer selects 'Pay Rent'; (2) Chooses a payment method (SSLCommerz/bKash/Nagad sandbox); (3) Completes the sandbox payment flow.	The system creates a Payment record linked to the RentalAgreement with status 'Completed', updates the rent tracking summary, and displays a confirmation receipt to the Tenant/Buyer.

Test Case ID	Reference	Test Description & Steps	Expected Result
TC-FT-009	FR-SA-004 / UC-07	Preconditions: A Tenant/Buyer has submitted a VerificationRequest with required documents. Steps: (1) System Administrator opens the Verification queue; (2) Reviews submitted documents; (3) Selects 'Approve'.	The system updates the VerificationRequest status to 'Approved' per the User Verification Lifecycle (Section 6.7), and the user's profile displays a 'Verified' badge.
TC-FT-010	FR-SA-009 / UC-12	Preconditions: A PropertyReview has been flagged by a user as inappropriate. Steps: (1) System Administrator opens the Reports queue; (2) Selects the flagged review; (3) Chooses 'Remove Review'.	The system removes the PropertyReview from public view, records the moderation action in the AuditLog, and notifies the review's author.

10.3 Integration Testing

Integration Testing verifies the correct interaction between THIKANA's internal modules and the external services defined in Sections 1–2, including Firebase Authentication, Firebase Storage, Firebase Cloud Messaging, Google Maps Platform/Mapbox, and the MySQL database. Particular attention is given to data consistency across module boundaries, as required by Validation Rules VR-008 and VR-009 (Section 8.3).

Test Case ID	Reference	Test Description & Steps	Expected Result
TC-IT-001	FR-CS-001, FR-CS-002	Verify integration between the Node.js/Express backend and Firebase Authentication: (1) Register a user via the API; (2) Confirm a corresponding Firebase Authentication record is created; (3) Confirm a corresponding User record is created in MySQL with matching unique identifier.	The Firebase Authentication UID is correctly stored as a foreign key reference in the MySQL User table, and both records remain consistent after registration.

Test Case ID	Reference	Test Description & Steps	Expected Result
TC-IT-002	FR-PO-002, FR-PO-003	Verify integration between Property Listing Management and Media Management: (1) Create a PropertyListing via the API; (2) Upload multiple images via Firebase Storage; (3) Retrieve the listing via the API.	The retrieved PropertyListing includes correctly ordered references (URLs) to all uploaded PropertyImage records stored in Firebase Storage.
TC-IT-003	FR-TB-007, FR-CS-011	Verify integration between the Map-Based Property Discovery module and Google Maps Platform/Mapbox: (1) Request property markers for a bounding box; (2) Confirm the map component renders markers at the correct PropertyLocation coordinates.	All PropertyListing records with PropertyLocation coordinates inside the requested bounding box are rendered as markers at the correct geographic positions on the map.
TC-IT-004	FR-CS-010, FR-CS-013	Verify integration between the Direct Communication System and Firebase Cloud Messaging: (1) User A sends a Message to User B while User B is offline; (2) User B comes online.	User B receives a push notification via Firebase Cloud Messaging referencing the new Message, and the Notification record is marked as delivered once the app is opened.
TC-IT-005	FR-TB-015, FR-PO-008	Verify integration between the Payment module and the Rental Agreement module: (1) Submit a rent payment via the sandbox payment gateway for an Active RentalAgreement; (2) Query the RentalAgreement's payment history.	The new Payment record is correctly linked to the RentalAgreement via foreign key, and the payment history reflects the updated

Test Case ID	Reference	Test Description & Steps	Expected Result
			total amount paid and remaining balance.
TC-IT-006	FR-SA-001, FR-CS-003	Verify integration between Role-Based Access Control middleware and the Express API: (1) Authenticate as a Tenant/Buyer; (2) Attempt to call an administrator-only endpoint (e.g., Manage Users).	The API returns an HTTP 403 Forbidden response, and no AnalyticsRecord, User, or AuditLog modification occurs as a result of the unauthorized call.

10.4 System Testing

System Testing executes complete end-to-end scenarios corresponding to the use cases defined in Section 6.3, exercising the system as a whole across multiple actors and modules. These tests validate that the Main Flows of related use cases interoperate correctly and that data created in one use case is correctly consumed in a subsequent use case.

Test Case ID	Reference	Test Description & Steps	Expected Result
TC-ST-001	UC-01, UC-02	End-to-end: A new Tenant/Buyer registers, verifies their account, and logs in for the first time, then is redirected to their dashboard.	All steps complete without errors; the user's session persists across page navigation, and the dashboard displays role-appropriate widgets (e.g., Favorites, Search shortcuts).
TC-ST-002	UC-03, UC-04	End-to-end: A Tenant/Buyer searches for properties using multiple filters, selects a result from the list, and views the full Property Details page including images, map location, and area intelligence statistics.	Search results, listing details, images, map marker position, and area statistics are all displayed consistently

Test Case ID	Reference	Test Description & Steps	Expected Result
			and correspond to the same PropertyListing record.
TC-ST-003	UC-05, UC-06	End-to-end: A Property Owner creates a new listing, saves it as Draft, edits the price, uploads additional images, and publishes the listing.	The final published listing reflects all edits (updated price, additional images) and is discoverable via Smart Property Search immediately after publication.
TC-ST-004	UC-07, UC-14	End-to-end: A Property Owner submits a VerificationRequest with documents; a System Administrator reviews and approves the request from the Manage Users interface.	The Property Owner's profile updates to show a 'Verified' badge, and the AuditLog records the administrator's action with a timestamp and administrator identifier.
TC-ST-005	UC-08, UC-09, UC-10	End-to-end: A Tenant/Buyer messages a Property Owner about a listing, both parties generate and accept a Rental Agreement, and the Tenant/Buyer submits the first rent payment.	The Conversation, RentalAgreement, and Payment records are all correctly linked to the same Tenant/Buyer, Property Owner, and PropertyListing, and each entity's lifecycle status updates as expected.
TC-ST-006	UC-11, UC-13	End-to-end: After a RentalAgreement is completed, the Tenant/Buyer submits a PropertyReview; the Property Owner	The PropertyReview is visible on the listing page and the Owner Analytics dashboard

Test Case ID	Reference	Test Description & Steps	Expected Result
		subsequently views updated Owner Analytics reflecting the new review.	reflects the updated average rating and review count without requiring a manual refresh of underlying data.
TC-ST-007	UC-12, UC-15	End-to-end: A user reports a listing as fraudulent; a System Administrator reviews the Report, removes the listing, and the affected Property Owner receives a notification explaining the action.	The PropertyListing status transitions to 'Removed' per the Property Listing Lifecycle (Section 6.7), the Report status updates to 'Resolved', and a Notification record is created for the Property Owner.

10.5 Security Testing

Security Testing targets the security risks identified in Section 9.2 (Security Risks) and verifies compliance with the Security category of Non-Functional Requirements in Section 4.3. Testing includes both automated scanning (where tooling permits) and manual verification of authentication, authorization, input validation, and data protection controls, informed by the OWASP Top Ten guidance referenced in Appendix E.

Test Case ID	Reference	Test Description & Steps	Expected Result
TC-SEC-001	RS-01	Send an API request to a protected endpoint (e.g., 'Create Listing') using an expired or malformed Firebase Authentication token.	The API rejects the request with HTTP 401 Unauthorized and does not create any PropertyListing record.

Test Case ID	Reference	Test Description & Steps	Expected Result
TC-SEC-002	RS-02	Submit a search filter value containing SQL injection patterns (e.g., "" OR '1'='1") and a messaging input containing a script tag.	The search query is executed safely using parameterized queries with no unintended results returned; the script tag is stored/escaped such that it is rendered as plain text, not executed, in the recipient's browser.
TC-SEC-003	RS-03	As an authenticated Tenant/Buyer, attempt to directly access the Firebase Storage URL of another user's uploaded NID verification document.	Access is denied by Firebase Storage security rules; only the document owner and System Administrators can retrieve the file.
TC-SEC-004	RS-04	Log in as a user, capture the issued session token, log out, and then attempt to reuse the captured token to call a protected endpoint.	The reused token is rejected with HTTP 401 Unauthorized, confirming that logout invalidates the active session token.
TC-SEC-005	RS-05	While authenticated as a Real Estate Agency, attempt to call the 'Manage Users' and 'Moderate Reports' endpoints reserved for System Administrators (UC-12, UC-14).	Both requests are rejected with HTTP 403 Forbidden, and the attempt is recorded in the AuditLog with the requesting user's identity.
TC-SEC-006	NFR (Security)	Attempt to register five user accounts using the same email address with minor variations, and	Duplicate-email registration is rejected after the first successful

Test Case ID	Reference	Test Description & Steps	Expected Result
	category, Section 4.3)	attempt 10 consecutive failed login attempts for a single account within one minute.	registration; repeated failed login attempts trigger a temporary account lockout or rate-limiting response as defined by the relevant Security NFR.

10.6 Performance Testing

Performance Testing verifies that the system meets the measurable thresholds defined in the Performance, Scalability, and Availability categories of Section 4.3, under load conditions appropriate to the academic scope defined in Sections 1–2. Load levels are deliberately scaled to a realistic number of concurrent users for a university project rather than enterprise-scale traffic.

Test Case ID	Reference	Test Description & Steps	Expected Result
TC-PERF-001	NFR (Performance category, Section 4.3)	Measure response time for the Smart Property Search endpoint when the PropertyListing table contains at least 500 records, executed 50 times with varying filter combinations.	At least 95% of search requests return results within the maximum response time specified in the corresponding Performance NFR (Section 4.3).
TC-PERF-002	NFR (Performance / Scalability categories, Section 4.3)	Simulate 50 concurrent users performing a mix of search, listing view, and messaging operations for a sustained 10-minute period using a load-testing tool.	The system maintains an average response time within the defined Performance NFR threshold and does not return HTTP 5xx errors for more than 1% of requests during the test.

Test Case ID	Reference	Test Description & Steps	Expected Result
TC-PERF-003	NFR (Performance category, Section 4.3)	Measure the time taken to render the Map-Based Property Discovery page when displaying 200 PropertyListing markers within the visible map bounding box.	The map renders all visible markers within the maximum load-time threshold specified in the corresponding Performance NFR, using marker clustering where applicable.
TC-PERF-004	NFR (Performance category, Section 4.3)	Measure end-to-end time from submitting a sandbox payment request (UC-10) to receiving a confirmation response from the system.	The payment confirmation is received and displayed to the user within the maximum response time specified in the corresponding Performance NFR, excluding delays attributable to the external payment sandbox itself.
TC-PERF-005	NFR (Availability category, Section 4.3)	Monitor system uptime over a continuous 7-day period during the testing phase using an automated health-check endpoint polled every 5 minutes.	Measured uptime meets or exceeds the percentage specified in the corresponding Availability NFR (Section 4.3).

10.7 User Acceptance Testing

User Acceptance Testing (UAT) is conducted with representative users (or proxies, such as classmates or the course instructor acting in a given role) for each of the four primary user classes defined in Section 1. UAT focuses on whether users can successfully complete their primary workflows using the user interface described in Section 4.2.1, and whether the Usability NFRs in Section 4.3 are perceived as met.

Test Case ID	Reference	Test Description & Steps	Expected Result
TC-UAT-001	Tenant/Buyer	A representative Tenant/Buyer user registers an account, searches for a rental property in a specific area within a given budget, saves a listing to Favorites, and contacts the Property Owner.	The user completes all steps without assistance within a reasonable time, and reports that the search filters and messaging flow are clear and intuitive (Usability NFR, Section 4.3).
TC-UAT-002	Property Owner	A representative Property Owner creates a new listing with images and a virtual tour link, publishes it, and reviews the Owner Analytics dashboard after simulated views and inquiries.	The user successfully publishes the listing without external help and correctly interprets the analytics dashboard metrics (views, inquiries) as intended.
TC-UAT-003	Real Estate Agency	A representative Real Estate Agency user manages multiple property listings under a single agency profile, including creating, editing, and unpublishing listings for different owners.	The agency user can distinguish between listings belonging to different properties and complete bulk-style management tasks without confusion about which listing is being edited.
TC-UAT-004	System Administrator	A representative System Administrator reviews pending VerificationRequests, approves or rejects them, and reviews a flagged Report, taking an appropriate moderation action.	The administrator can locate pending items, take action, and confirm that the AuditLog accurately reflects each action taken, within the

Test Case ID	Reference	Test Description & Steps	Expected Result
			workflow described in UC-07 and UC-12.
TC-UAT-005	Tenant/Buyer & Property Owner	Both a Tenant/Buyer and a Property Owner participate in generating a Rental Agreement and completing a sandbox rent payment, then review the resulting agreement and payment history.	Both users confirm that the agreement terms and payment history displayed match their expectations, and that the rental agreement and payment statuses update visibly without requiring a page reload to appear current.

APPENDIX A - GLOSSARY

This glossary provides authoritative plain-language definitions for the domain-specific terms, technical concepts, acronyms, and abbreviations used throughout the **Thikana** Software Requirements Specification. It is intended to create a shared understanding among the development team, project manager, QA tester, academic instructor, and future contributors.

Each entry identifies the term, its domain category, and its meaning within the context of the **Thikana** project. Where a term has both a general meaning and a Thikana-specific meaning, the Thikana-specific meaning takes precedence throughout this SRS. Terms are organized alphabetically for quick look-up and cross-referenced through the acronym quick-reference table in Section A.2.

A.1 Glossary Table

Table A.1 presents the complete glossary of 75 terms and abbreviations used in the Thikana SRS. The Category column identifies the area from which the term originates, such as property domain, technical architecture, security, maps, payments, localization, or requirements engineering.

Term / Acronym	Category	Definition
360-Degree Virtual Tour	Feature / Media	An interactive panoramic representation of a property interior that allows a user to explore rooms remotely. In Thikana, virtual tours are image-based and may be rendered using Three.js, Pannellum, or a similar client-side viewer.
Accessibility	Usability / Standards	The quality of making the platform usable by people with diverse abilities, including keyboard-only users and screen-reader users. Thikana targets WCAG 2.1 Level AA for user-facing interfaces.
Address Hierarchy	Localization / Data	The Bangladeshi location structure used in property addresses: Division, District, Upazila, Union, Area, and Road or House Number. Thikana uses this hierarchy for filtering, maps, and listing details.
Admin Panel	System / Interface	The restricted interface used by platform administrators to verify users, moderate listings and reviews, manage reports, and view platform-wide analytics.
Agency Dashboard	Interface / Analytics	The role-specific dashboard through which real estate agencies manage multiple listings, monitor enquiries, and access weekly or monthly trend reports.

Term / Acronym	Category	Definition
AI Recommendation	Feature / Analytics	A future enhancement that suggests properties based on user preferences, saved listings, viewing behavior, and search history. In the academic prototype this is treated as a low-priority or simulated feature.
Amenity	Domain / Property	A facility or feature associated with a property listing, such as lift, generator, parking, security guard, gas line, balcony, Wi-Fi, or furnished status.
API (Application Programming Interface)	Technical	A defined interface that allows software components or external services to exchange data. Thikana uses APIs for maps, authentication, payments, notifications, and backend data access.
Area Intelligence	Feature / Location	Location-contextual information displayed around a property, such as nearby schools, hospitals, pharmacies, markets, mosques, transport stops, safety rating, and commute estimates.
Authentication	Security	The process of verifying that a user is who they claim to be. Thikana supports phone OTP, password-based login, JWT sessions, and optional Google OAuth login.
Authorization	Security	The process of determining what an authenticated user is allowed to do. Thikana applies role-based permissions for tenants, buyers, owners, agencies, and administrators.
Backup	Operations	A copy of database and system data created for recovery in case of failure. Thikana requires automated daily backups of MySQL data and appropriate cloud storage backup for media and Firebase data.
bKash	Payment / External Service	A Bangladeshi mobile financial service that may be integrated with Thikana as a BDT payment gateway, initially through sandbox or simulated payment flows.
BDT (Bangladeshi Taka)	Localization / Financial	The official currency of Bangladesh. Thikana displays all rent, price, fee, and utility amounts in BDT or the  symbol.
Buyer	Actor	A public user searching for property to purchase. In Thikana, buyer capabilities overlap with tenant capabilities, including search, property detail viewing, area intelligence, favorites, and messaging.

Term / Acronym	Category	Definition
Cloud Storage	Technical / Infrastructure	Remote file storage used to keep property photos, floor plans, 360-degree images, and verification documents. Firebase Storage or AWS S3 may be used.
Commute Estimate	Feature / Maps	An approximate travel-time calculation from a property to a user-provided destination, generated using a directions service such as Google Maps Directions API.
Content Moderation	Safety / Administration	The administrative review process for removing or restricting fake listings, suspicious accounts, abusive messages, and inappropriate reviews.
Coverage Data	External Data	Information about approximate internet or mobile network availability in an area. Thikana may use such data to inform tenants about connectivity around a property.
CRUD	Technical	Create, Read, Update, and Delete. These are the core data operations used throughout Thikana for users, listings, photos, reviews, agreements, payments, reports, and favorites.
Dashboard	Interface	A role-specific landing area that summarizes relevant actions and metrics. Thikana includes dashboards for tenants/buyers, owners, agencies, and administrators.
Data Dictionary	Requirements / Database	A structured definition of key fields, data types, allowed values, and meanings for Thikana entities such as User, Listing, Photo, Review, Message, Agreement, Payment, Report, and Favorite.
Digital Rental Agreement	Domain / Document	A system-generated agreement document containing property, owner, tenant, rent amount, duration, and acceptance status. The agreement supports review and digital acceptance or rejection by both parties.
Direct Communication	Feature / Messaging	The in-app communication capability that allows tenants and owners to exchange messages without relying entirely on external phone calls or social media.
Email Server / SMTP	Technical / External Service	The email delivery service used to send registration confirmations, password resets, agreement notifications, and rent reminders.

Term / Acronym	Category	Definition
Entity-Relationship Diagram (ERD)	Modeling / Database	A diagram showing the main data entities and relationships in Thikana, including User, Listing, Photo, Review, Message, Agreement, Payment, Report, and Favorite.
Favorite	Feature / Data Entity	A saved property listing associated with a user account. Favorites allow users to revisit listings and receive alerts when a saved listing changes status or price.
FCM (Firebase Cloud Messaging)	Technical / Notification	Google's push notification service used by Thikana to deliver message alerts, listing status updates, rent reminders, and other real-time notifications.
Firestore Auth	Technical / Authentication	A Firestore service used for phone OTP verification and optional Google OAuth login in the Thikana platform.
Firestore Realtime Database	Technical / Database	A cloud-hosted NoSQL database service used for real-time chat messages, notification counters, and online status indicators.
Firestore Storage	Technical / Storage	A cloud storage service used to store property photos, floor plans, 360-degree images, and NID or student ID document uploads.
Floor Plan	Property / Media	An image or PDF showing the layout of a property. Owners may attach a floor plan to help tenants and buyers understand room arrangement before visiting.
Geocoding	Maps / Technical	The process of converting a textual address into latitude and longitude coordinates for map display, search, and area intelligence features.
Google Maps Platform	External Service / Maps	The external mapping service used for map display, property pins, geocoding, nearby places, and commute-time calculations.
i18n (Internationalization)	Localization / Technical	The architectural preparation needed to support multiple languages and regional formats. Thikana is initially English-first but designed for future Bengali support.
JWT (JSON Web Token)	Security / Technical	A signed token used for stateless authentication in Thikana API requests. JWT tokens carry user identity, role information, and expiry time.

Term / Acronym	Category	Definition
Listing	Core Entity / Property	A property advertisement created by an owner or agency. A listing contains title, description, type, location, price, availability status, photos, contact settings, and optional floor plan or virtual tour.
Map Pin	Maps / Interface	A clickable marker shown on the map view to represent an active property listing at its geographic location.
Mapbox	External Service / Maps	An alternative mapping platform that may be used in place of or alongside Google Maps for map-based search and location visualization.
Masked Phone Number	Privacy / Communication	A partially hidden or controlled phone number display that reduces direct exposure of an owner's contact details. Thikana may reveal phone numbers only to verified tenants.
Media Asset	Data / Storage	Any uploaded file associated with the system, including property photos, floor plans, virtual tour images, chat attachments, and verification documents.
Messaging Inbox	Interface / Messaging	The interface where users view current and previous conversations related to property enquiries, roommate requests, agreements, or support interactions.
Mobile-First Design	Usability / Interface	A design approach that ensures the platform works effectively on smartphones before scaling up to tablet and desktop layouts.
MVP (Minimum Viable Product)	Project Management	The first working version of Thikana that demonstrates the core value of the platform. The academic MVP focuses on registration, property listing, search with filters, and map view.
Nagad	Payment / External Service	A Bangladeshi mobile financial service that may be supported as a payment option for BDT transactions in future or sandbox payment flows.
NID (National Identity Document)	Identity / Bangladesh	The Bangladeshi National Identity Document used for manual tenant or owner verification. In the student prototype, real government API verification is not available and is simulated through uploaded document images.

Term / Acronym	Category	Definition
OAuth 2.0	Security / Authentication	An authorization framework used to support third-party login, such as Google sign-in, without requiring the platform to handle the external account password.
OTP (One-Time Password)	Security	A temporary code sent to a registered phone number or email address to verify identity during registration, login, or password reset.
Owner Reputation	Trust / Review	The trust indicator derived from tenant ratings and reviews. It helps future tenants and buyers assess owner reliability and past interaction quality.
Payment Gateway	Financial / External Service	A third-party service such as SSLCommerz, bKash, or Nagad that processes digital BDT transactions. The prototype uses sandbox or simulated payment behavior.
PII (Personally Identifiable Information)	Privacy / Compliance	Personal data that can identify a user, including name, phone number, email address, NID image, and profile information. Thikana must protect PII and restrict disclosure.
Places API	Maps / External Service	A location service used to retrieve nearby facilities such as schools, hospitals, pharmacies, markets, mosques, and transport stops around a listing.
Property Owner	Actor	A user who owns one or more properties and uses Thikana to create listings, upload media, respond to enquiries, manage availability, and view analytics.
Property Type	Domain / Listing	The category of a property, such as flat, house, bachelor unit, sublet, office, or buy/sell listing.
Push Notification	Notification	A real-time alert delivered to the user's browser or future mobile application for messages, favorites, rent reminders, verification updates, and listing status changes.
Rate Limiting	Security	A control that limits repeated requests to protect authentication and API endpoints from brute-force attacks or abuse. Thikana locks authentication attempts after repeated failures.

Term / Acronym	Category	Definition
Real Estate Agency	Actor	A professional organization or agent that manages multiple property listings and uses the platform for bulk management, enquiries, and agency-level analytics.
Report	Moderation / Data Entity	A user-submitted complaint against a fake listing, suspicious account, inappropriate review, or fraudulent activity. Reports are reviewed by administrators.
REST API	Technical	A web API style in which Thikana backend resources are accessed using HTTPS methods and JSON responses.
Review	Trust / Data Entity	A rating and written comment submitted by a tenant or buyer after a completed interaction. Reviews contribute to owner reputation and area safety ratings.
Role-Based Access Control (RBAC)	Security	A permission model that assigns capabilities based on user roles such as tenant/buyer, property owner, agency, and administrator.
Roommate Matching	Feature / Future Enhancement	A feature that allows users to post roommate requests and find compatible people based on area, budget, lifestyle preference, and gender preference.
Search Filter	Feature / Interface	A user-selected search constraint such as price range, property type, bedroom count, bathroom count, furnished status, or amenities.
Soft Delete	Technical / Data Management	A deletion approach in which records are marked inactive or deleted rather than being immediately removed from the database, supporting dispute handling and auditability.
SSLCommerz	Payment / External Service	A Bangladeshi payment gateway that can process online BDT transactions. Thikana uses sandbox mode for academic prototype demonstrations.
SRS (Software Requirements Specification)	Requirements Engineering	The formal document that defines the functional requirements, non-functional requirements, interfaces, data requirements, quality attributes, and analysis models for Thikana.
Sublet	Domain / Property	A rental arrangement where an existing tenant rents part or all of their rented property to another person. Thikana treats sublet as one supported property category.

Term / Acronym	Category	Definition
Tailwind CSS	Technical / Frontend	A utility-first CSS framework used to implement consistent, responsive, and maintainable front-end styling in the Thikana web application.
Tenant / Buyer	Actor	A general public user searching for rental or purchase properties. Tenant/buyer users can search, view listings, save favorites, message owners, submit reviews, and access area intelligence.
TLS (Transport Layer Security)	Security	A cryptographic protocol used to secure HTTPS communication between the client browser and the Thikana backend. TLS 1.2 or higher is required.
Utility Expense Tracking	Feature / Finance	A low-priority feature allowing tenants to track monthly utility costs such as electricity, gas, internet, and service charges with summaries.
Verification Badge	Trust / Interface	A visual label displayed on a user profile or listing to show that the account has completed phone verification or admin-approved document verification.
Verification Request	Identity / Workflow	A user-submitted request containing NID or student ID documents for manual review by administrators. Requests may be approved, rejected, or returned for additional documents.
WebSocket	Technical / Communication	A persistent communication mechanism used for real-time features such as chat and notification updates. Thikana may implement this through Firebase services.
WCAG 2.1	Accessibility / Standards	The Web Content Accessibility Guidelines version 2.1, an international standard for making web content accessible. Thikana targets Level AA compliance.

A.2 Acronym Quick-Reference

The following table provides a condensed quick-reference index of acronyms and abbreviations appearing in the Thikana SRS, cross-referenced to the glossary entries where the full definitions are provided.

Acronym	Expansion	Acronym	Expansion
AI	Artificial Intelligence	API	Application Programming Interface
BDT	Bangladeshi Taka	CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets	ERD	Entity-Relationship Diagram
FCM	Firebase Cloud Messaging	FR	Functional Requirement
GPS	Global Positioning System	HTTPS	Hypertext Transfer Protocol Secure
i18n	Internationalization	JWT	JSON Web Token
MVP	Minimum Viable Product	NFR	Non-Functional Requirement
NID	National Identity Document	OAuth	Open Authorization
OTP	One-Time Password	PII	Personally Identifiable Information
PWA	Progressive Web Application	RBAC	Role-Based Access Control
REST	Representational State Transfer	RTM	Requirements Traceability Matrix
SRS	Software Requirements Specification	SQL	Structured Query Language
SSL / TLS	Secure Sockets Layer / Transport Layer Security	TC	Test Case
UC	Use Case	UI	User Interface
UML	Unified Modeling Language	VPS	Virtual Private Server
WCAG	Web Content Accessibility Guidelines	WEBP	Web Picture Image Format
XSS	Cross-Site Scripting		

APPENDIX B — OPEN ISSUES LIST

This appendix documents unresolved issues, open questions, and pending decisions identified during the preparation of this Software Requirements Specification. Each issue is assigned a unique identifier, a description of the open question or concern, the requirement(s) or section(s) it affects, an assigned priority, its current resolution status, and a proposed path toward resolution. Issues listed here do not block the current baseline of Sections 1 through 10 but must be addressed before or during the corresponding implementation phase identified in the Incremental Development Model (Section 2.6). This list shall be reviewed and updated at the start of each development increment.

Issue ID	Description	Affected Area / Requirement(s)	Priority	Status	Proposed Resolution Path
ISS-01	The choice of SMS/OTP gateway provider for Bangladeshi mobile numbers (e.g., Firebase Authentication's native SMS, or a local provider such as Banglalink/Robi SMS API) has not been finalized. Firebase phone authentication has known cost and quota limitations for non-trial use.	FR-TB-002, FR-CS-001, ACT-05, NFR-PF	High	Open	Increment 1
ISS-02	It is not yet decided whether property listing publication will require mandatory administrator pre-approval (moderation-first) or will be published immediately with post-hoc moderation (publish-first, FR-SA-004/FR-SA-005). This affects the Listing Lifecycle state diagram (Section 6.7) and decision table (Section 6.11).	FR-PO-006, FR-SA-004, FR-SA-005, BR-005	High	Open	Increment 1
ISS-03	The exact set of accepted identity document types for verification (NID only, or NID + Student ID + Passport) and the validation method (manual visual review only, vs. automated OCR/NID-API integration) is undecided. Government NID API integration is listed only as a future enhancement (Appendix C).	FR-TB-015, FR-SA-002, FR-SA-003, BR-007, M-08	High	Open	Increment 2

ISS-04	The specific Map Service Provider (Google Maps Platform vs. Mapbox) has not been finally selected. Google Maps Platform offers richer Places API data for area intelligence (FR-TB-009) but has a usage-based billing model that may exceed the free tier during heavy testing; Mapbox offers a more generous free tier but with less comprehensive Bangladesh POI coverage.	FR-TB-007, FR-TB-009, ACT-06, M-04, M-07	Medium	Open	Increment 1
ISS-05	The sandbox payment gateway to be integrated (SSLCommerz, bKash, or Nagad) has not been finalized. Each sandbox has different registration requirements, callback formats, and supported test card/wallet credentials, which will affect the implementation of UC-10 and the Payment entity's gatewayResponse-equivalent fields.	FR-CS-011, UC-10, ACT-09, M-11.1	Medium	Open	Increment 3
ISS-06	The minimum review-eligibility condition in BR-013 ('verified interaction') is defined conceptually but the precise system-detectable trigger (e.g., existence of any Conversation vs. existence of an Active or Completed RentalAgreement) has not been agreed upon. This affects FR-TB-016 and the Review and Reputation Management module (M-13).	FR-TB-016, BR-013, M-13.1	Medium	Open	Increment 2
ISS-07	The retention period referenced in BR-014 and FR-CS-015 for soft-deleted listings and deactivated accounts is currently a placeholder value (e.g., 30 days) and has not been validated against any specific institutional or legal guidance, since no formal Bangladeshi data-retention regulation specific to this domain was identified during elicitation (Section 1.5).	FR-CS-015, FR-PO-009, BR-014, NFR-CO-03	Low	Open	Increment 3

ISS-08	It is unresolved whether the Real Estate Agency role requires a separate staff-account hierarchy (i.e., multiple individual login accounts under one AgencyProfile with permission tiers) or whether a single shared agency login is sufficient for the academic prototype scope. FR-RA-006 (assigning staff contacts) currently assumes a single-login model with a lightweight 'assigned contact' field.	FR-RA-002, FR-RA-006, AgencyProfile (5.2)	Low	Open	Increment 2
ISS-09	The exact mechanism for the 'contact-reveal' rule in BR-010 — whether the owner's phone number is revealed permanently once conditions are met, or only for the duration of an active conversation/session — has not been decided, which affects the implementation of FR-TB-014 and the Conversation/Message data model.	FR-TB-014, BR-010, M-09.5	Medium	Open	Increment 2
ISS-10	The frequency and exact scheduling mechanism for the Scheduler actor (ACT-11) — whether implemented via a Node.js cron library (e.g., node-cron) within the existing backend process, or as a separate scheduled task/serverless function — has not been finalized, which may affect deployment architecture (Section 2.4) and NFR-DR-01/NFR-LG-03.	ACT-11, FR-CS-017, FR-PO-016, M-11.2	Low	Open	Increment 3
ISS-11	The exact set of 'amenities' and 'property type' enumeration values (e.g., whether 'Bachelor Mess', 'Sublet', and 'Office Space' are distinct property types or sub-categories) has not been finalized with stakeholder input, which affects the PropertyListing data dictionary (Section 5.2) and search filter design (FR-TB-006).	FR-TB-006, FR-PO-002, PropertyListing (5.2)	Low	Open	Increment 1

ISS-12	Accessibility testing (NFR-AC-01 to NFR-AC-03) has been specified with target metrics but no accessibility testing tool or environment has yet been set up by the development team; the academic timeline may constrain the depth of accessibility validation that can realistically be performed.	NFR-AC-01, NFR-AC-02, NFR-AC-03	Low	Open	Increment 3
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Issue priorities follow the same High/Medium/Low scale defined in Section 1.3. High-priority issues (ISS-01, ISS-02, ISS-03) directly affect core registration, listing publication, and verification workflows and must be resolved before the corresponding increment's design activities begin. The 'Proposed Resolution Path' column references the Increment (per the Incremental Development Model, Section 2.6) during which the issue is expected to be resolved through team discussion, supervisor consultation, or technical proof-of-concept, and this appendix shall be revised accordingly as decisions are made.

APPENDIX C — FUTURE ENHANCEMENTS

This appendix identifies features and capabilities that are outside the scope of the current academic implementation of THIKANA but represent realistic directions for future development, extension, or commercialization of the platform. These enhancements are not part of the baseline requirements specified in Sections 4 and 5 of this document and shall not be considered mandatory for the current project deliverable.

C.1 Mobile Application (Android/iOS)

A dedicated native or cross-platform (e.g., React Native or Flutter) mobile application shall be developed to provide Tenants/Buyers, Property Owners, Real Estate Agencies, and System Administrators with access to core platform features such as property search, messaging, notifications, and listing management on mobile devices. The mobile application would leverage Firebase Cloud Messaging for push notifications and the device's GPS for location-based property discovery.

C.2 AI-Based Property Recommendation Engine

A machine learning-based recommendation system shall be introduced to suggest properties to Tenants/Buyers based on their search history, saved favorites, budget range, preferred locations, and behavioral patterns. The recommendation engine could employ collaborative filtering or content-based filtering techniques to improve match relevance over time.

C.3 Fraud Detection System

An automated fraud detection module shall be implemented to identify suspicious activities such as duplicate listings, fake property images, price manipulation, or repeated fraudulent verification attempts. This module could utilize rule-based heuristics in early stages and machine learning-based anomaly detection in later iterations to flag high-risk listings or users for administrator review.

C.4 Digital Signature Integration

The Rental Agreement Management module shall be extended to support legally recognized digital signatures, allowing Tenants/Buyers and Property Owners to sign rental agreements electronically through an integrated digital signature service (e.g., a DocuSign-equivalent or a locally compliant Bangladeshi digital signature provider), thereby reducing reliance on physical paperwork.

C.5 Government National ID (NID) Verification Integration

The Tenant and Owner Verification module shall be enhanced to integrate directly with the Bangladesh National Identity Registration Wing (NID) verification API or an equivalent government-approved identity verification service, enabling automated, real-time validation of user-submitted NID information rather than relying solely on manual administrator review.

C.6 Roommate Matching System

A roommate-matching feature shall be introduced to allow Tenants/Buyers seeking shared accommodation to create roommate profiles specifying preferences such as budget, lifestyle, gender preference, and location, and to be matched with compatible roommates seeking similar arrangements.

C.7 Smart Property Valuation Tool

A property valuation module shall be developed to provide Property Owners and Real Estate Agencies with an estimated fair market rental or sale price for a property based on historical pricing data, area intelligence statistics, property attributes, and comparable listings, assisting users in setting competitive and realistic prices.

C.8 Cloud Deployment and Scalability Enhancements

The platform's deployment architecture shall be migrated from a single-server or local development environment to a cloud-based infrastructure (e.g., AWS, Google Cloud Platform, or Microsoft Azure), incorporating load balancing, auto-scaling, containerization (Docker/Kubernetes), and managed database services to support a larger user base and ensure higher availability and fault tolerance.

Note: The enhancements listed in this appendix are provided for forward-planning purposes only and do not impose any implementation obligations on the current project scope defined in Sections 1 through 10 of this document.

APPENDIX D — ASSUMPTIONS

This appendix lists the assumptions made during the preparation of this Software Requirements Specification for THIKANA — Smart Real Estate & Rental Management Platform. These assumptions describe conditions that are believed to be true for the purposes of defining the requirements in Sections 4 through 7 and the supporting models in Section 6. If any assumption listed below is later found to be invalid, the affected requirements shall be re-evaluated through the change control and review process described in Section 8.1, and updated accordingly.

ID	Assumption	Category
ASM-001	All users (Tenant/Buyer, Property Owner, Real Estate Agency) have access to a smartphone, tablet, or computer with a modern web browser and a stable internet connection sufficient to use the THIKANA web application.	General
ASM-002	The project will be developed and demonstrated primarily within Bangladesh, and all examples of locations, currency (BDT), addresses, and area intelligence data are based on Bangladeshi administrative divisions, districts, and cities.	General
ASM-003	The THIKANA platform is intended as an academic Software Engineering project and will be deployed in a development, staging, or limited demonstration environment rather than a full-scale commercial production environment.	General
ASM-004	The frontend will be implemented using React.js with Tailwind CSS, and the backend using Node.js with Express, as specified in Sections 1–2; no alternative frameworks will be substituted without a formal change request.	Technical
ASM-005	MySQL will be used as the relational database management system for all persistent application data described in Section 5; no NoSQL database will be introduced unless explicitly added as a Future Enhancement (Appendix C).	Technical
ASM-006	Firebase Authentication will be used as the primary identity provider for user registration and login, and the development team will have access to a Firebase project with sufficient free-tier quota for development and testing.	Technical

ID	Assumption	Category
ASM-007	Firestore Cloud Messaging and Firestore Storage will remain available, free-tier accessible, and functionally stable for the duration of the project; any breaking changes to these services are out of scope for this SRS.	Technical
ASM-008	Google Maps Platform or Mapbox will provide sufficient free-tier API quota for map rendering, geocoding, and marker display during development, testing, and demonstration of the Map-Based Property Discovery module.	Technical
ASM-009	Payment processing will be implemented and tested exclusively against the sandbox/test environments of SSLCommerz, bKash, or Nagad; no real monetary transactions will occur as part of this academic project.	Technical
ASM-010	All third-party services (Firestore, Google Maps Platform/Mapbox, payment sandboxes) will remain accessible from the development team's network without requiring institutional firewall exceptions or paid enterprise agreements.	Technical
ASM-011	The development team has, or will acquire through self-study, sufficient working knowledge of React.js, Node.js/Express, MySQL, and PlantUML to implement the requirements specified in Sections 4–6 within the academic timeline.	Project
ASM-012	The Incremental Development Model described in Section 1 will be followed, with each increment corresponding to a coherent subset of the functional areas listed in Section 1, and increments will be reviewed per the process in Section 8.1.	Project
ASM-013	The academic timeline (semester schedule) provides sufficient calendar time to complete all High-priority functional requirements (Section 4.1) and High-priority non-functional requirements (Section 4.3) before final submission.	Project
ASM-014	Team composition (number of members and their roles) will remain stable throughout the project; significant team changes, if they occur, will be handled per the mitigation strategies in Section 9.5 (Resource Risks).	Project

ID	Assumption	Category
ASM-015	All users register with truthful personal information; the system is not required to detect identity fraud beyond the manual and semi-automated verification described in the Tenant and Owner Verification module (Section 6.3, UC-07).	Data
ASM-016	Property data (listings, images, locations) entered by Property Owners and Real Estate Agencies is assumed to be accurate at the time of entry; the system validates data format and completeness but not the truthfulness of property descriptions.	Data
ASM-017	Area Intelligence statistics (e.g., average rent per area) are derived from aggregated data within the THIKANA database itself and do not depend on external government or third-party real-estate datasets unless added as a Future Enhancement.	Data
ASM-018	All monetary values within the system are recorded and displayed in Bangladeshi Taka (BDT); multi-currency support is not required for the current scope.	Data
ASM-019	All dates and timestamps recorded by the system use a single consistent timezone (Bangladesh Standard Time, UTC+6) and the ISO 8601 date format as referenced in Validation Rule VR-015 (Section 8.3).	Data
ASM-020	End users (Tenant/Buyer, Property Owner, Real Estate Agency) possess basic digital literacy sufficient to navigate a typical web application, including form entry, file uploads, and map interactions, without requiring extensive in-app tutorials.	Operational
ASM-021	System Administrators are trusted internal roles (e.g., assigned to the project team or instructor for demonstration purposes) and are not subject to the same identity verification workflow as Tenant/Buyer, Property Owner, or Real Estate Agency users.	Operational
ASM-022	The number of concurrent users during testing and demonstration will remain within a range realistic for an academic project (tens of users), and performance NFRs in Section 4.3 are defined accordingly rather than for enterprise-scale traffic.	Operational

ID	Assumption	Category
ASM-023	Rental agreements generated by the system (Section 6.3, UC-09) are treated as digital records for platform tracking purposes; they do not replace legally binding physical contracts unless Digital Signature Integration (Appendix C) is implemented.	Legal/Regulatory
ASM-024	The platform does not independently verify land ownership or property title with government land registries; the Tenant and Owner Verification module relies on user-submitted documents and administrator review only, unless Government NID Integration (Appendix C) is implemented.	Legal/Regulatory
ASM-025	Data protection practices implemented for this academic project (Section 4.3, Security and Compliance NFRs) are designed to demonstrate sound software engineering principles and are not represented as full compliance with any specific national data protection law.	Legal/Regulatory
ASM-026	Stakeholder needs captured in Sections 1–2 remain valid and unchanged for the duration of the project unless formally revised through the change control process described in Section 8.1.	Project
ASM-027	Hardware for hosting the application during development and demonstration (development laptops, a shared staging server, or a free-tier cloud instance) is sufficient to run the React.js frontend, Node.js/Express backend, and MySQL database simultaneously.	Technical

APPENDIX E — REFERENCES

This appendix lists the standards, technical documentation, textbooks, and online resources consulted during the preparation of this Software Requirements Specification for THIKANA — Smart Real Estate & Rental Management Platform. References are listed in the order in which their associated standards, tools, or concepts are referenced throughout the document.

E.1 Standards and Methodology References

- [1] IEEE Computer Society, IEEE Recommended Practice for Software Requirements Specifications, IEEE Std 830-1998, Institute of Electrical and Electronics Engineers, 1998.
- [2] International Organization for Standardization, Systems and Software Engineering — Life Cycle Processes — Requirements Engineering, ISO/IEC/IEEE 29148:2018, 2018.
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- [6] A. Cockburn, Writing Effective Use Cases. Boston, MA, USA: Addison-Wesley, 2000.

E.2 Modeling and Diagramming Tools

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E.3 Frontend Technology References

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E.5 Supporting Services and APIs

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E.6 Domain and Regulatory References

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- [22] Bangladesh Bureau of Statistics, “Administrative Divisions, Districts, and Upazilas of Bangladesh,” [Online]. Available: <http://www.bbs.gov.bd/>

E.7 Security and Quality References

- [23] OWASP Foundation, “OWASP Top Ten Web Application Security Risks,” [Online]. Available: <https://owasp.org/www-project-top-ten/>
- [24] National Institute of Standards and Technology, Security and Privacy Controls for Information Systems and Organizations, NIST Special Publication 800-53, Rev. 5, 2020.
- [25] International Organization for Standardization, Systems and Software Quality Requirements and Evaluation (SQuaRE) — System and Software Quality Models, ISO/IEC 25010:2011, 2011.